

(12) **United States Patent**
Stengel et al.

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(54) **GAZE ESTIMATION USING ONE OR MORE NEURAL NETWORKS**

2207/10016 (2013.01); G06T 2207/20081 (2013.01); G06T 2207/20084 (2013.01); G06T 2207/30201 (2013.01)

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(58) **Field of Classification Search**
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See application file for complete search history.

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(73) Assignee: **NVIDIA Corporation**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(51) **Int. Cl.**

G06K 9/00	(2022.01)
G06F 3/01	(2006.01)
G06N 3/08	(2023.01)
G06T 7/246	(2017.01)
G06T 7/70	(2017.01)
G06V 10/82	(2022.01)
G06V 20/20	(2022.01)
G06V 40/19	(2022.01)

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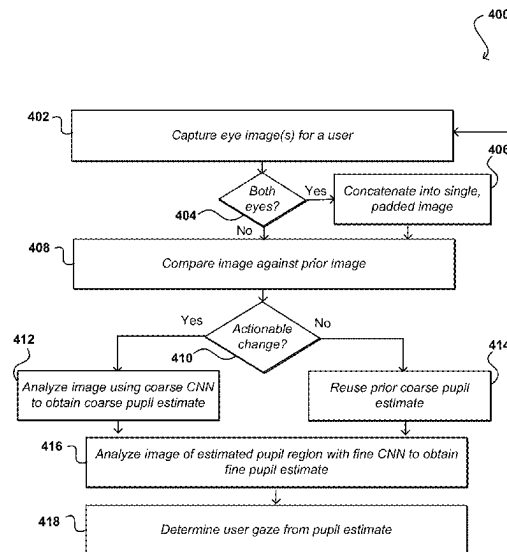
(52) **U.S. Cl.**

CPC **G06F 3/013** (2013.01); **G06N 3/08** (2013.01); **G06T 7/246** (2017.01); **G06T 7/70** (2017.01); **G06V 10/82** (2022.01); **G06V 20/20** (2022.01); **G06V 40/19** (2022.01); **G06T**

(57) **ABSTRACT**

Apparatuses, systems, and techniques are presented to estimate user gaze. In at least one embodiment, one or more neural networks are used to determine coarse and fine gaze estimates for one or more users.

31 Claims, 46 Drawing Sheets



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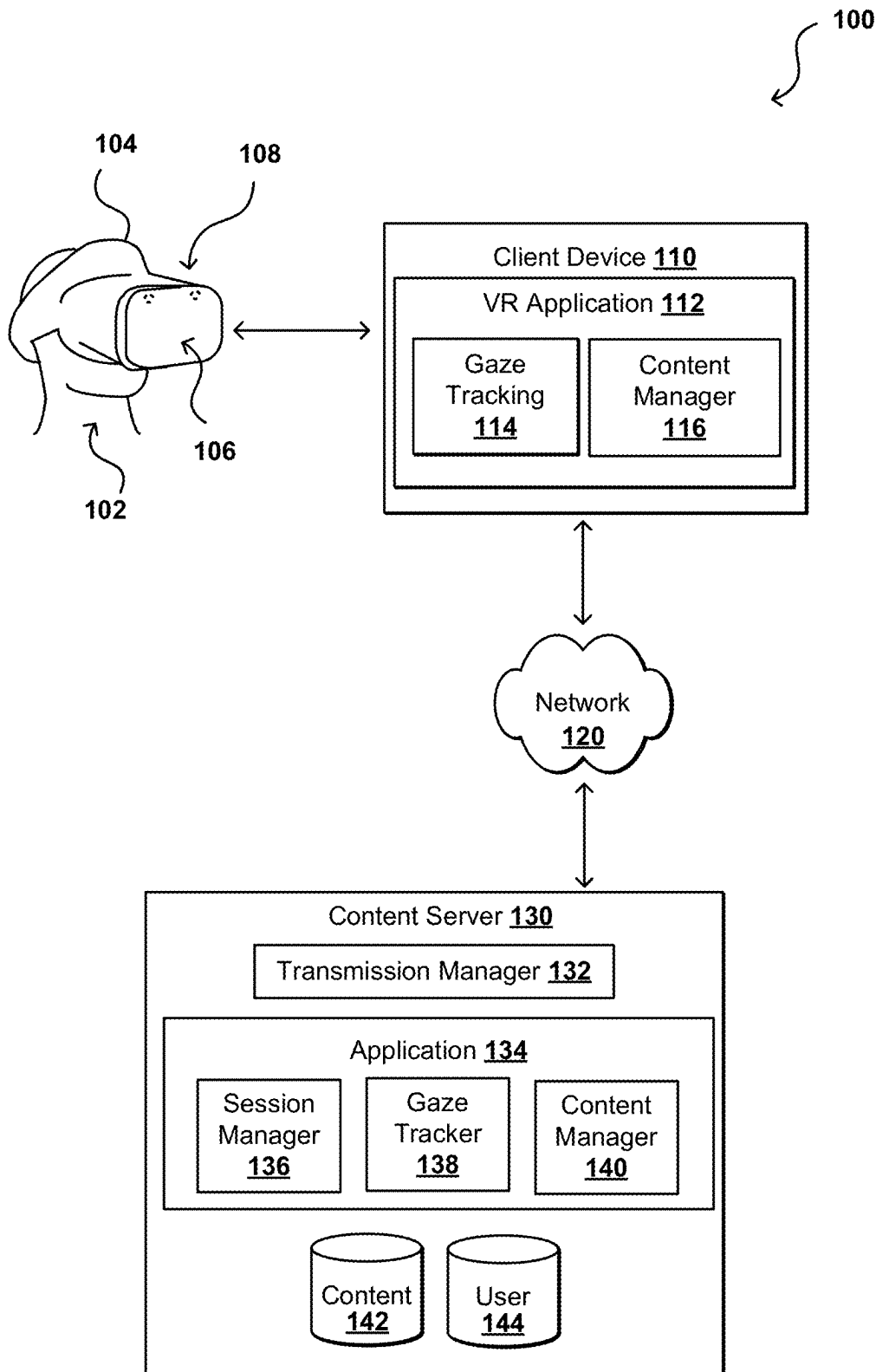


FIG. 1

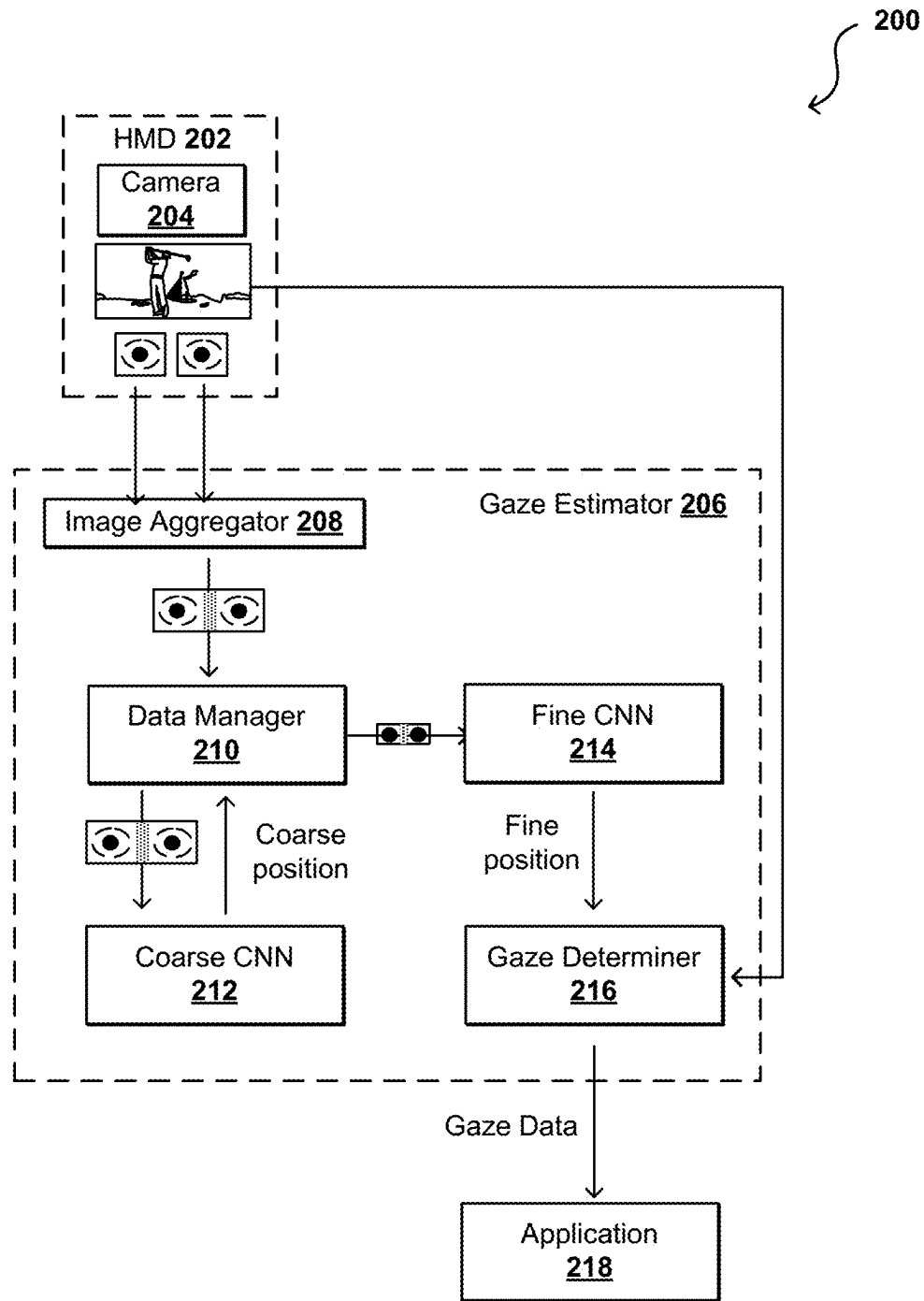


FIG. 2

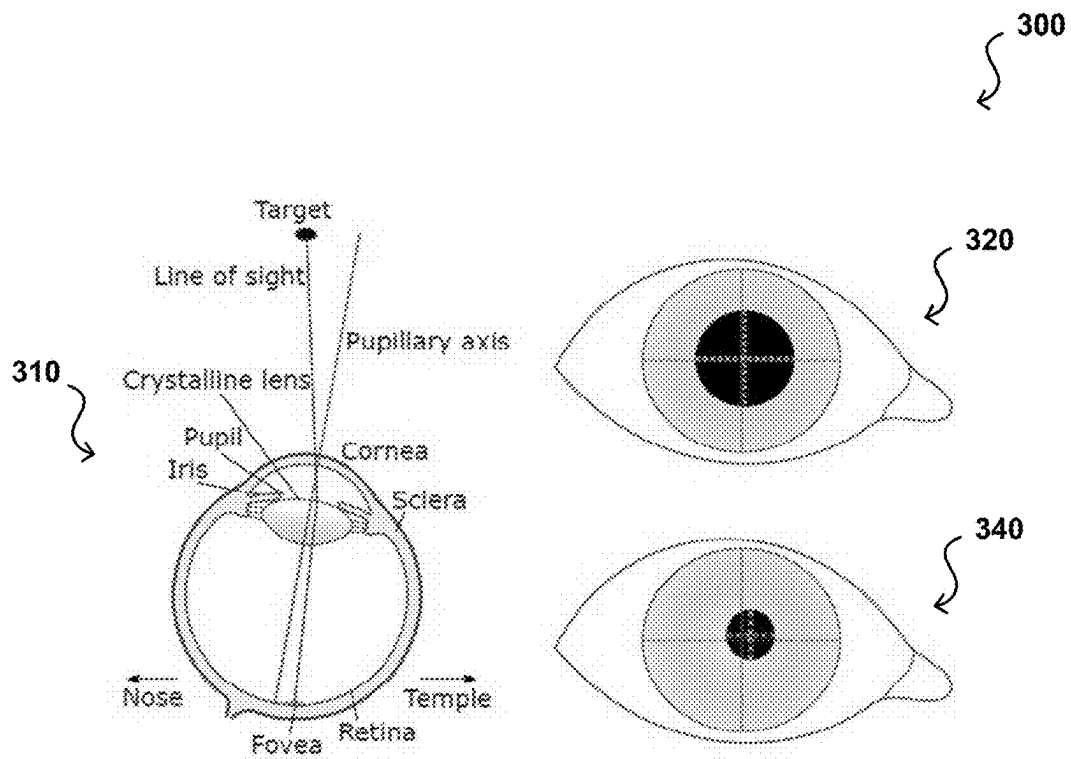


FIG. 3A

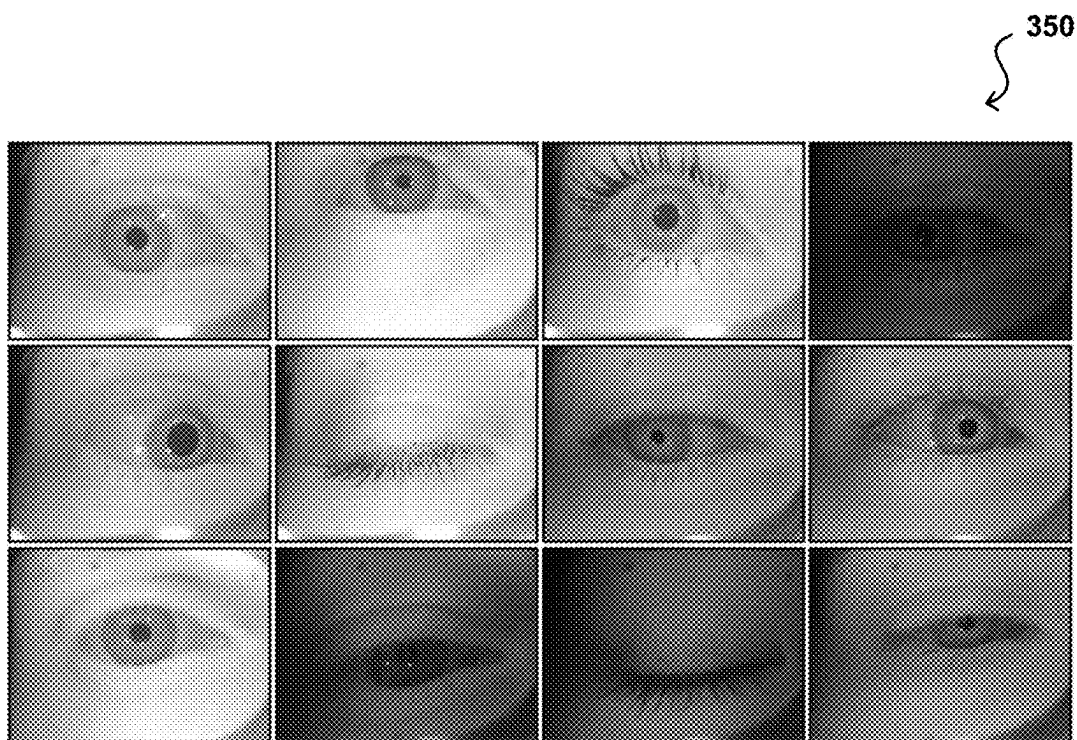


FIG. 3B

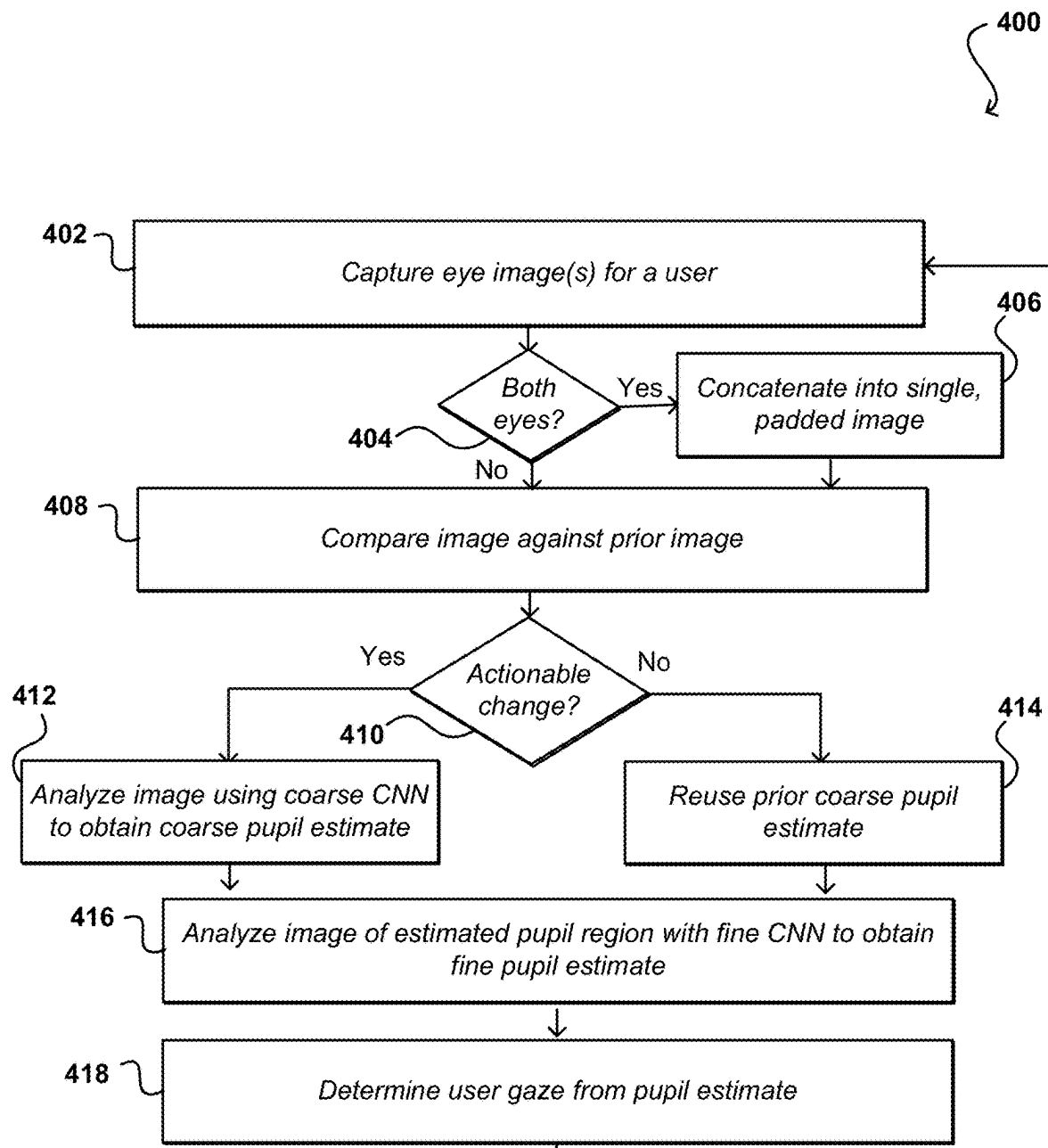


FIG. 4

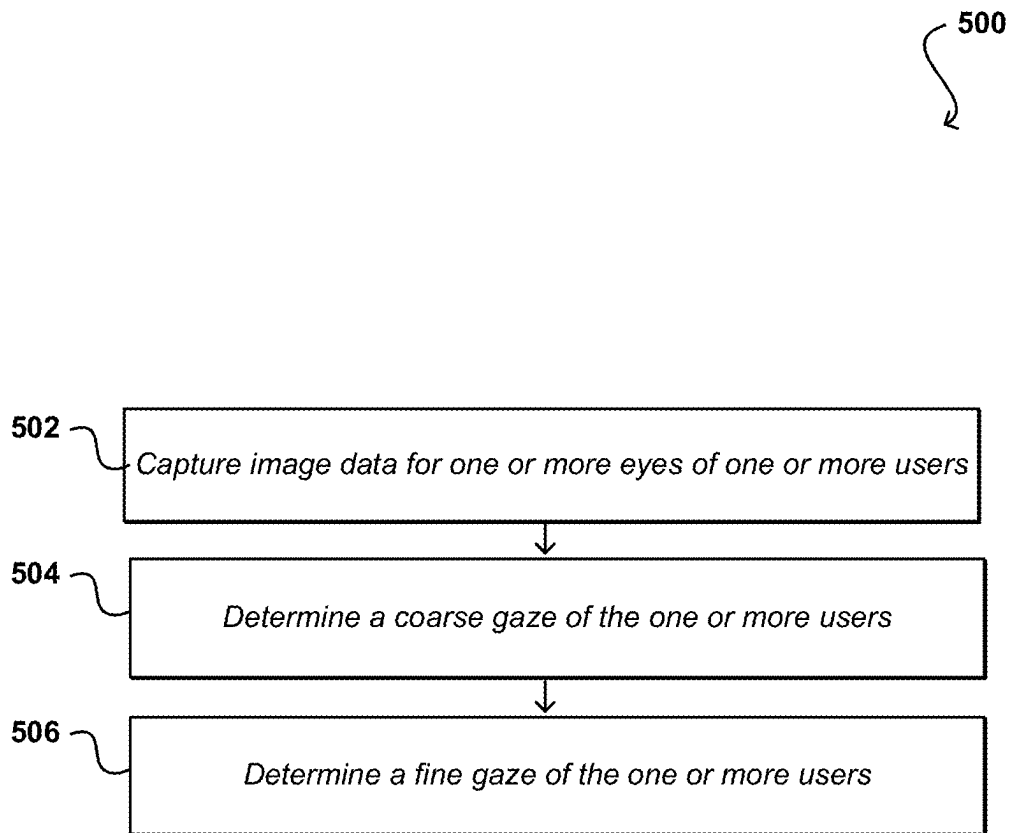


FIG. 5

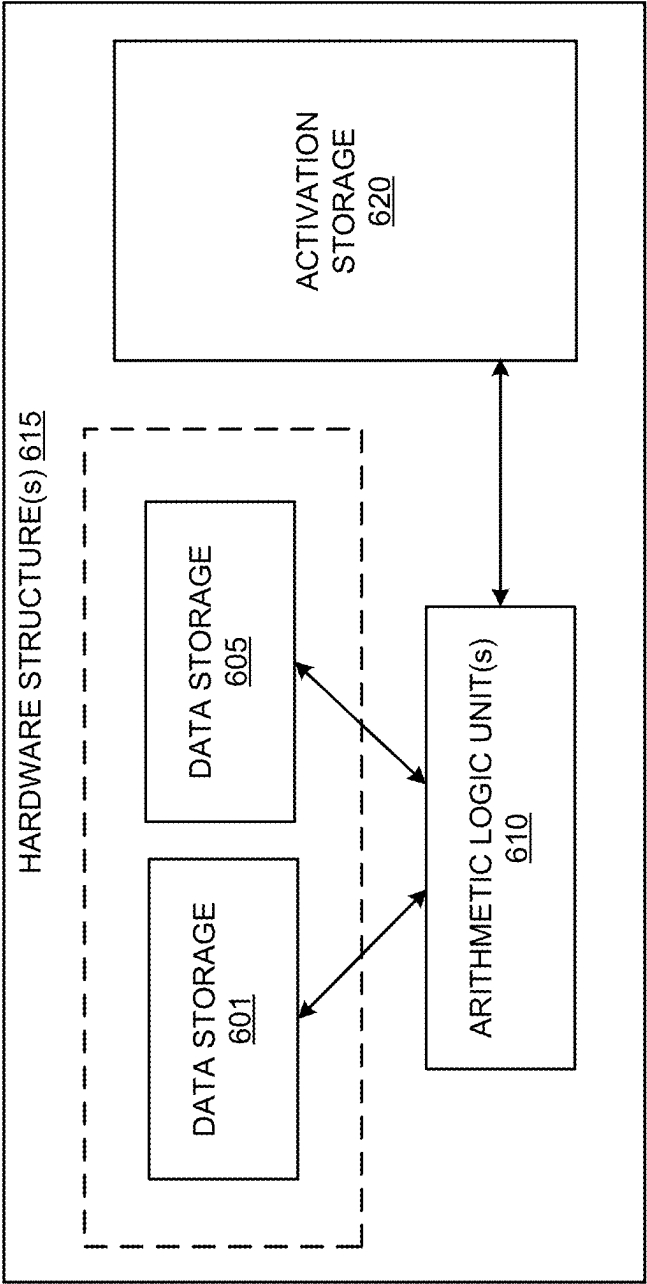


FIG. 6A

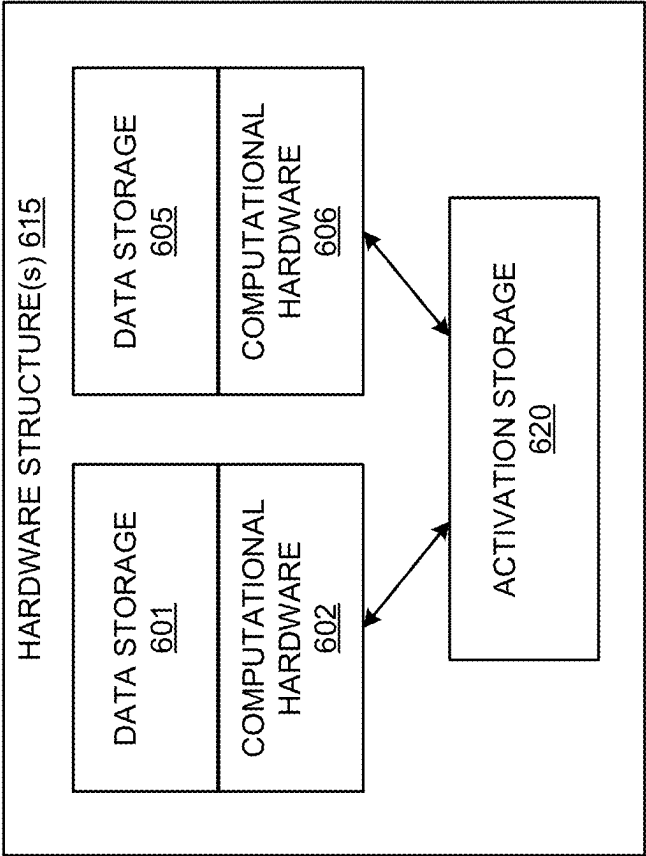


FIG. 6B

DATA CENTER
700

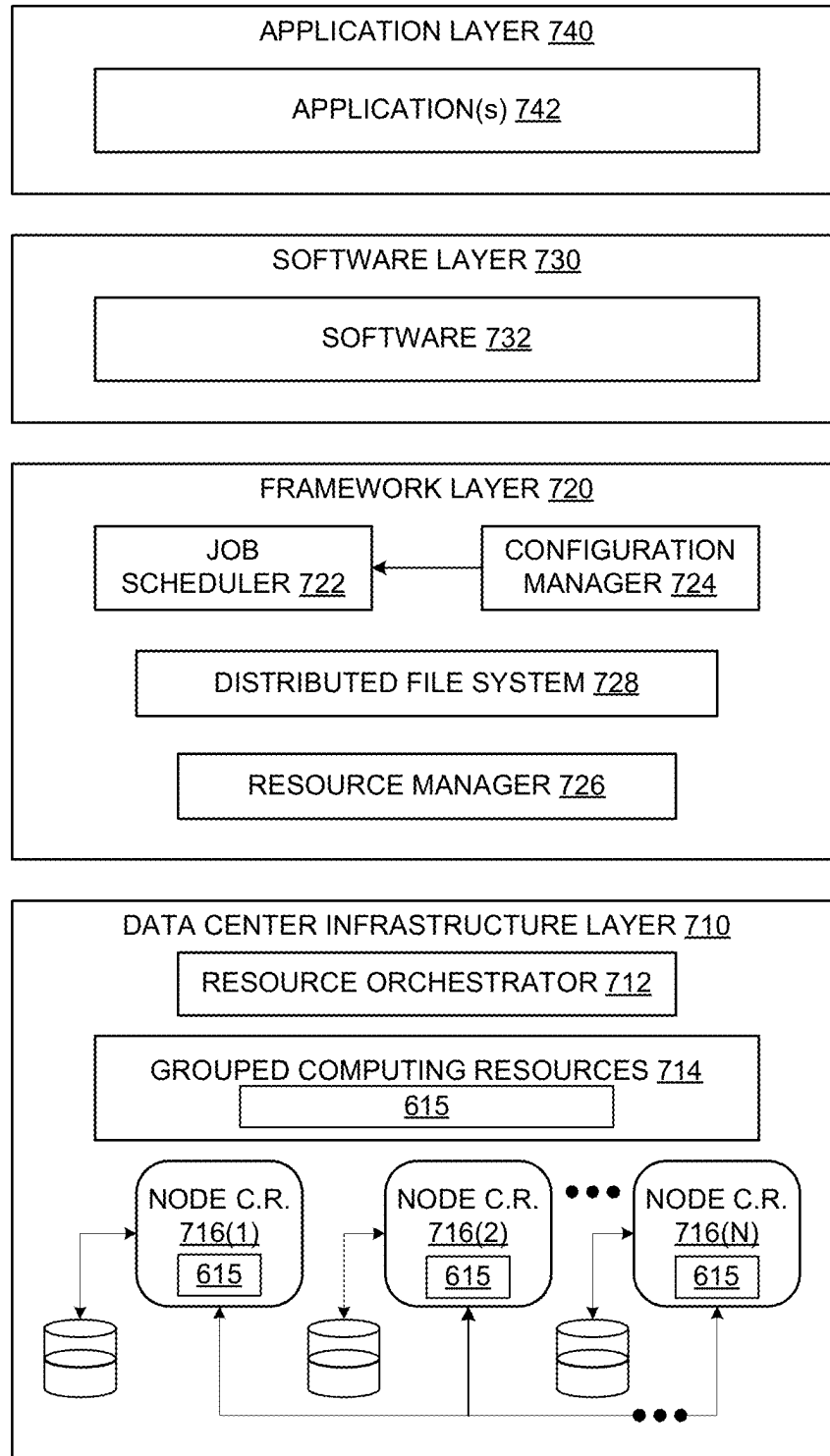


FIG. 7

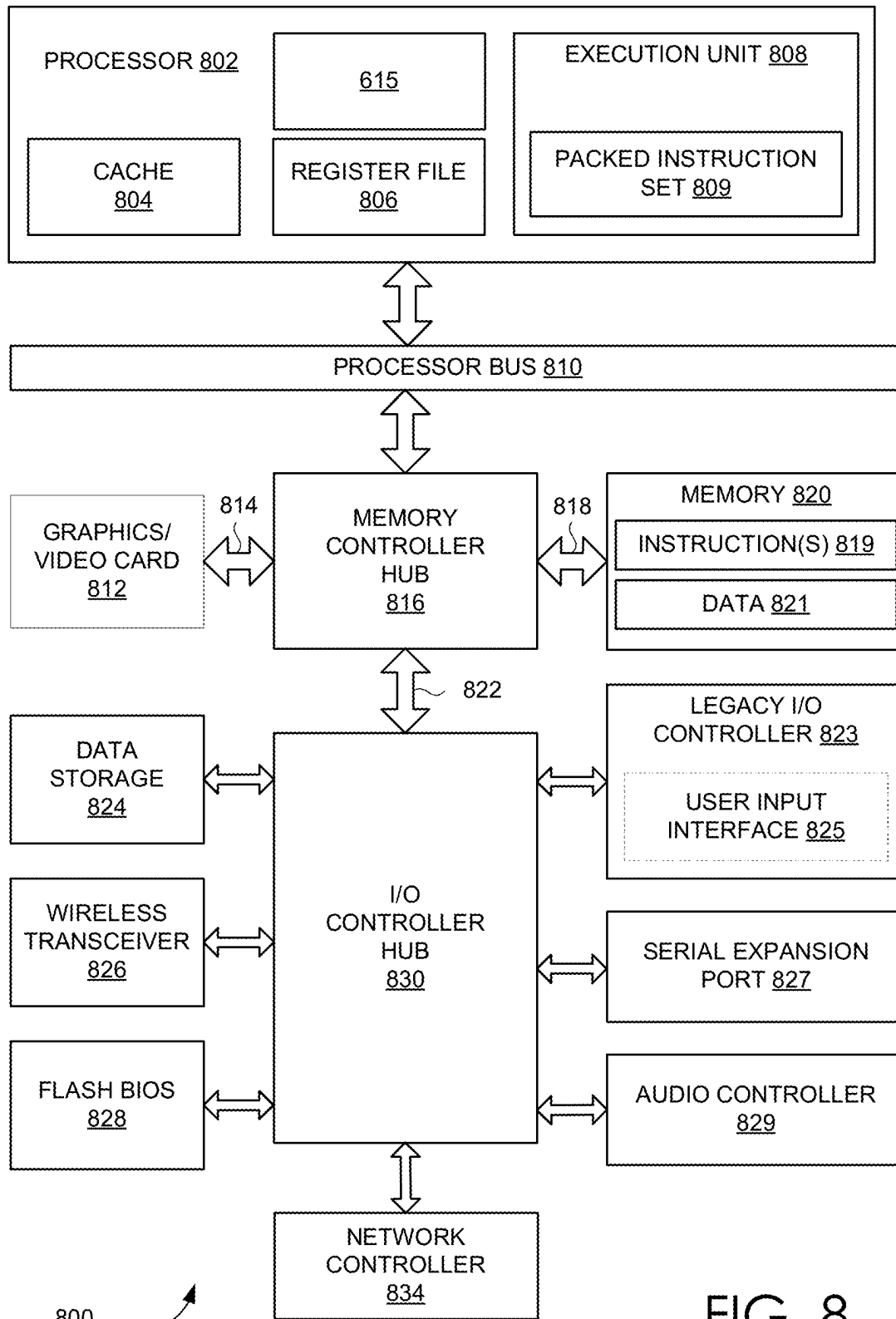


FIG. 8

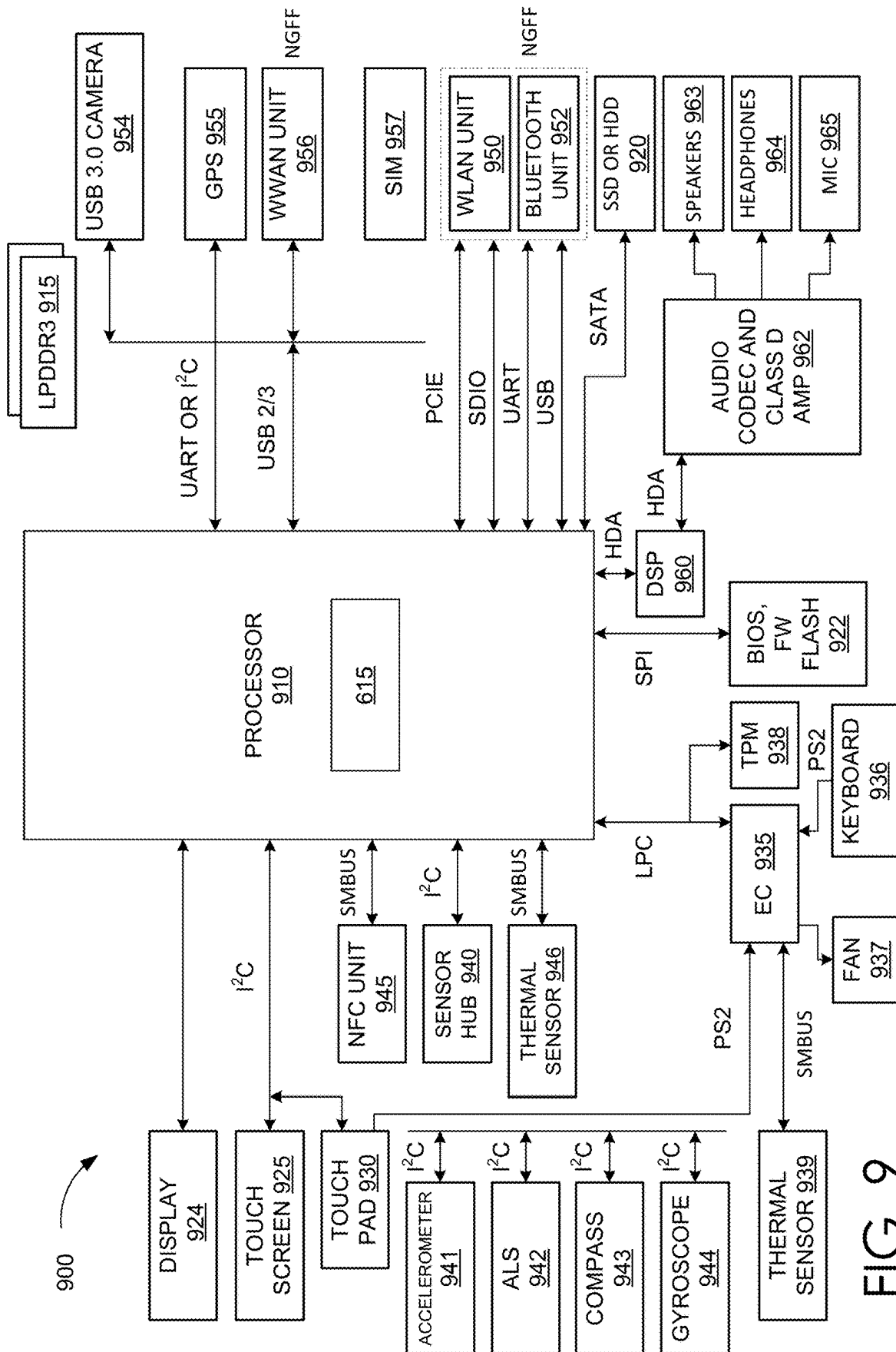


FIG. 9

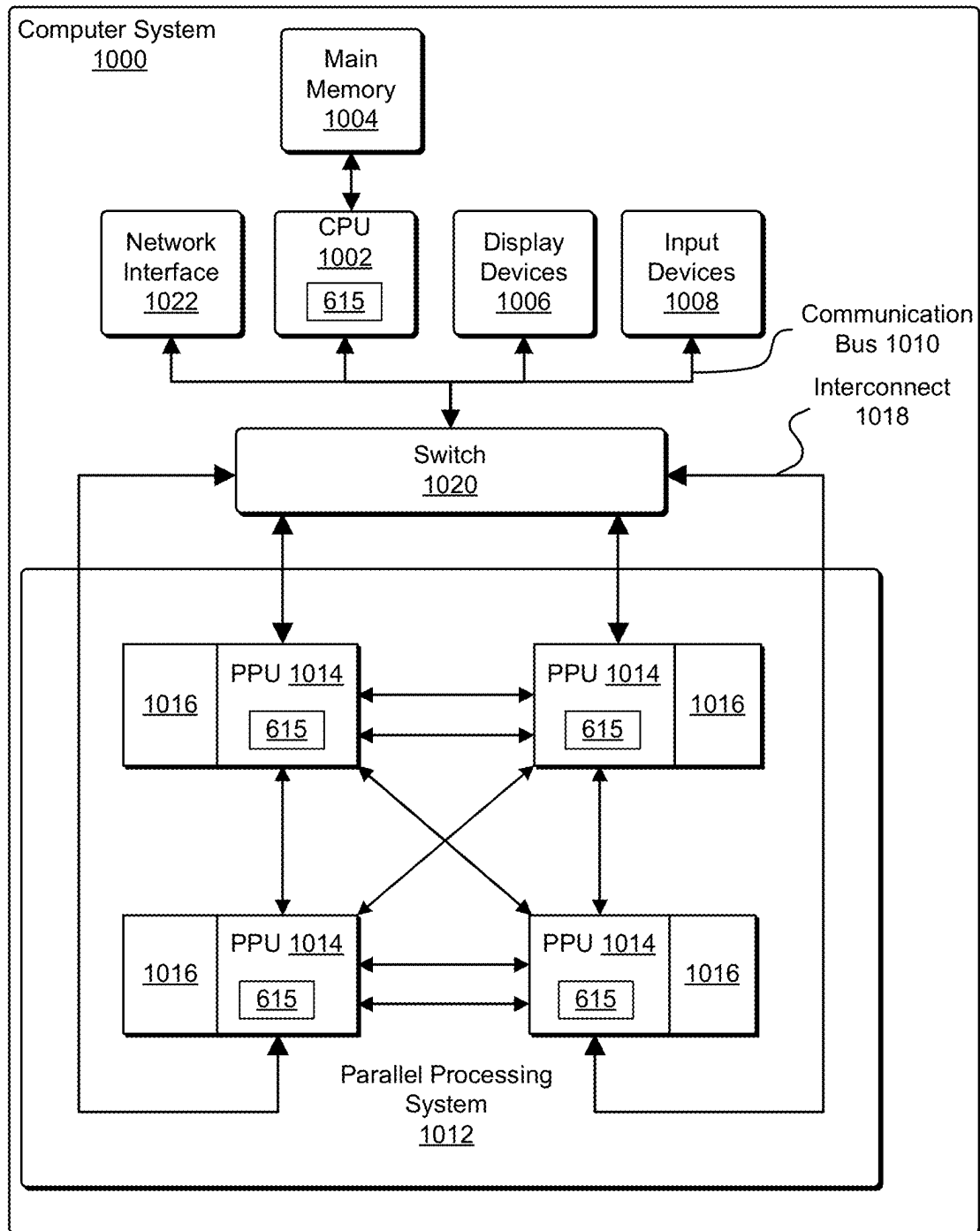


FIG. 10

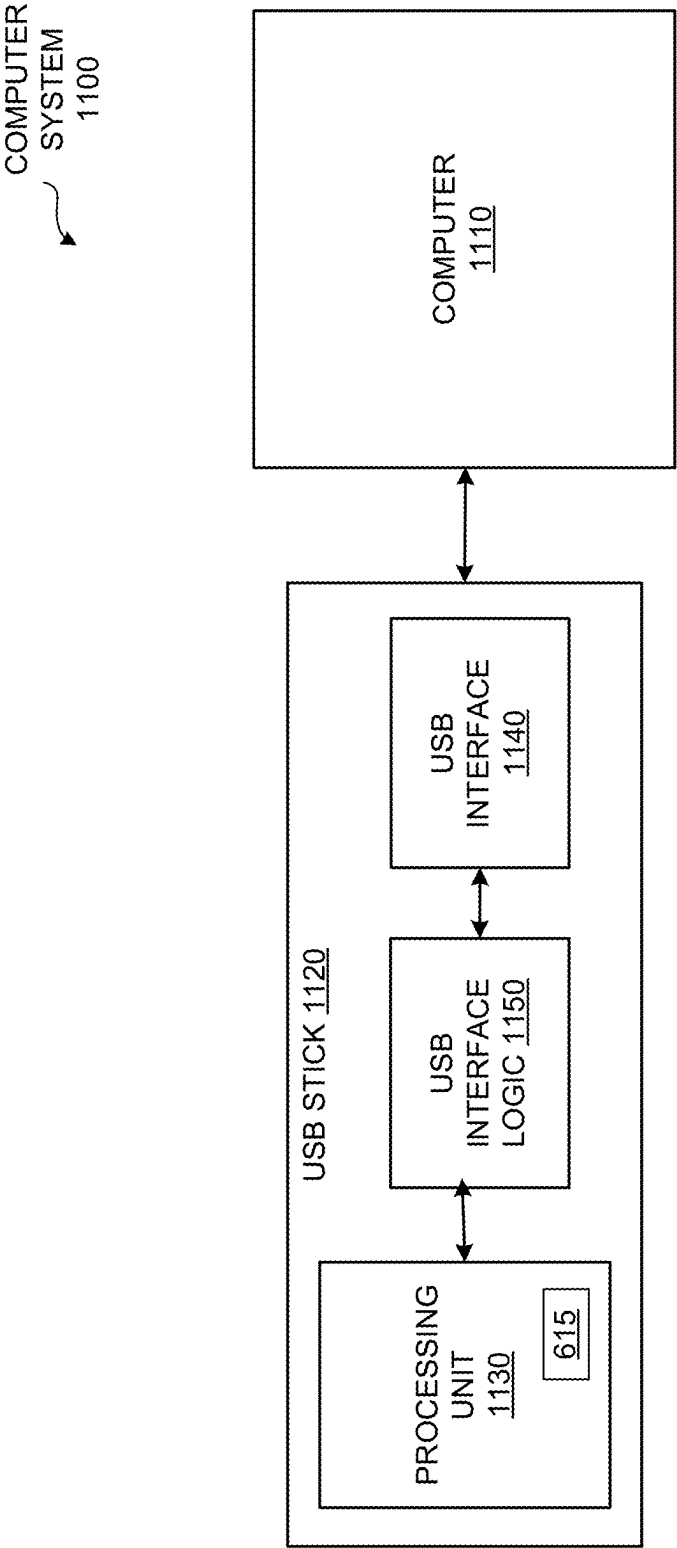


FIG. 11

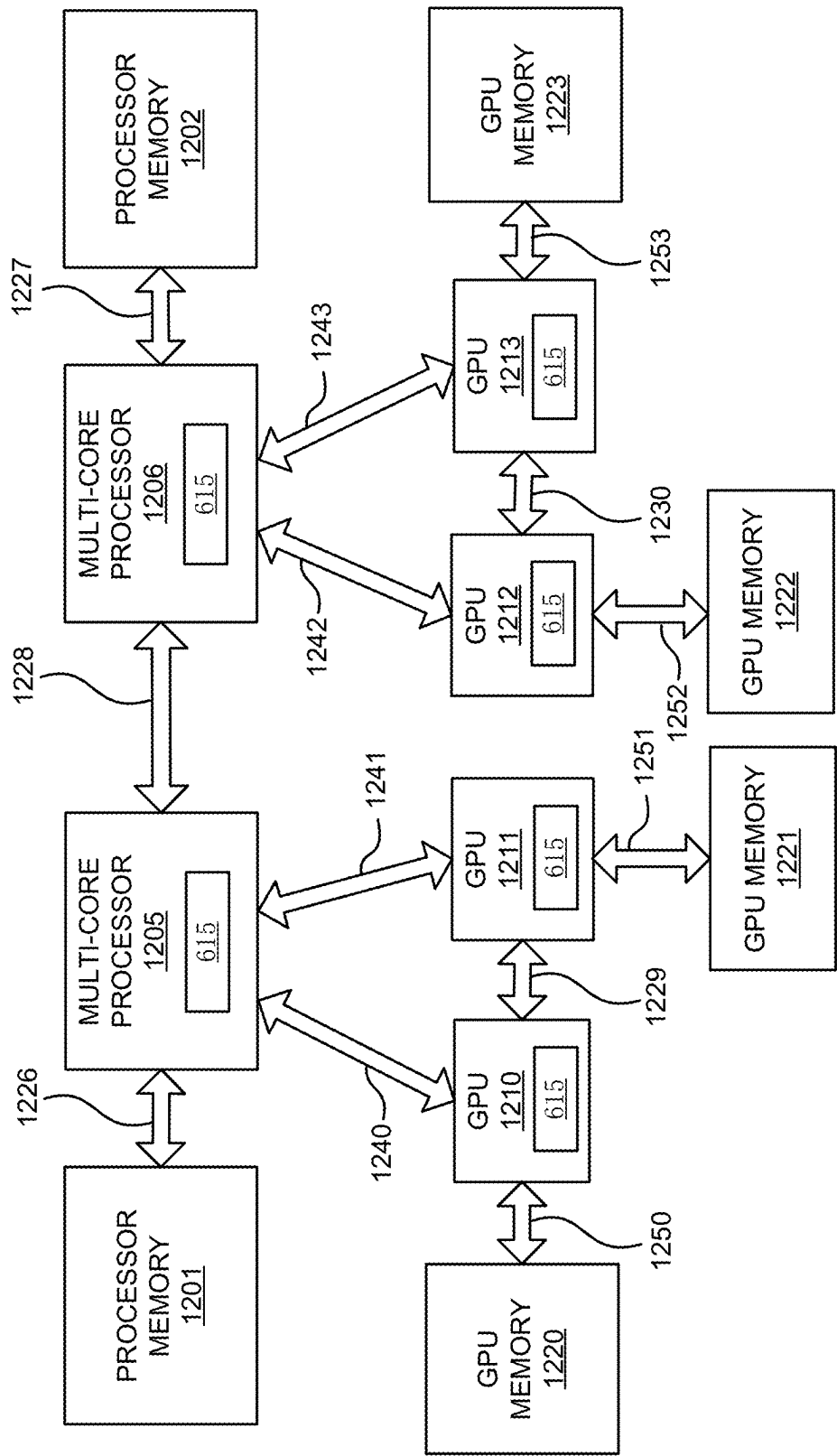


FIG. 12A

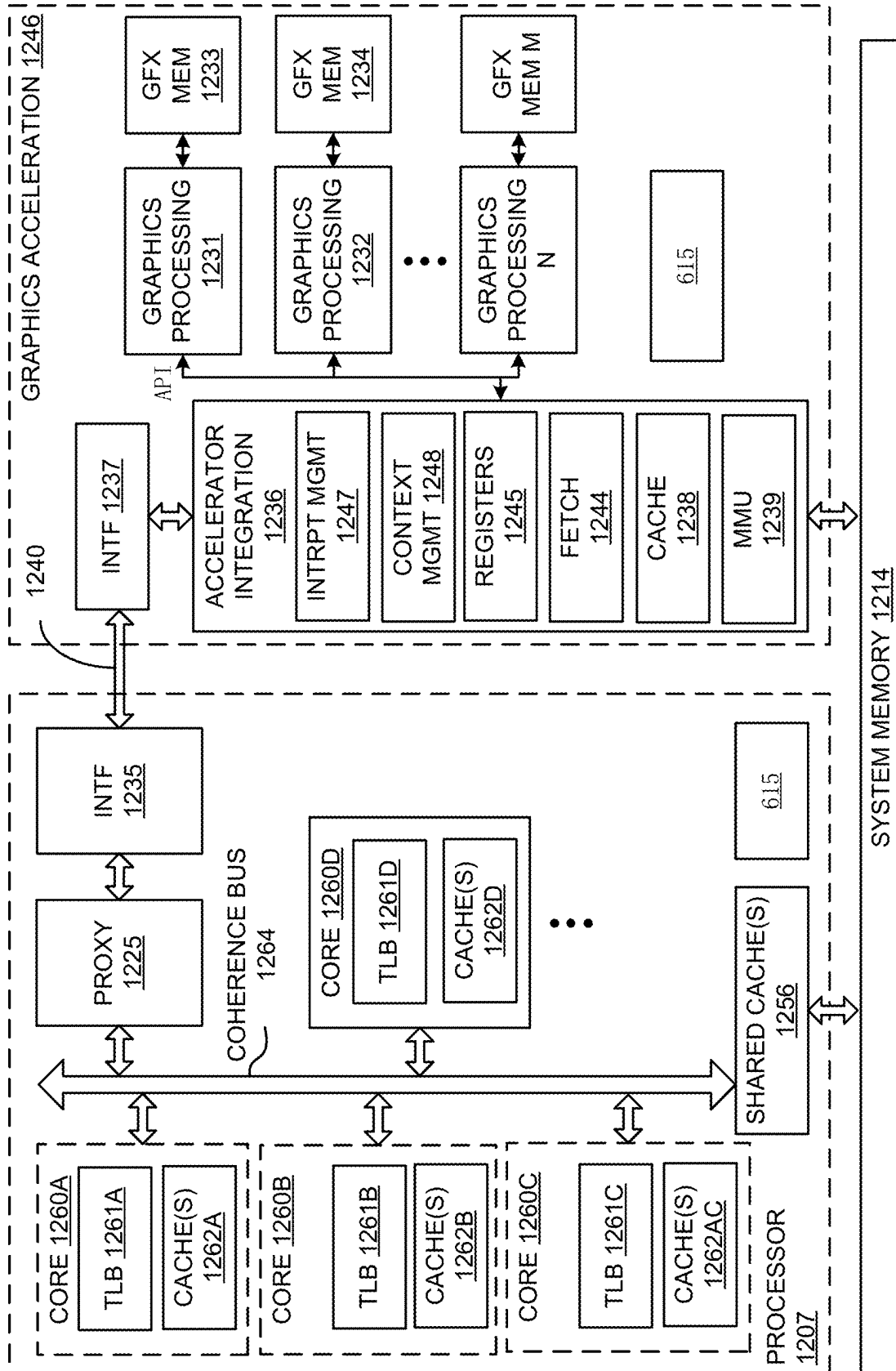


FIG. 12B

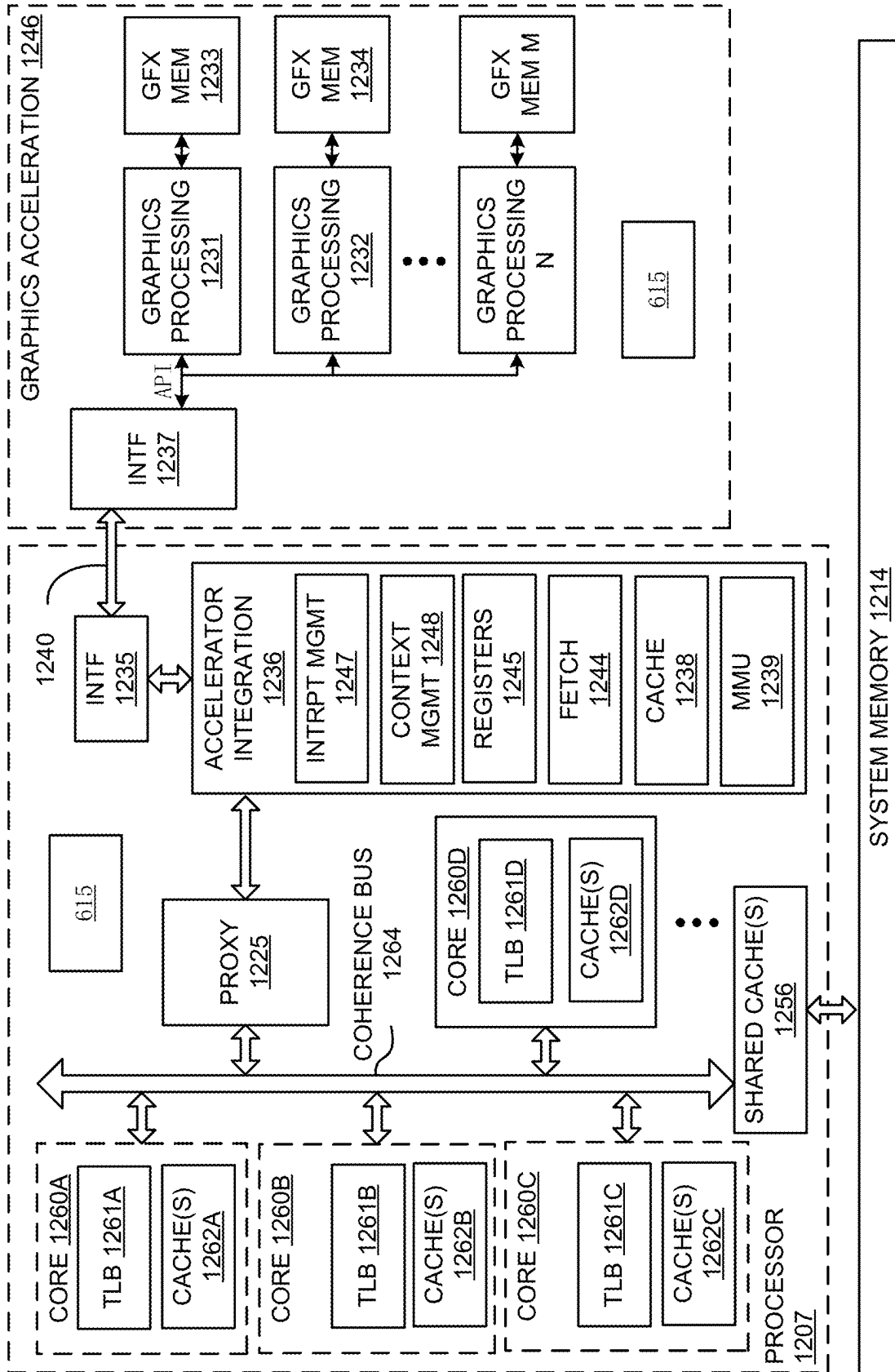


FIG. 12C

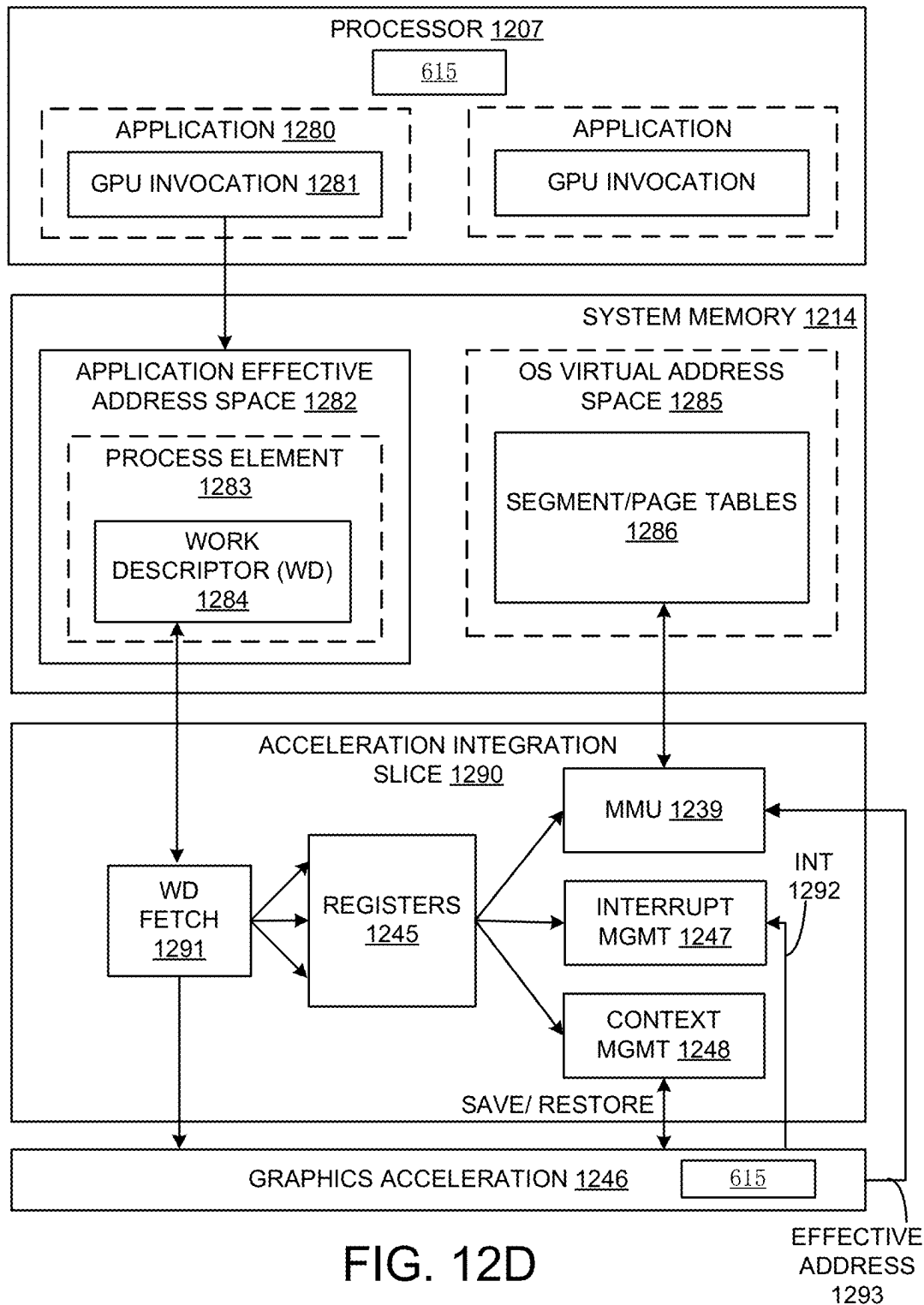


FIG. 12D

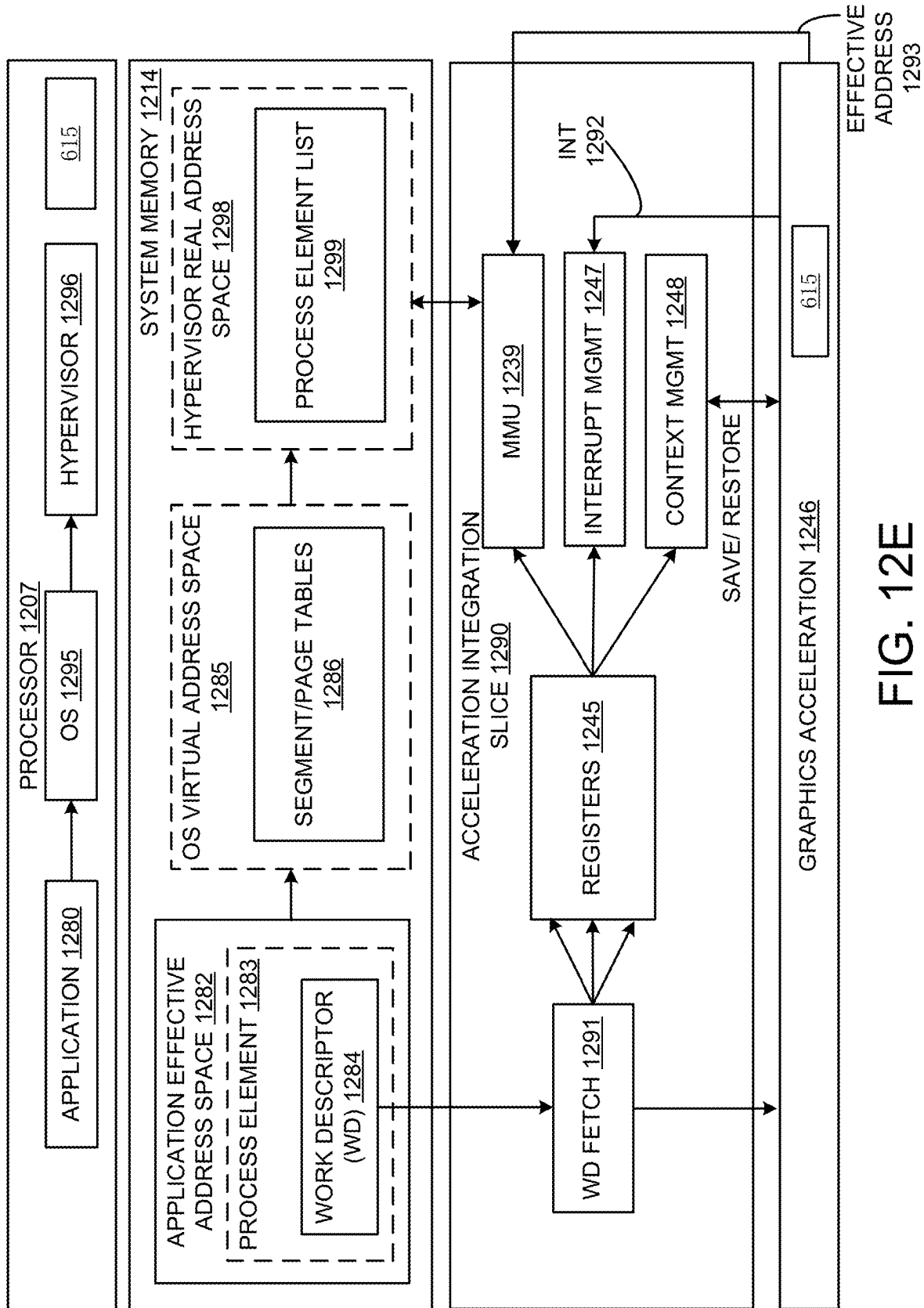


FIG. 12E

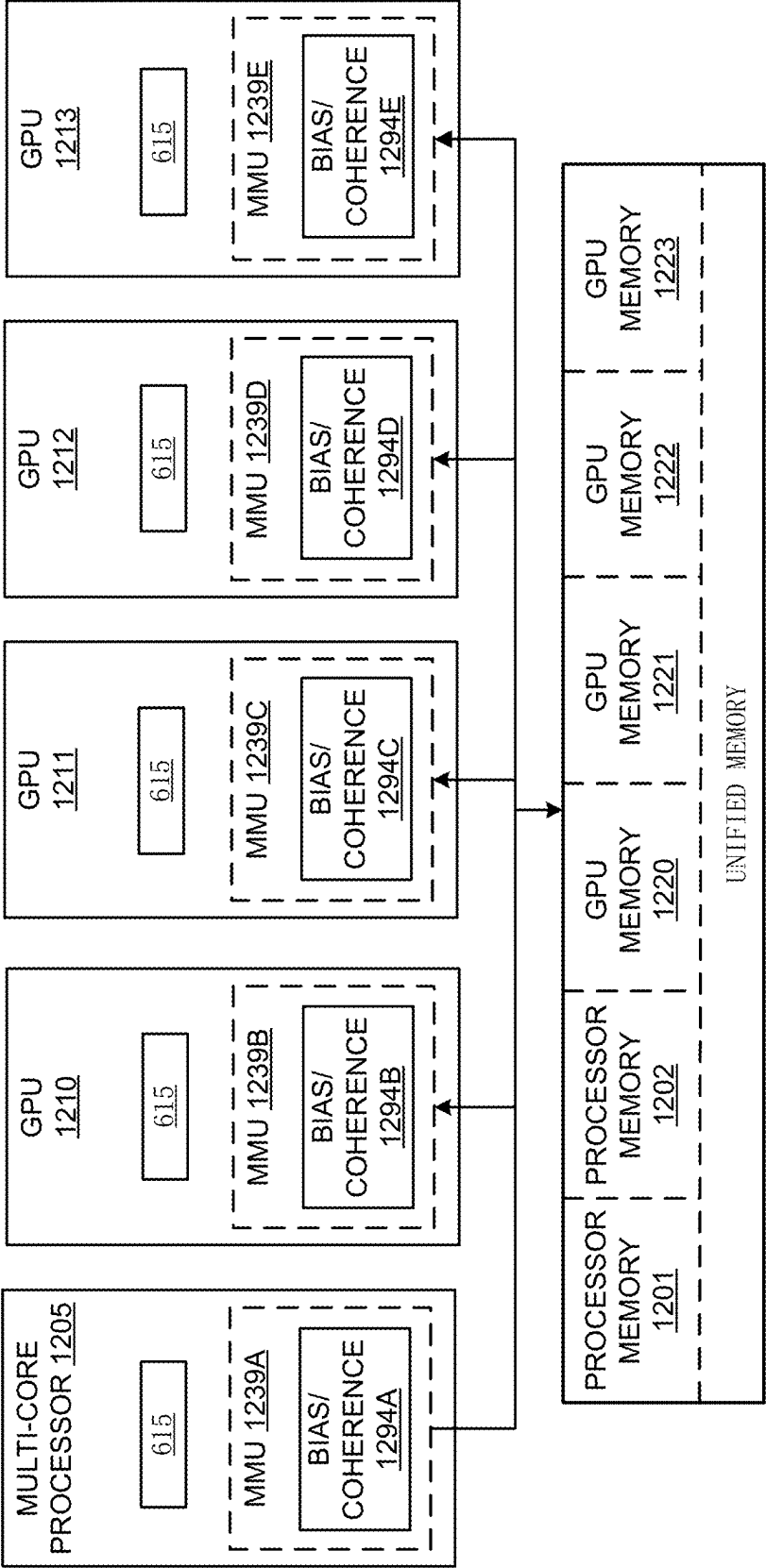


FIG. 12F

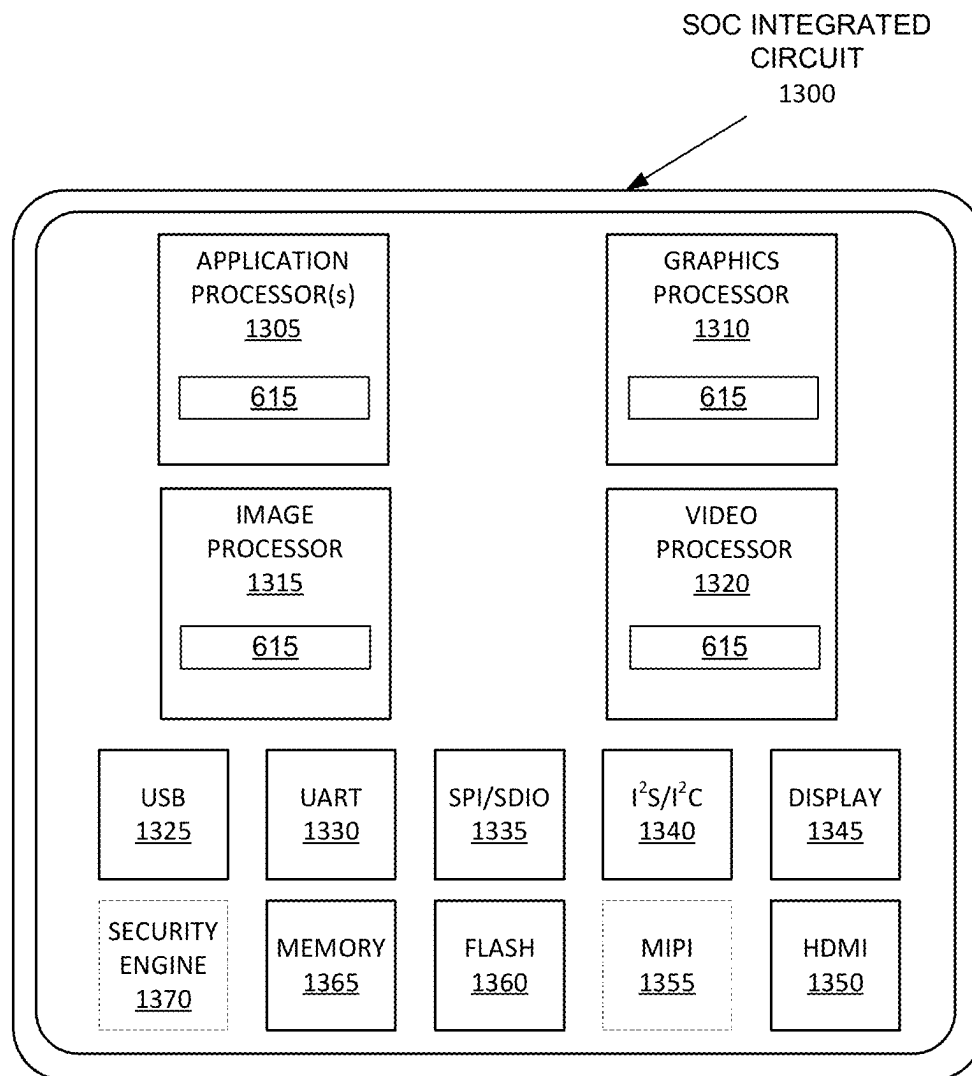


FIG. 13

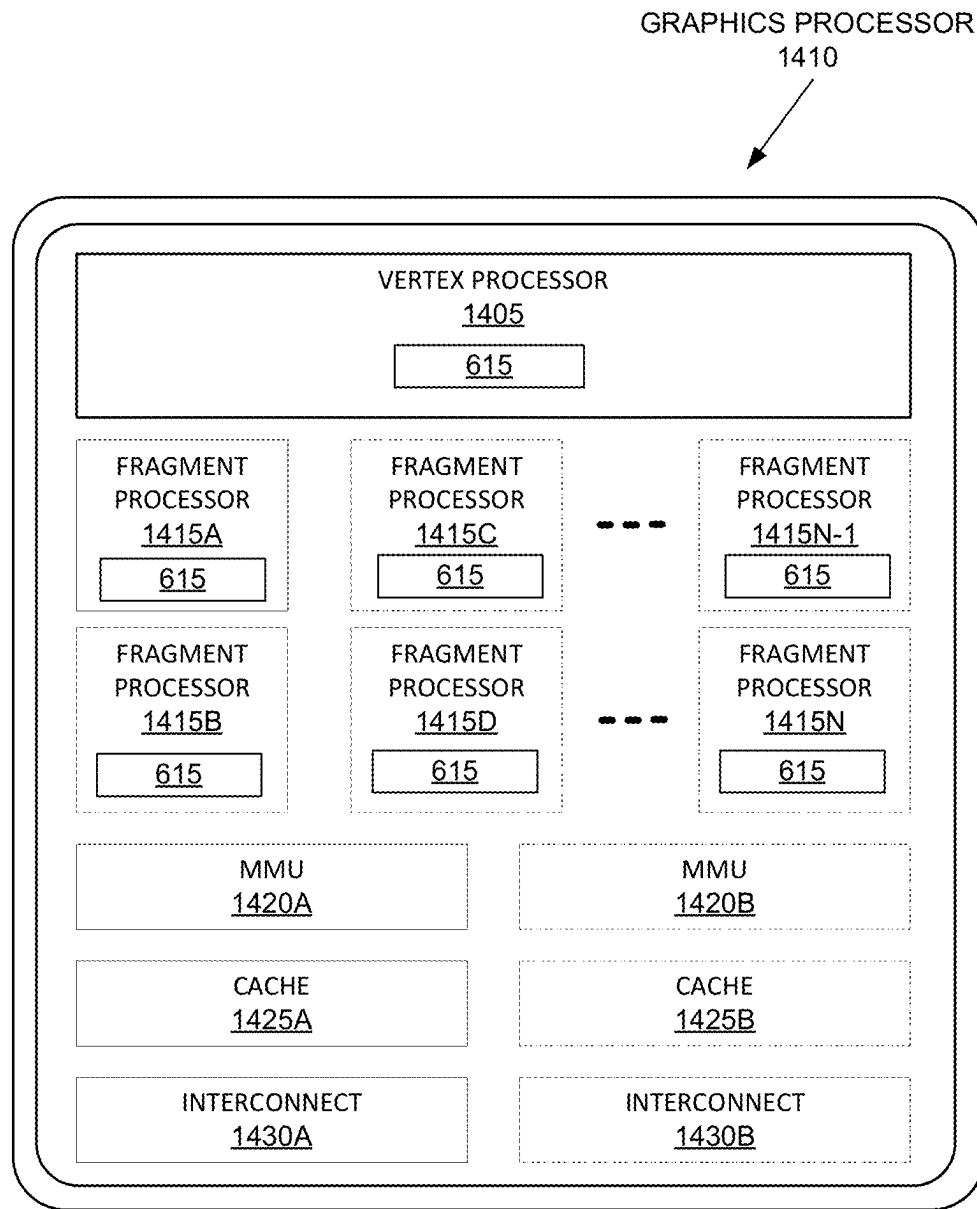


FIG. 14A

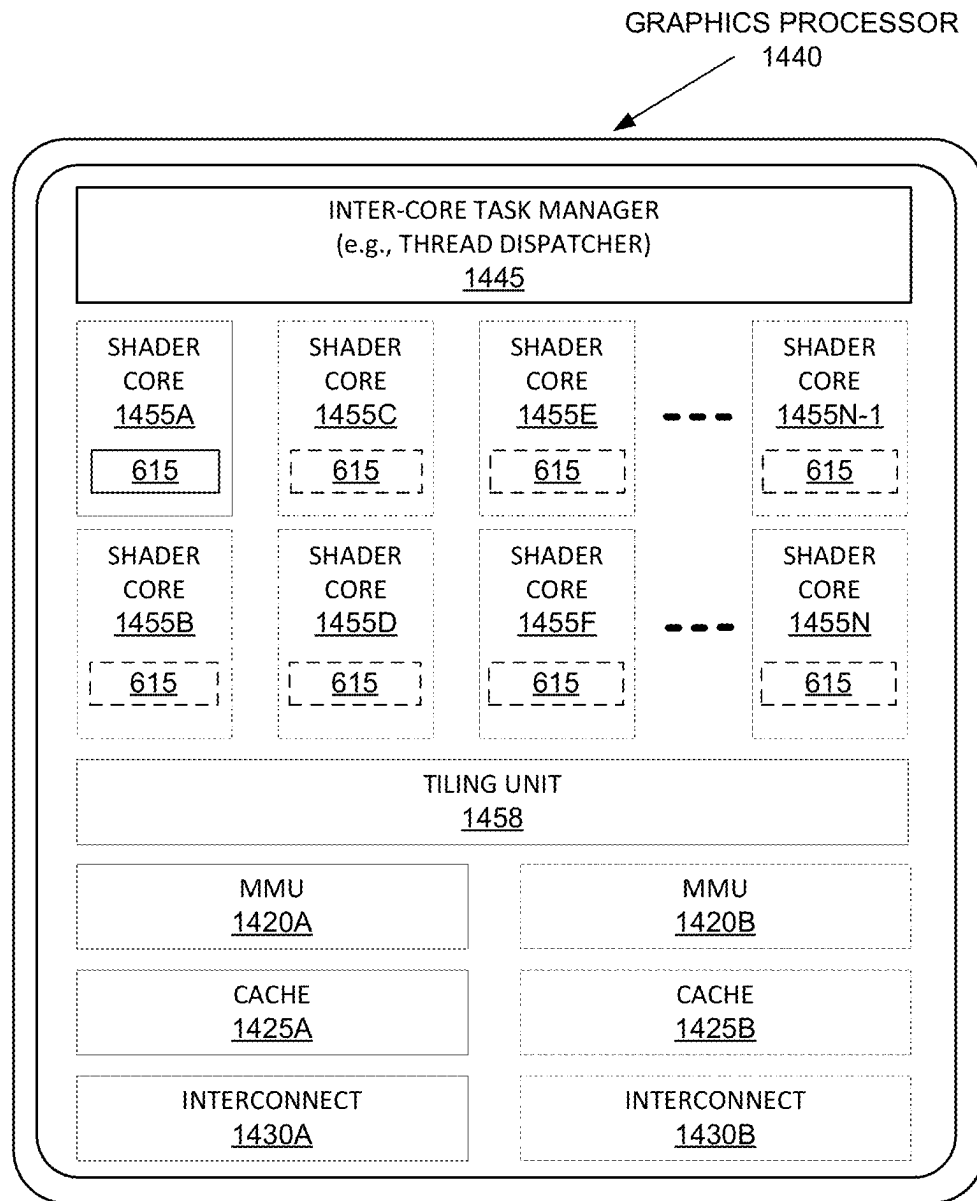


FIG. 14B

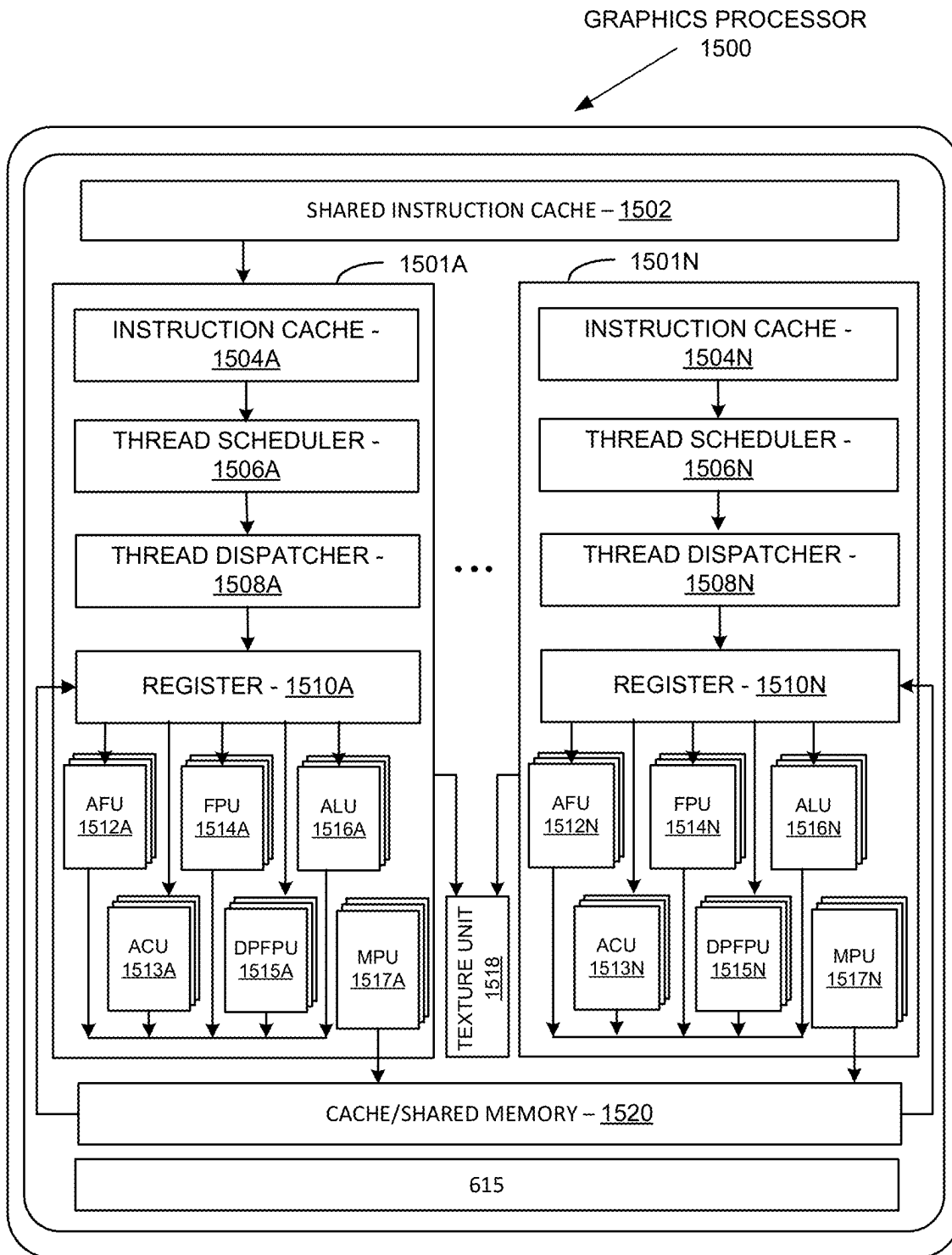


FIG. 15A

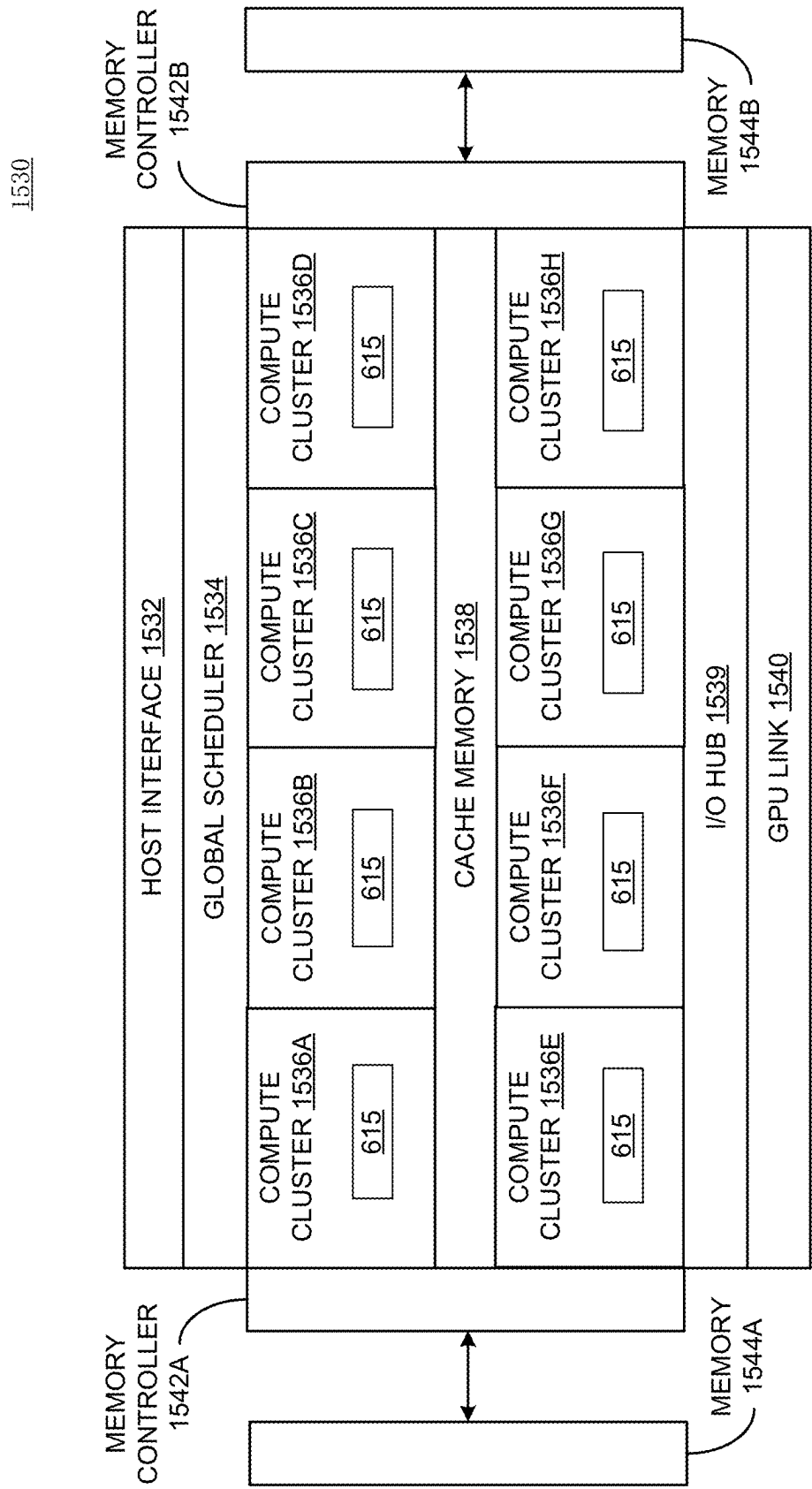


FIG. 15B

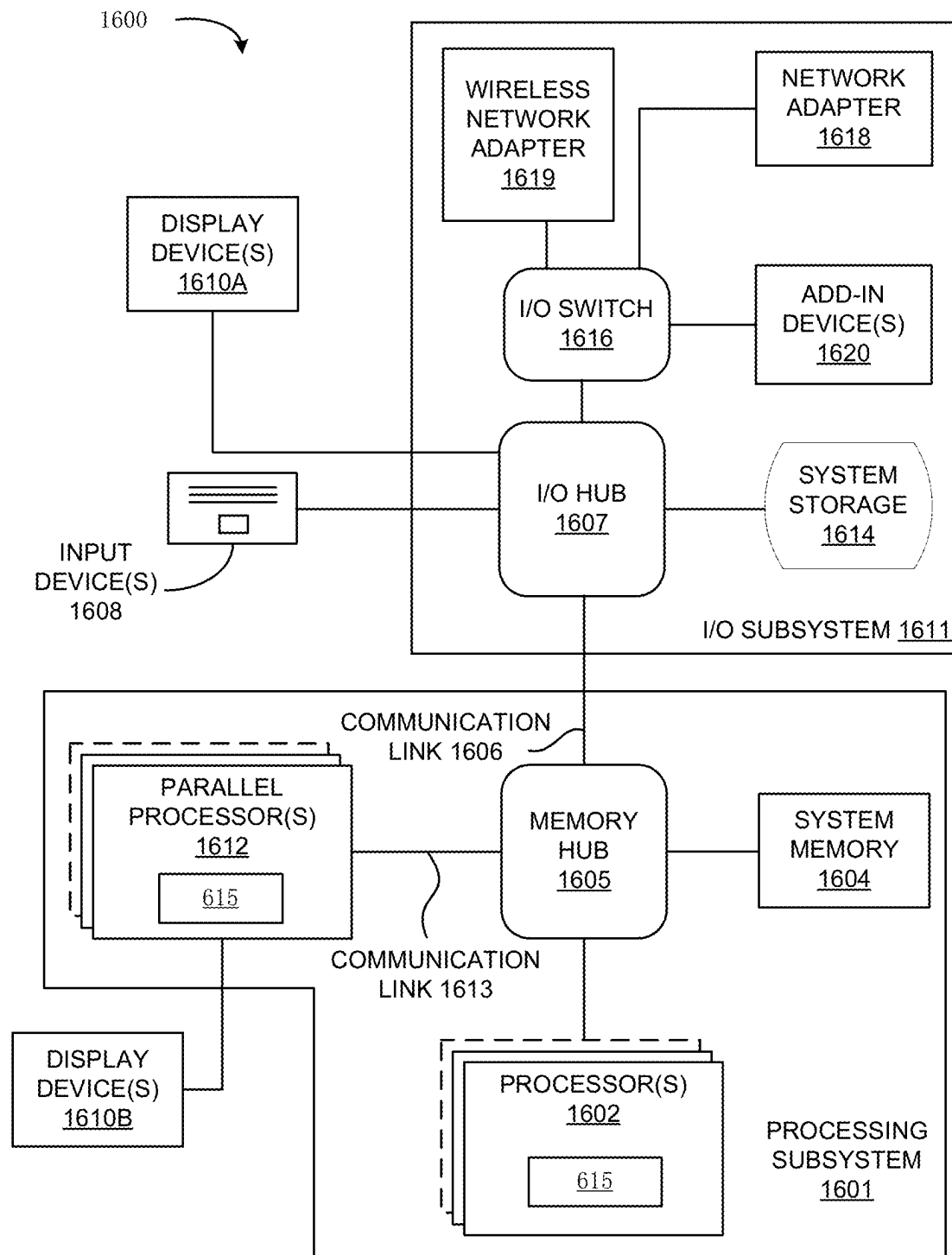


FIG. 16

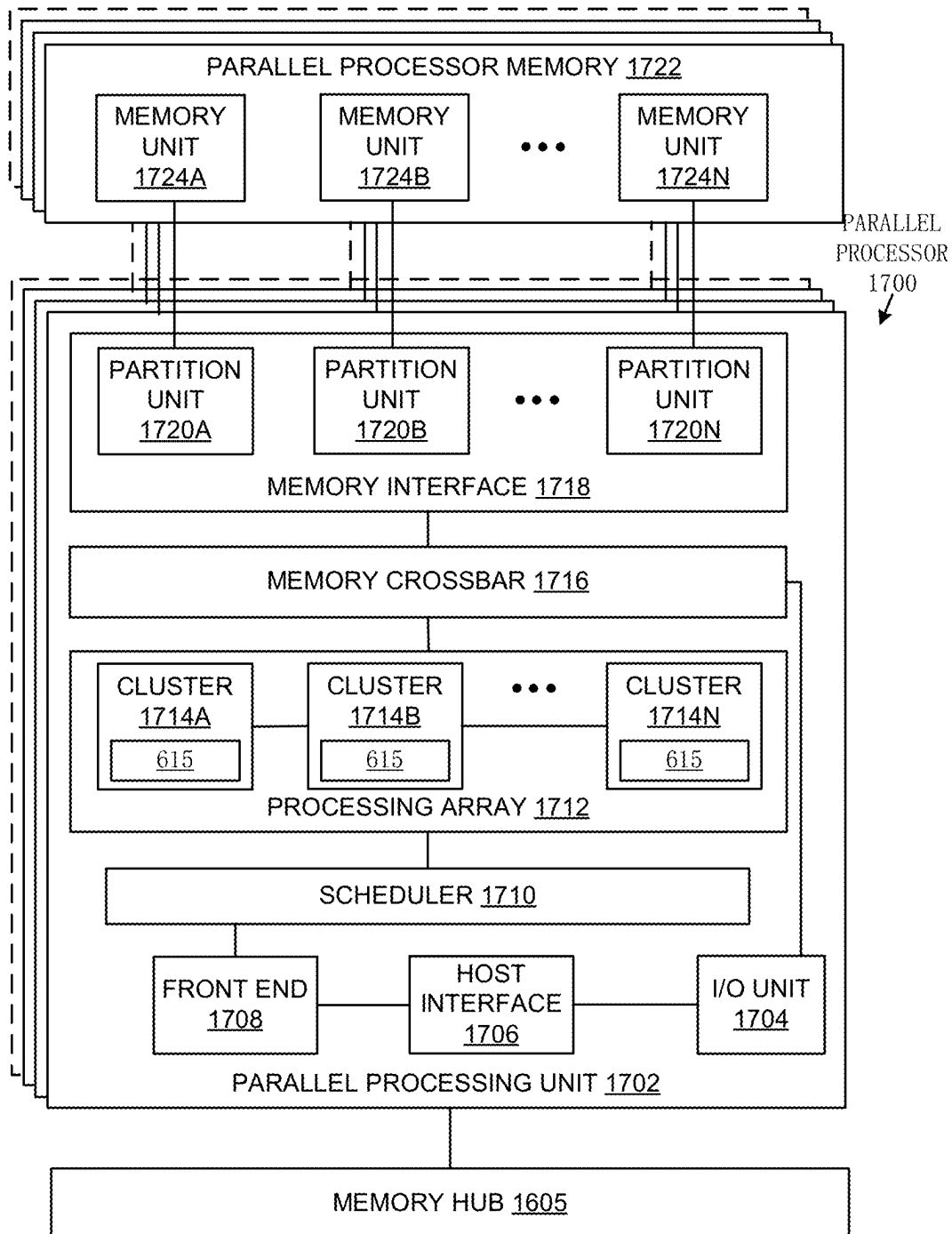


FIG. 17A

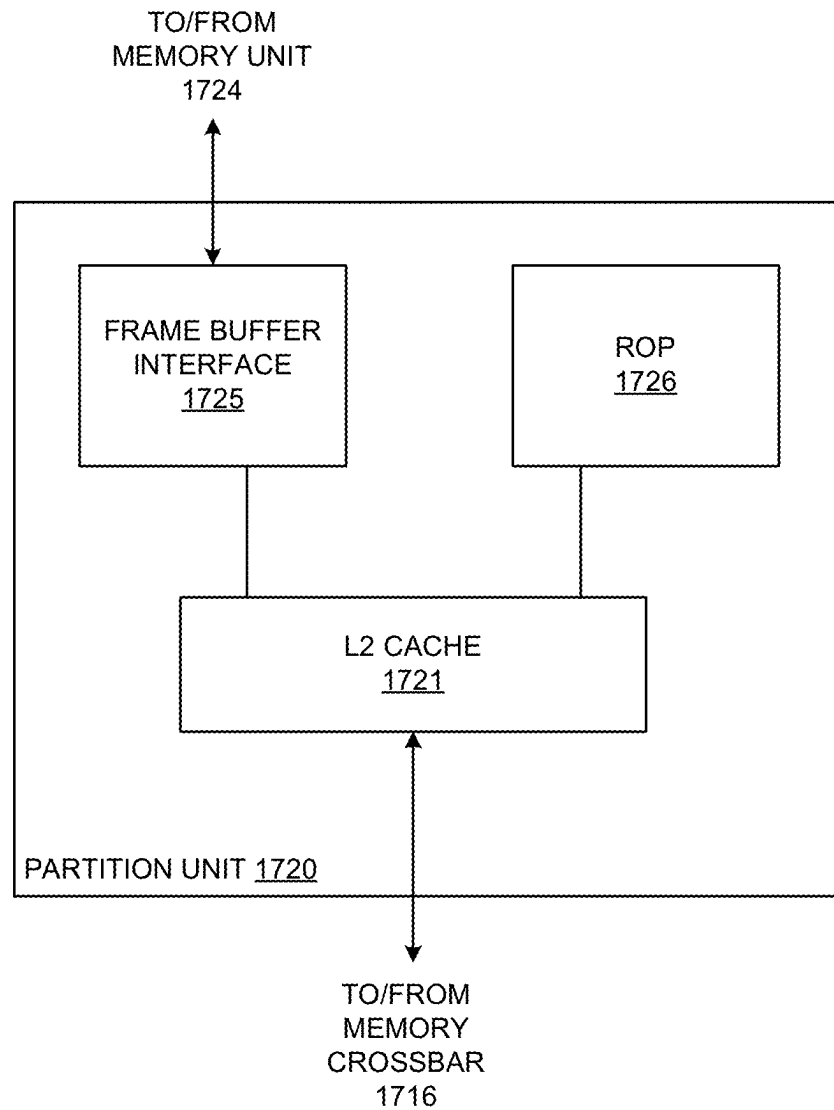


FIG. 17B

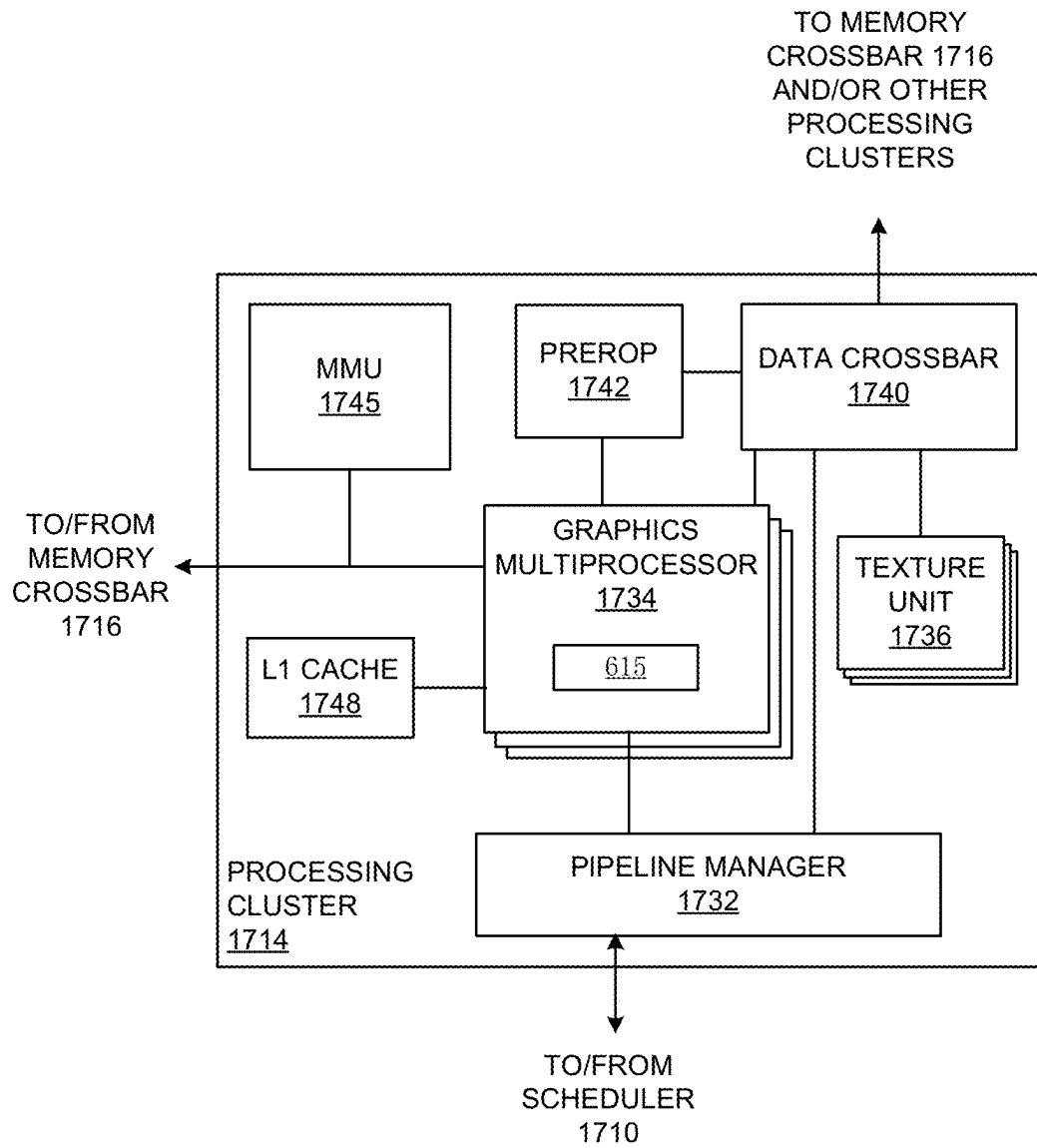


FIG. 17C

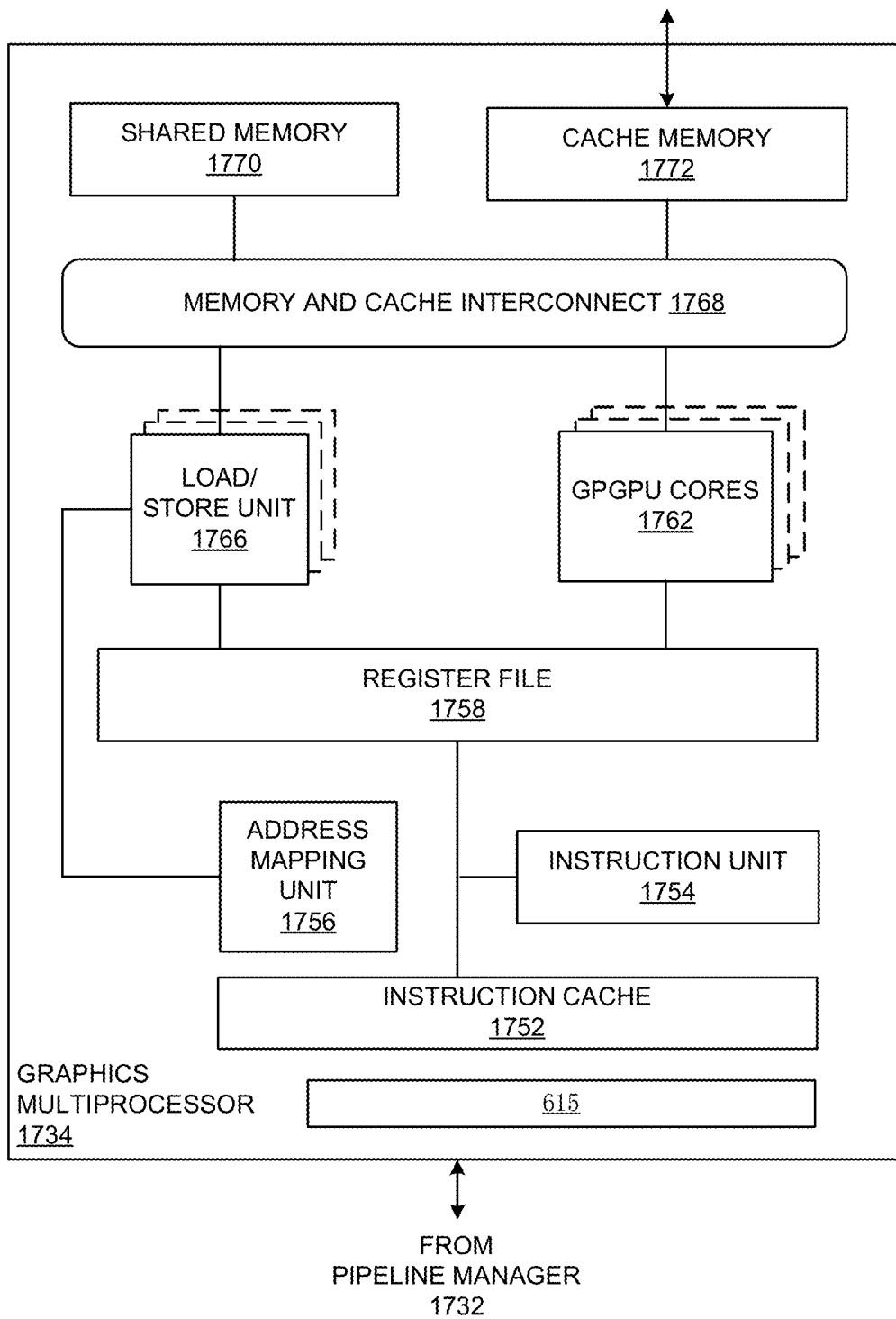


FIG. 17D

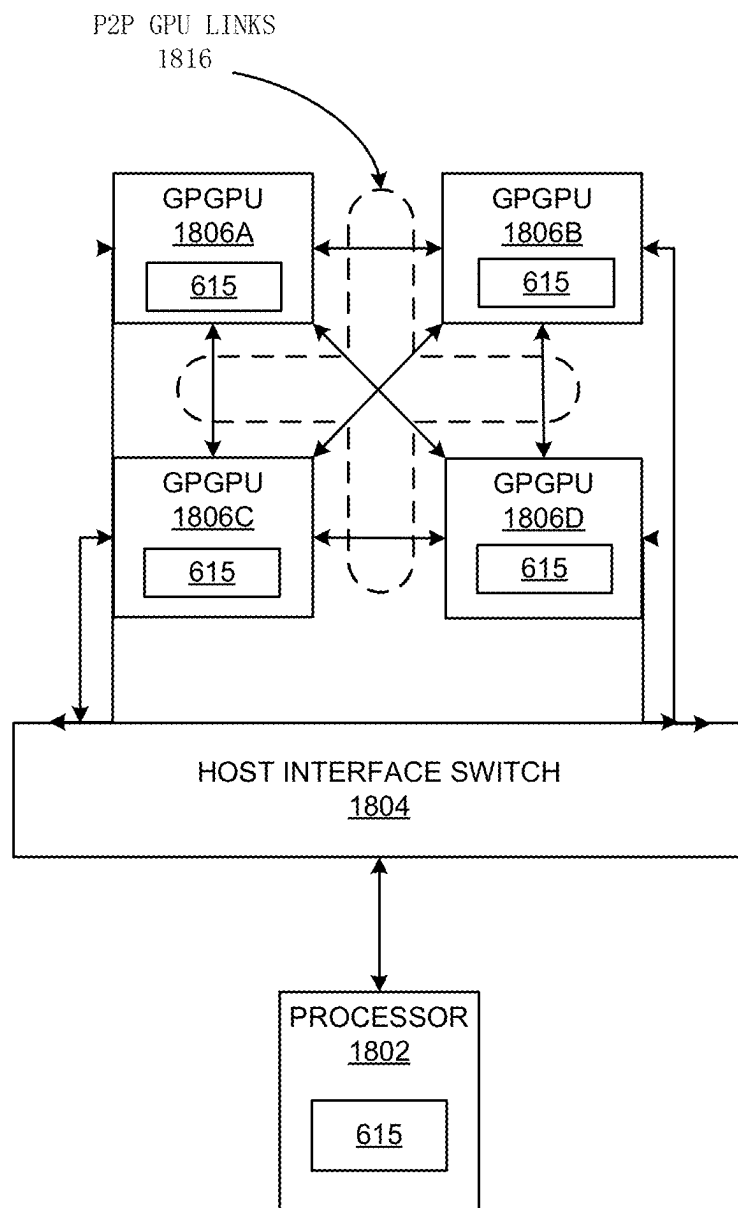
1800

FIG. 18

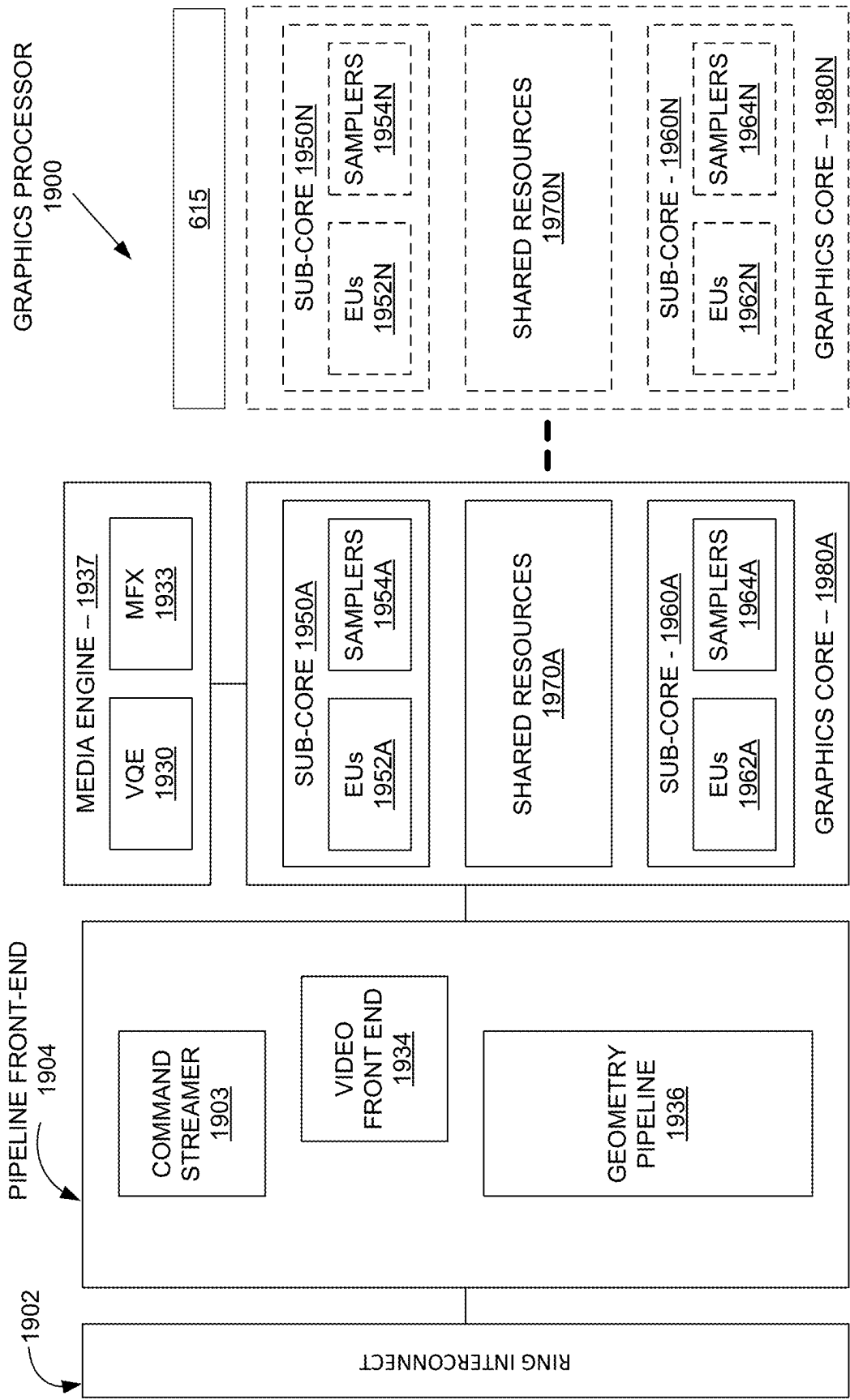


FIG. 19

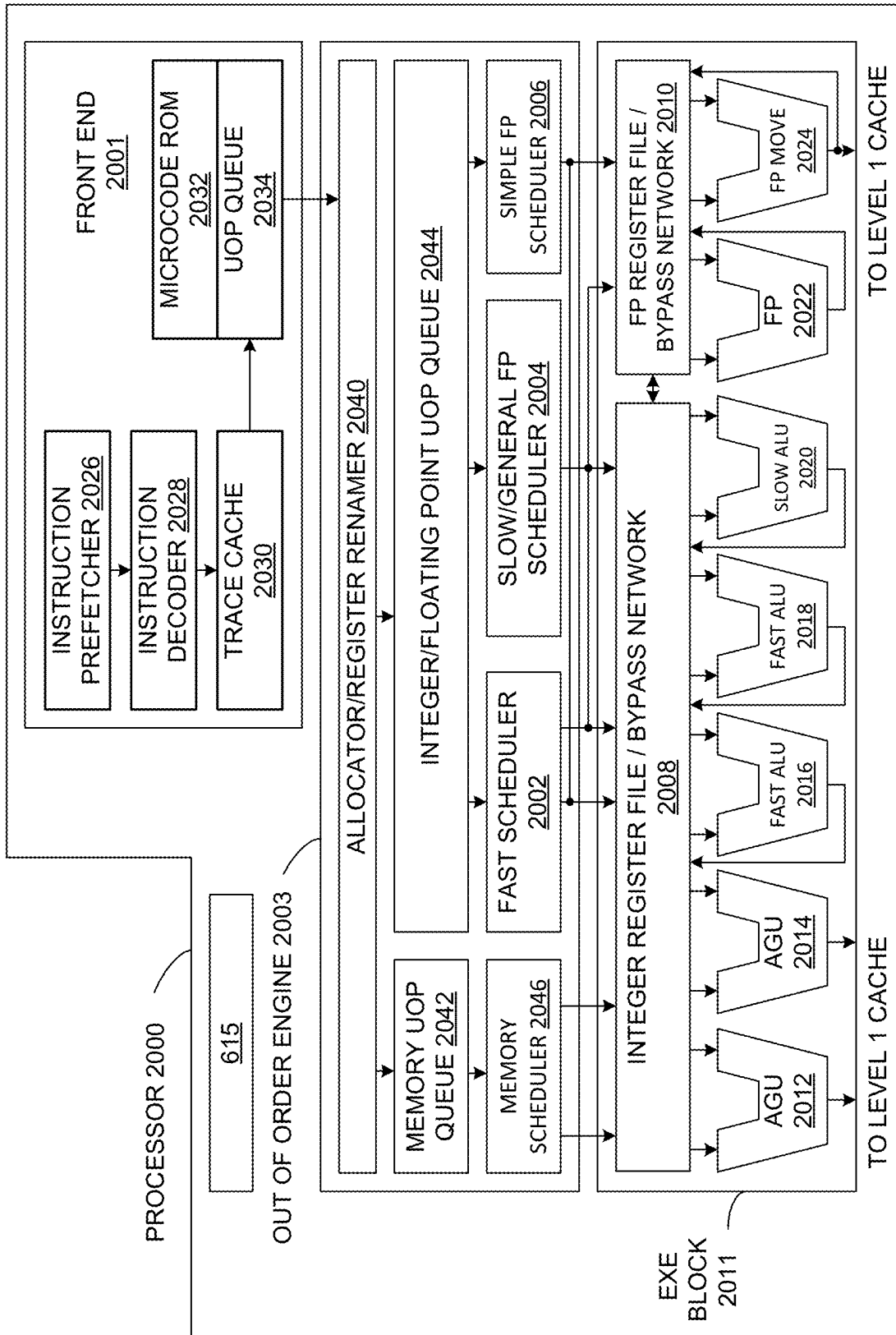


FIG. 20

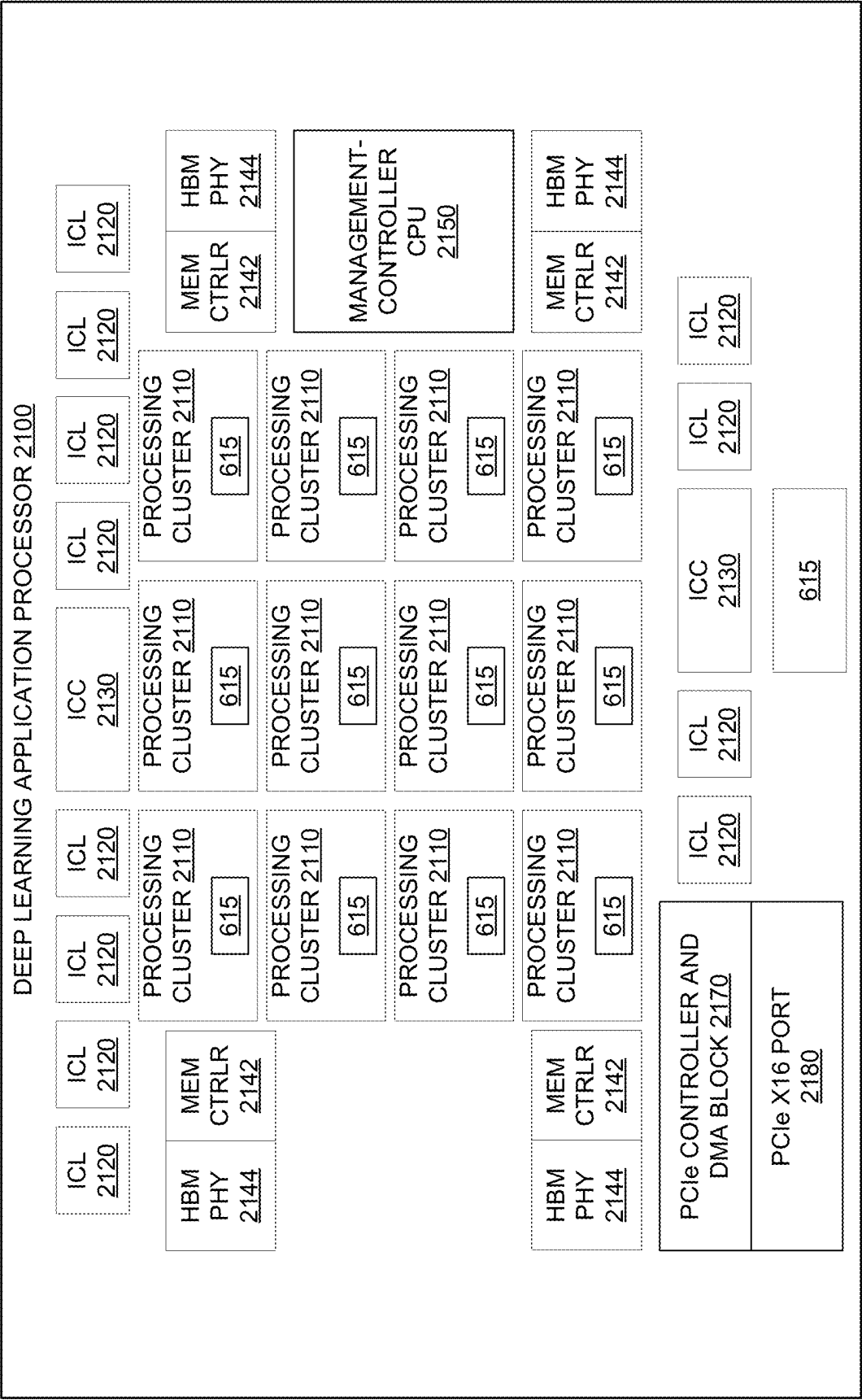


FIG. 21

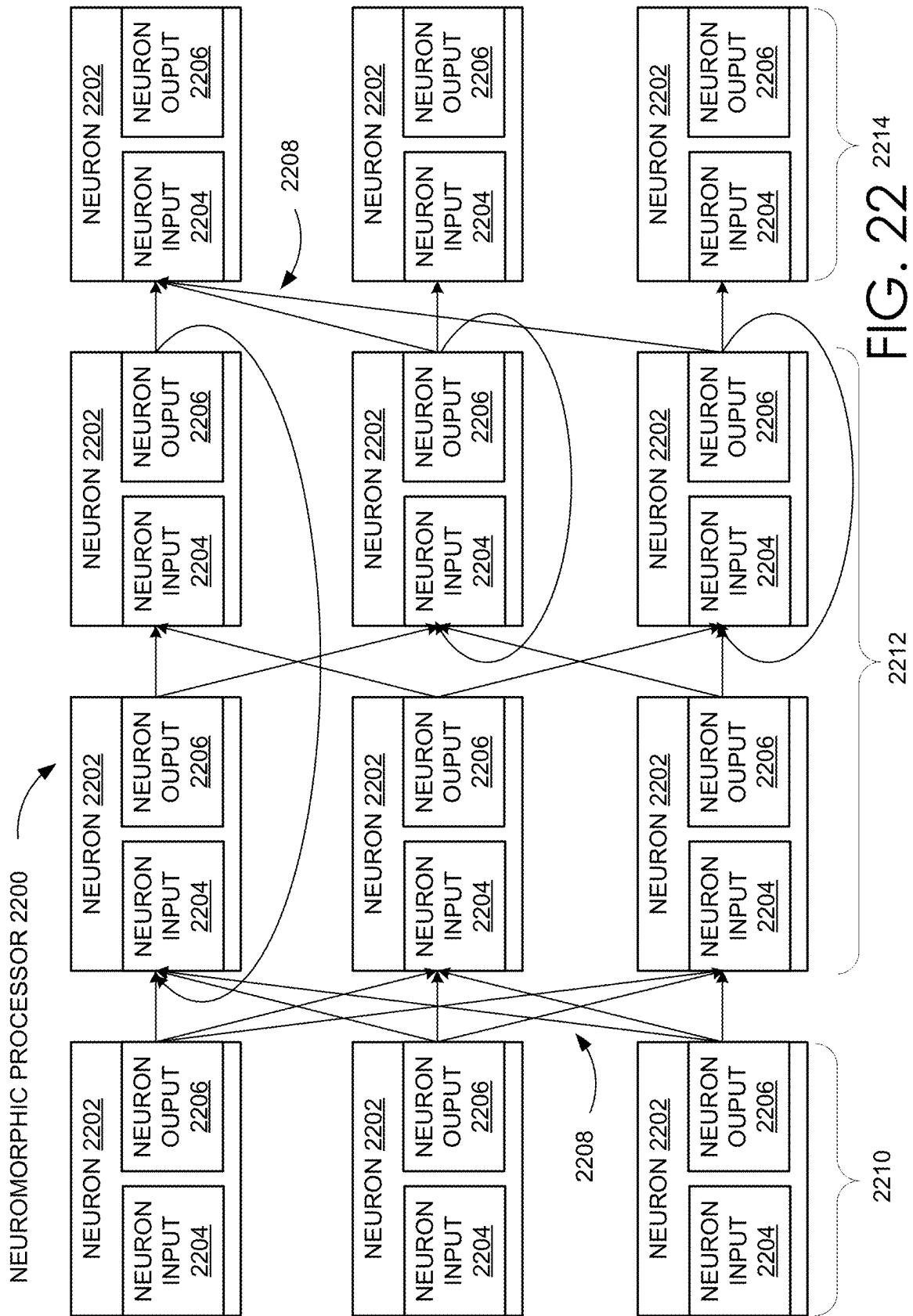


FIG. 22

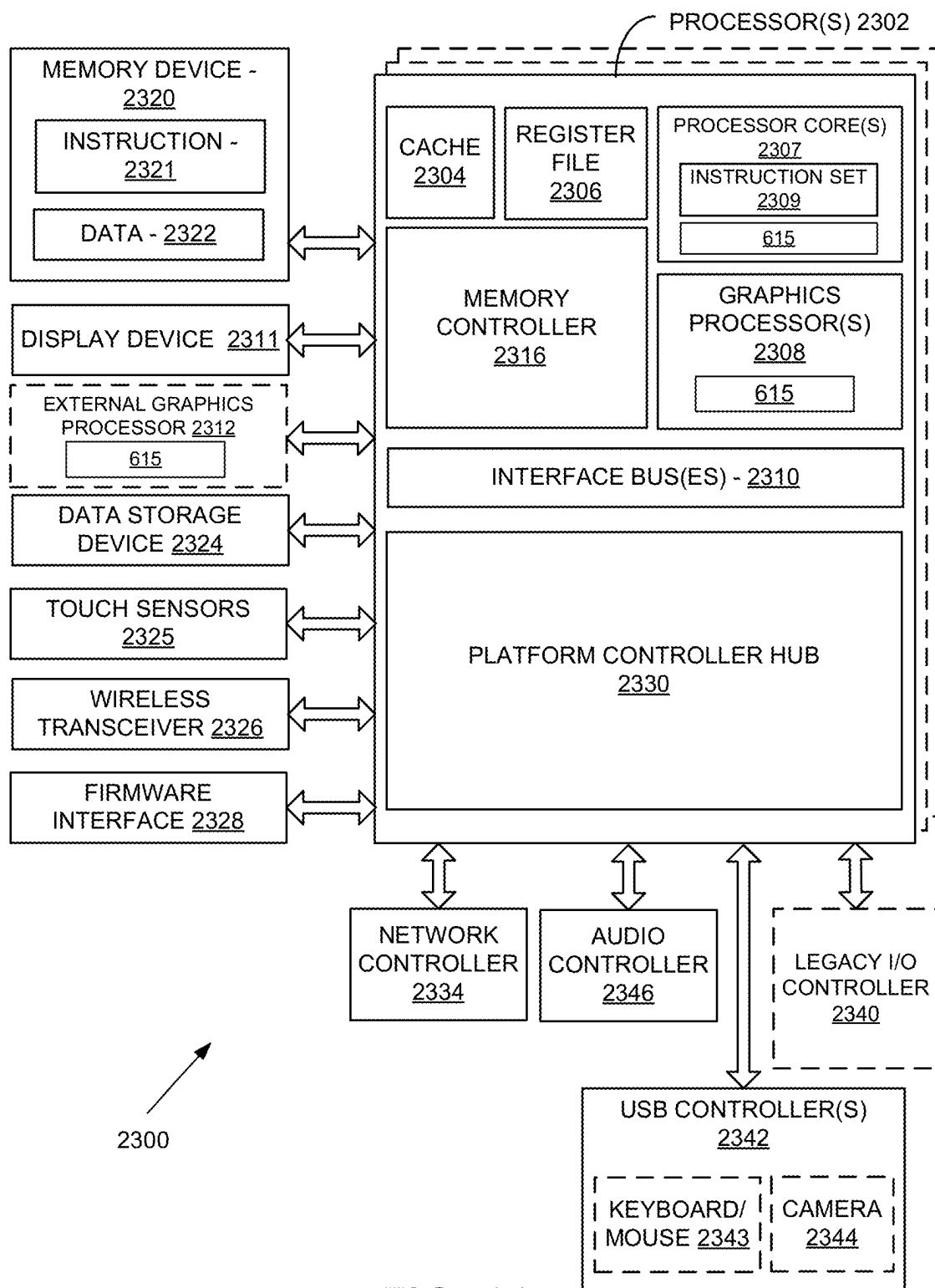


FIG. 23

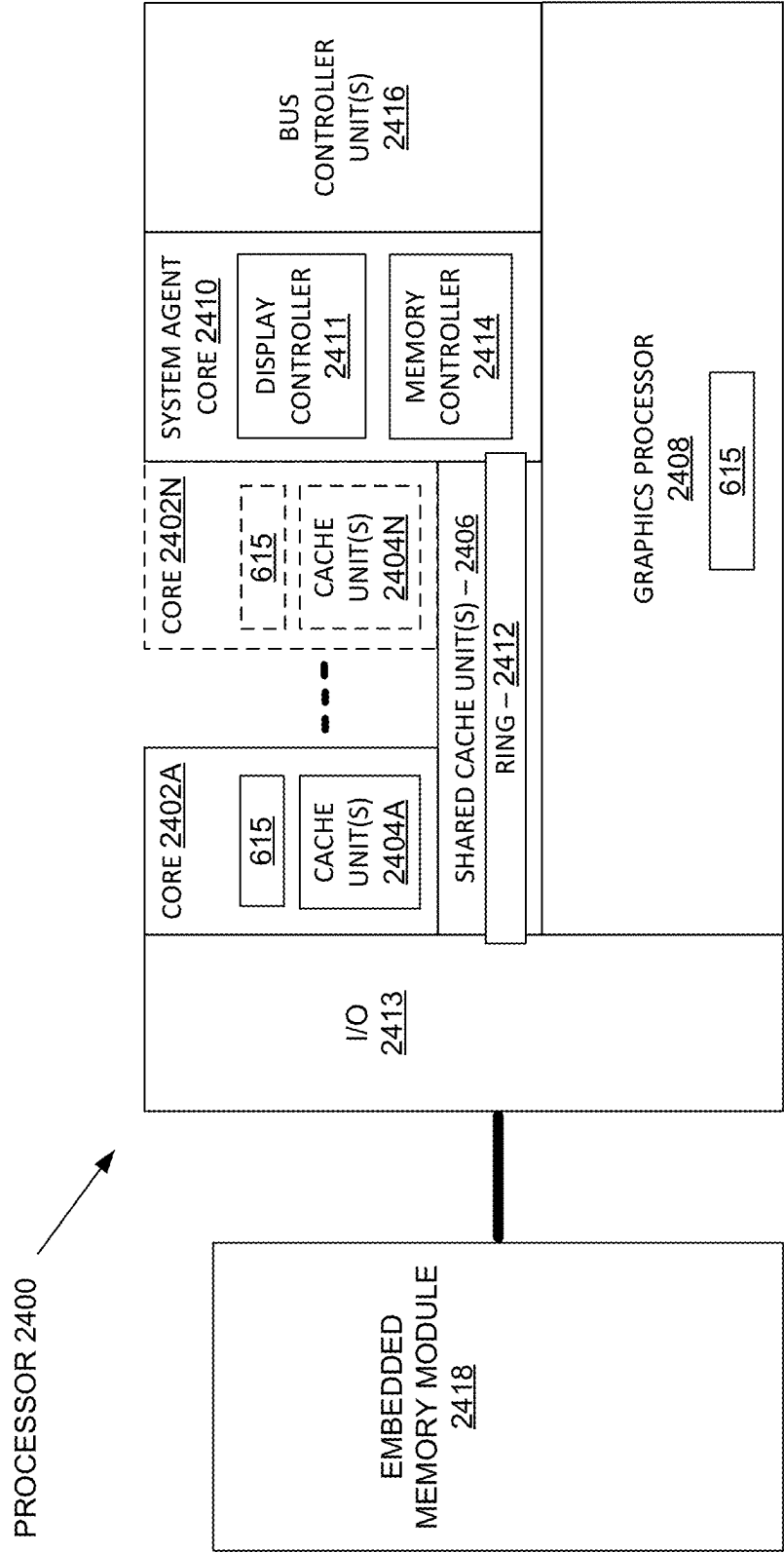


FIG. 24

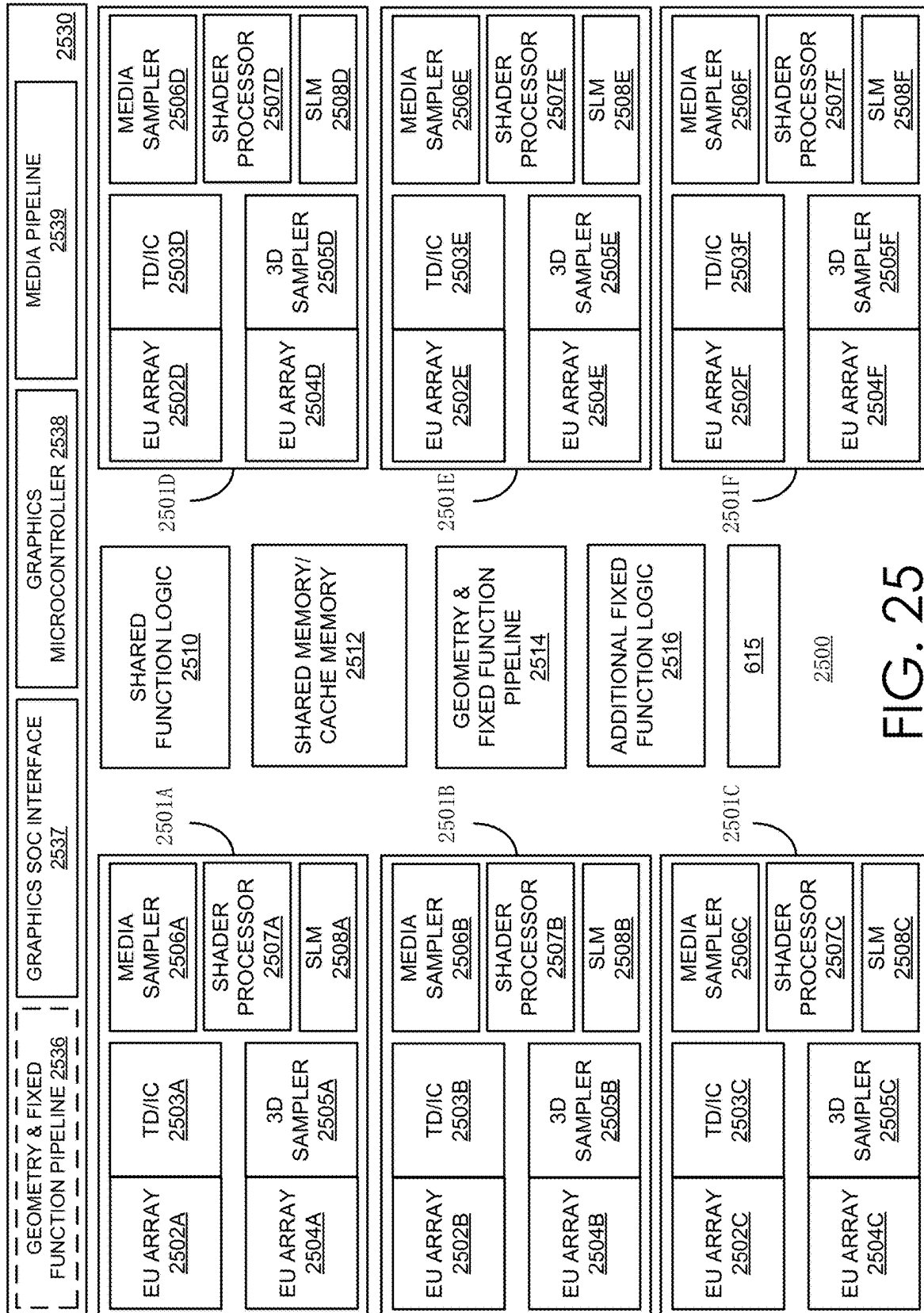


FIG. 25

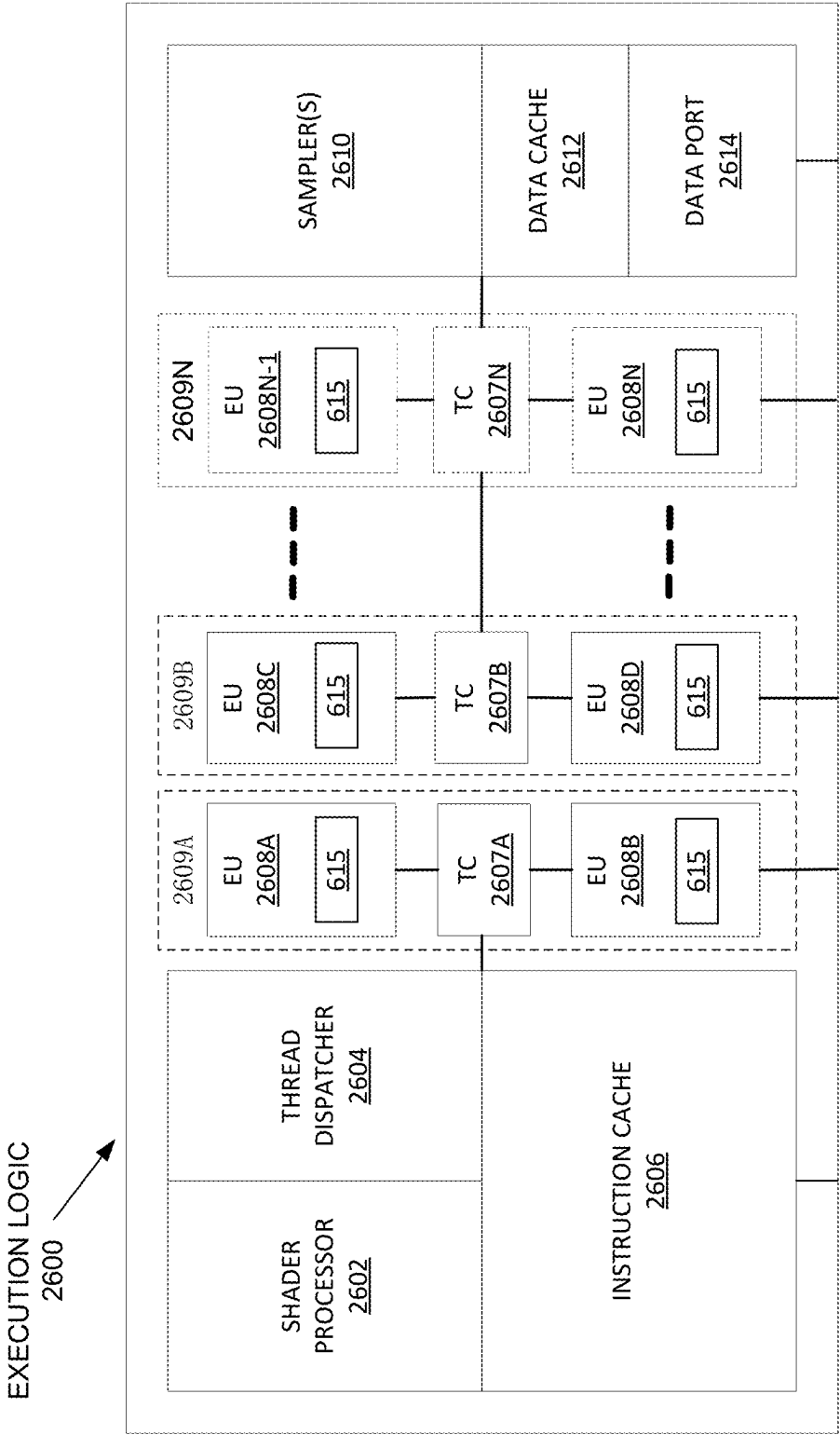


FIG. 26A

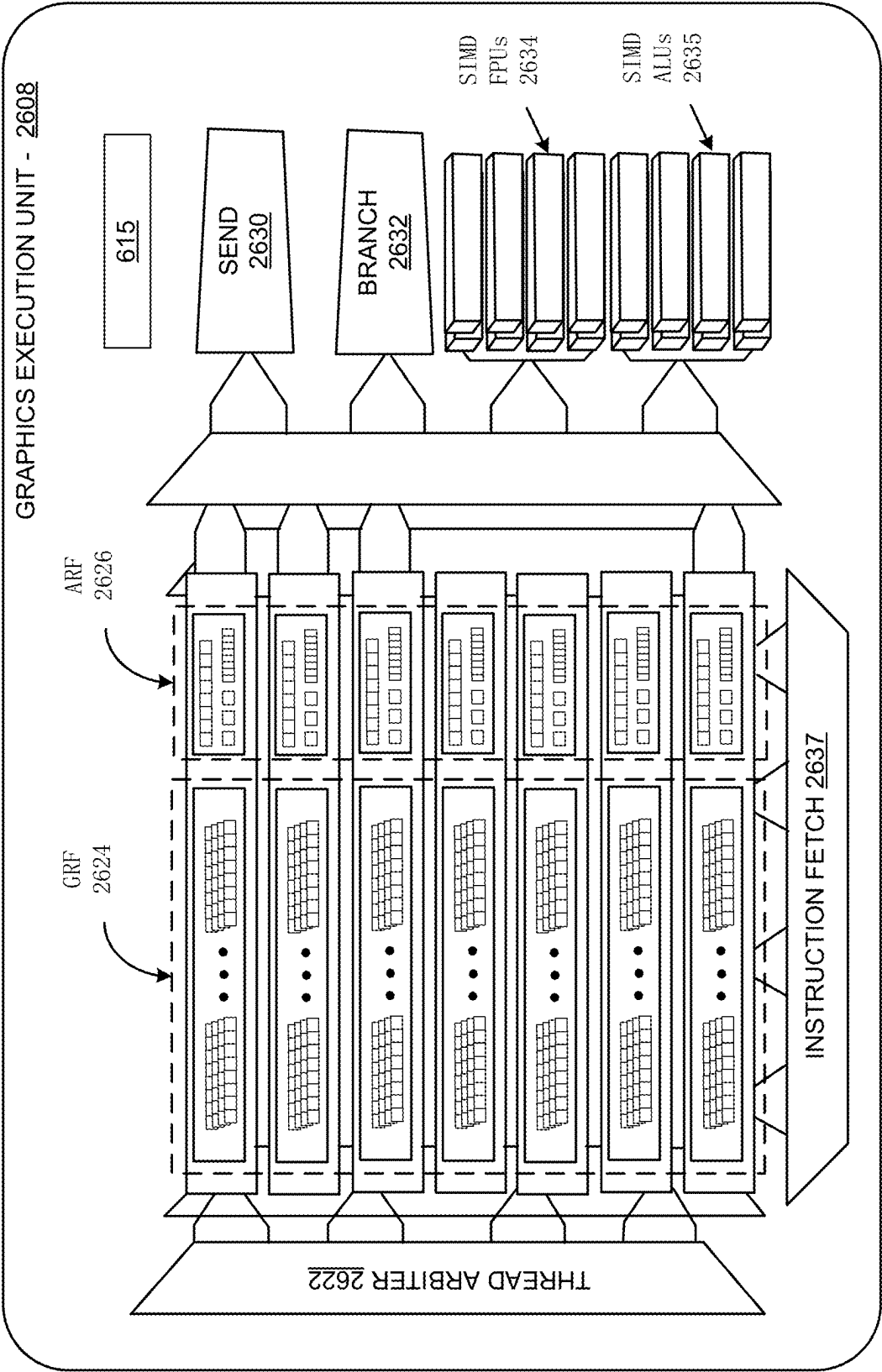


FIG. 26B

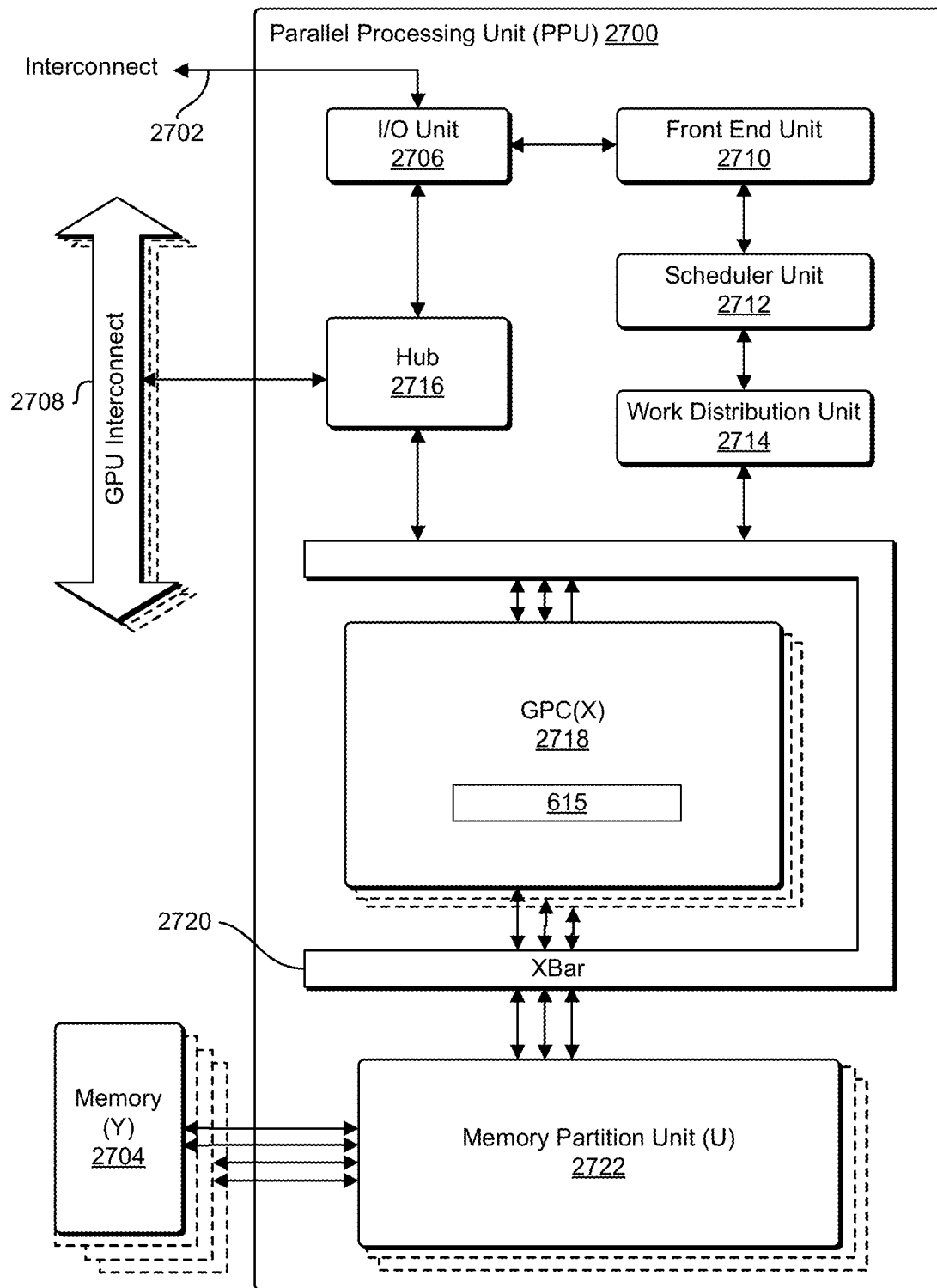


FIG. 27

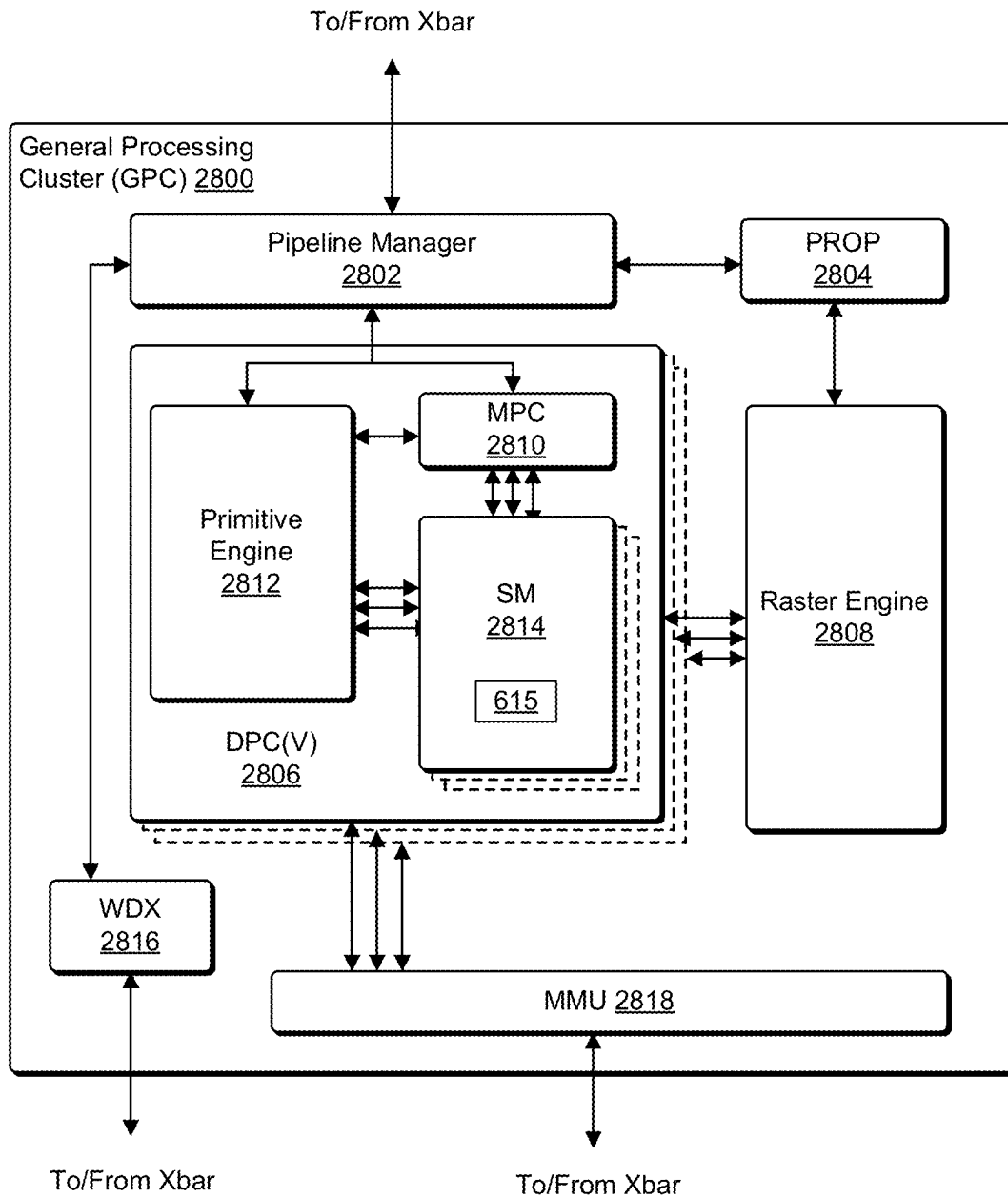


FIG. 28

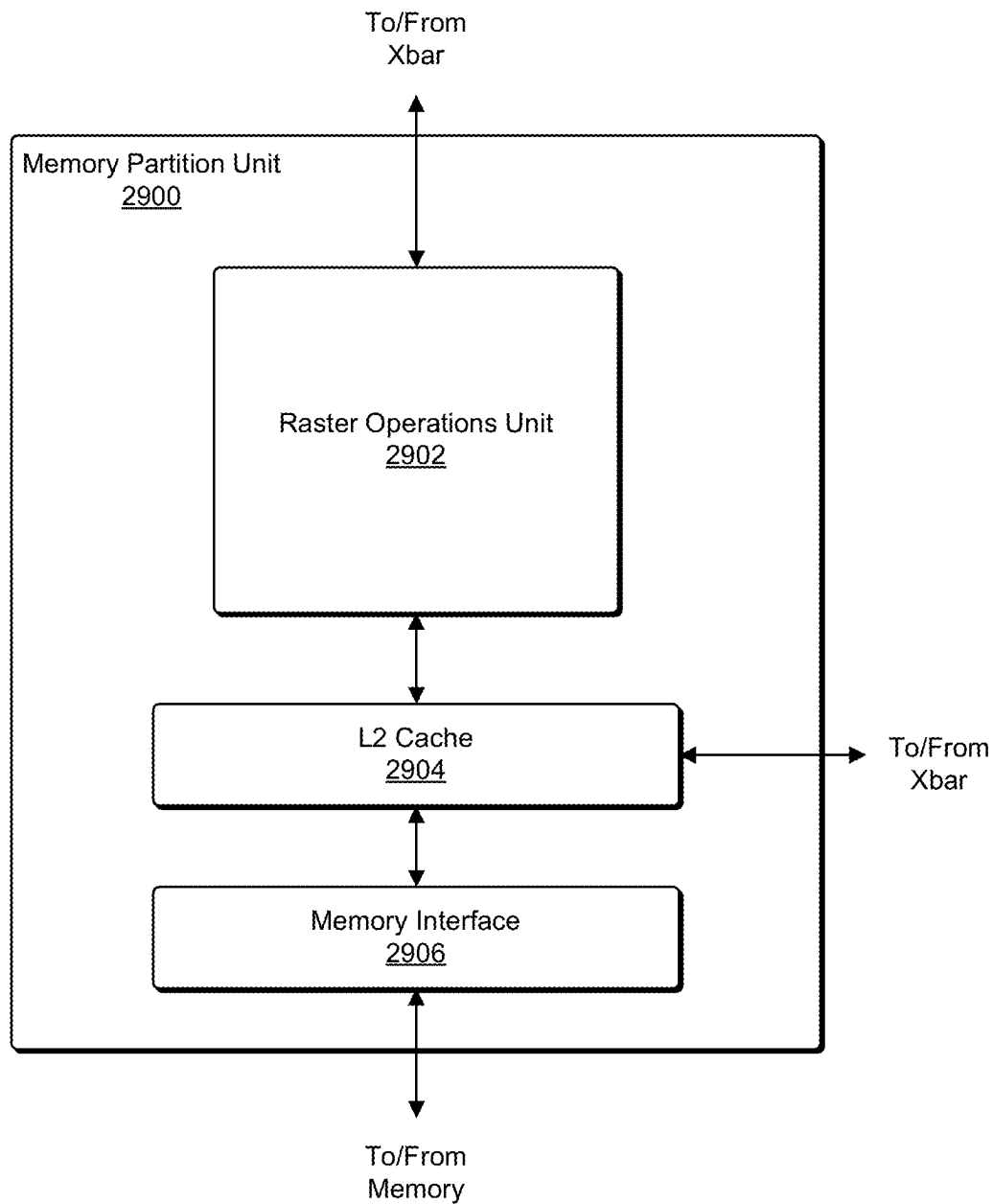


FIG. 29

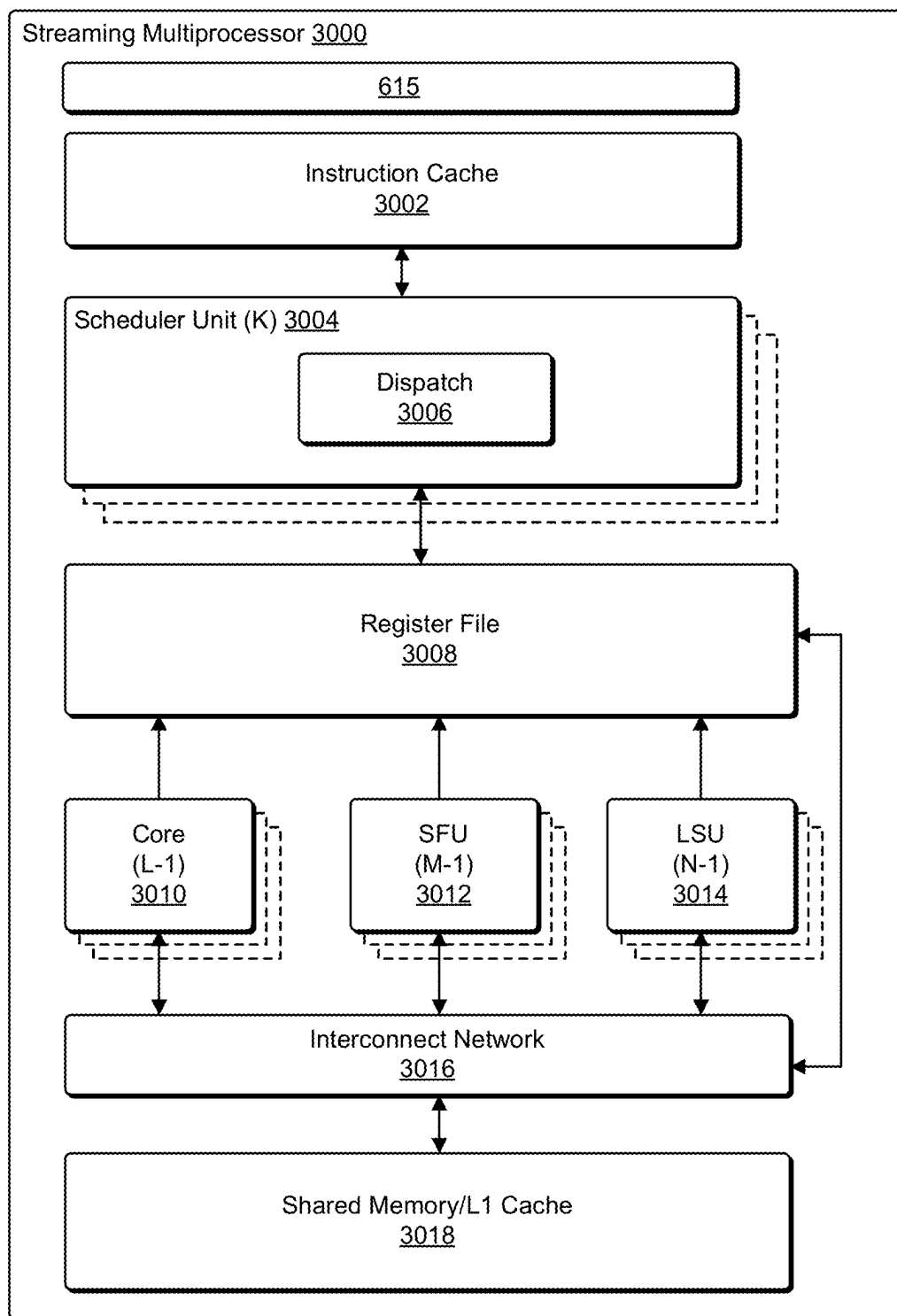


FIG. 30

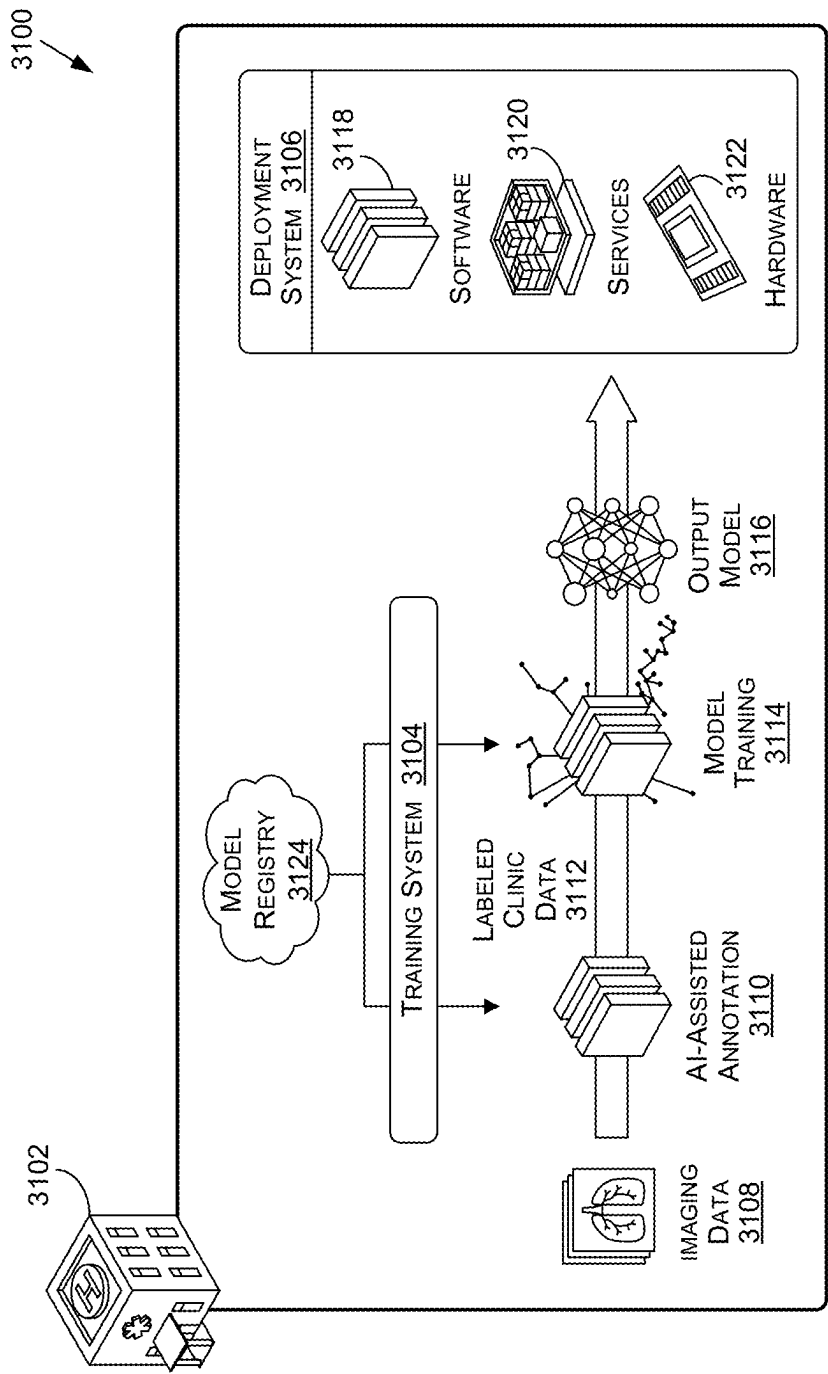


FIG. 31

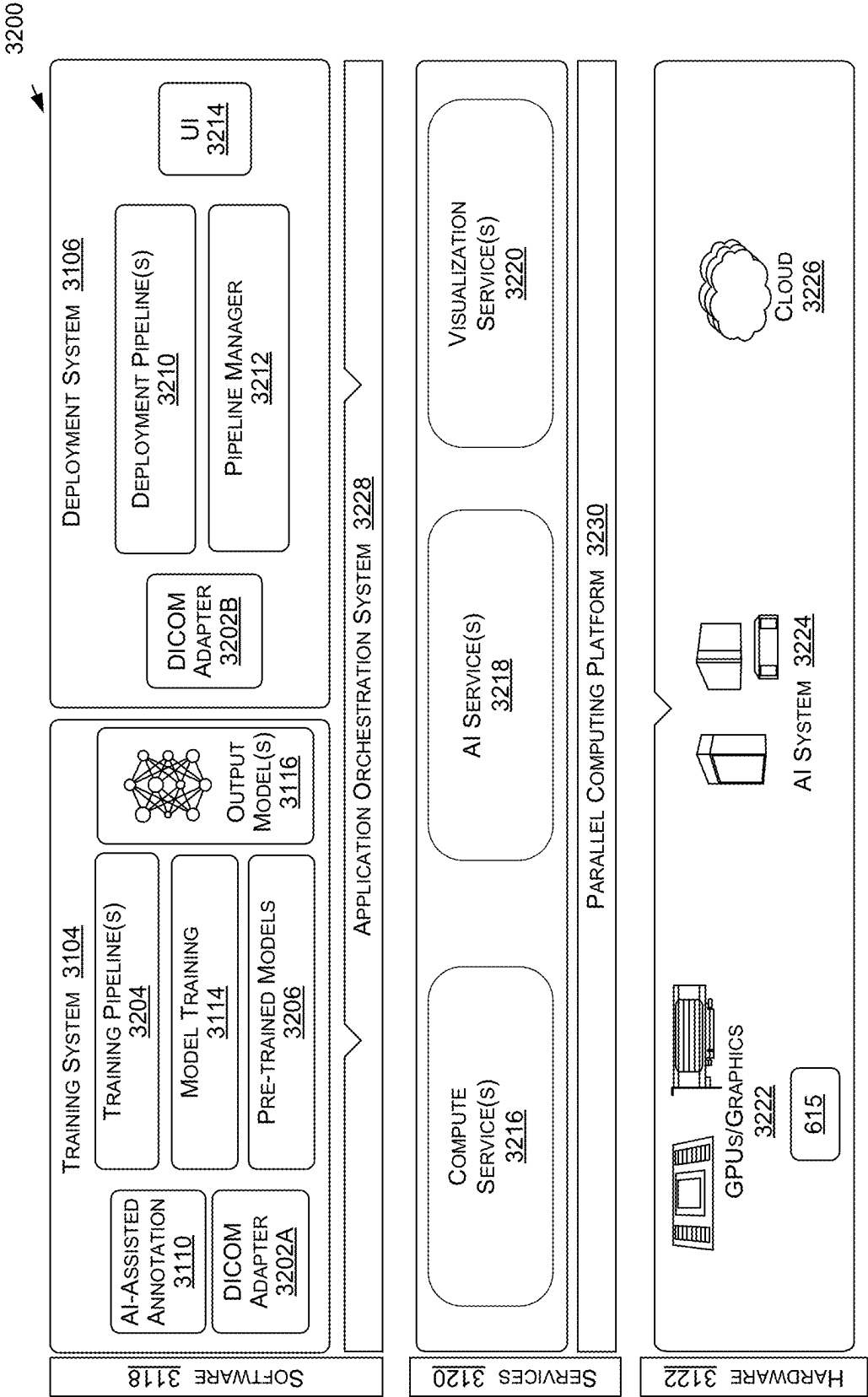


FIG. 32

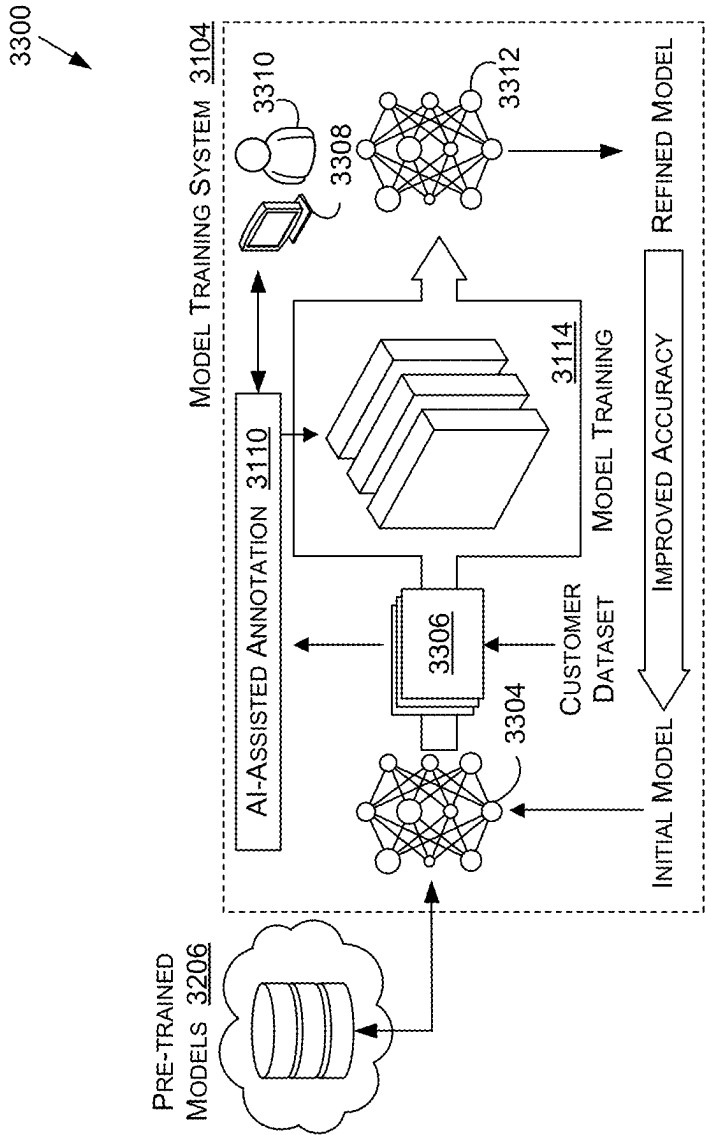


FIG. 33A

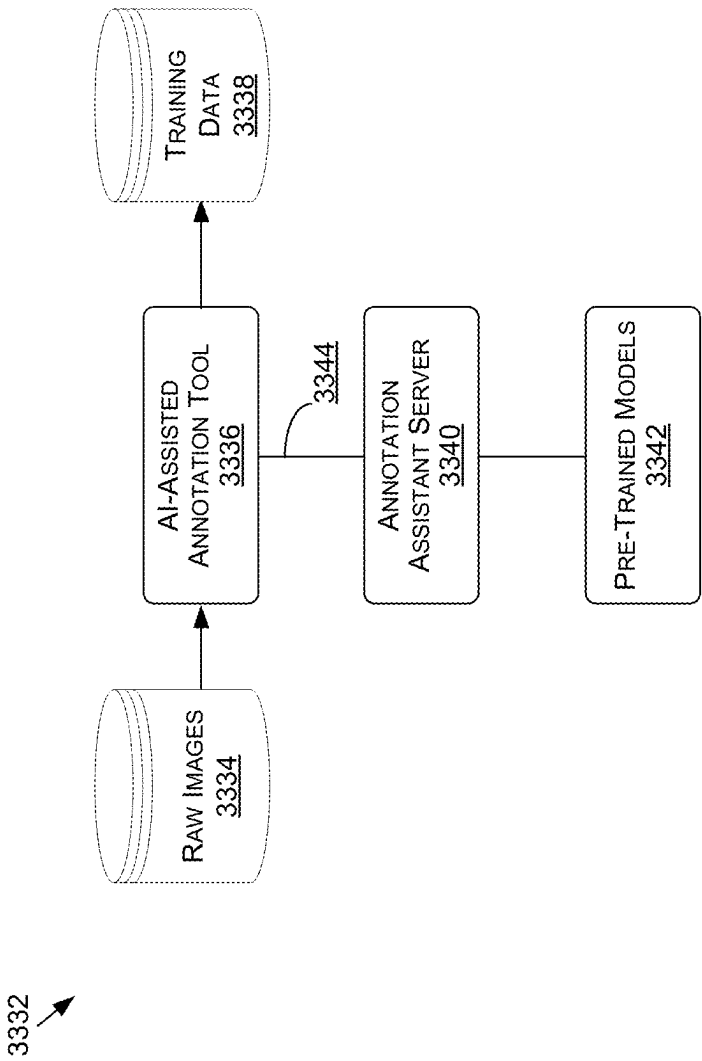


FIG. 33B

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GAZE ESTIMATION USING ONE OR MORE NEURAL NETWORKS

FIELD

At least one embodiment pertains to processing resources used to perform and facilitate artificial intelligence. For example, at least one embodiment pertains to processors or computing systems used to train neural networks according to various novel techniques described herein.

BACKGROUND

Gaze tracking and estimation are being utilized for an ever-increasing variety of applications, which are increasingly requiring higher levels of accuracy. Unfortunately, existing gaze systems that may be able to achieve these levels of accuracy are very resource intensive or have latencies that make them inappropriate for many of these intended uses.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments in accordance with the present disclosure will be described with reference to the drawings, in which:

FIG. 1 illustrates a system for managing presentation of virtual reality content, according to at least one embodiment;

FIG. 2 illustrates a gaze estimation system, according to at least one embodiment;

FIGS. 3A and 3B illustrate eye data, according to at least one embodiment;

FIG. 4 illustrates a process for estimating gaze for a user, according to at least one embodiment;

FIG. 5 illustrates a process for determining coarse and fine gaze, according to at least one embodiment;

FIG. 6A illustrates inference and/or training logic, according to at least one embodiment;

FIG. 6B illustrates inference and/or training logic, according to at least one embodiment;

FIG. 7 illustrates an example data center system, according to at least one embodiment;

FIG. 8 illustrates a computer system, according to at least one embodiment;

FIG. 9 illustrates a computer system, according to at least one embodiment;

FIG. 10 illustrates a computer system, according to at least one embodiment;

FIG. 11 illustrates a computer system, according to at least one embodiment;

FIG. 12A illustrates a computer system, according to at least one embodiment;

FIG. 12B illustrates a computer system, according to at least one embodiment;

FIG. 12C illustrates a computer system, according to at least one embodiment;

FIG. 12D illustrates a computer system, according to at least one embodiment;

FIGS. 12E and 12F illustrate a shared programming model, according to at least one embodiment;

FIG. 13 illustrates exemplary integrated circuits and associated graphics processors, according to at least one embodiment;

FIGS. 14A-14B illustrate exemplary integrated circuits and associated graphics processors, according to at least one embodiment;

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FIGS. 15A-15B illustrate additional exemplary graphics processor logic, according to at least one embodiment;

FIG. 16 illustrates a computer system, according to at least one embodiment;

FIG. 17A illustrates a parallel processor, according to at least one embodiment;

FIG. 17B illustrates a partition unit, according to at least one embodiment;

FIG. 17C illustrates a processing cluster, according to at least one embodiment;

FIG. 17D illustrates a graphics multiprocessor, according to at least one embodiment;

FIG. 18 illustrates a multi-graphics processing unit (GPU) system, according to at least one embodiment;

FIG. 19 illustrates a graphics processor, according to at least one embodiment;

FIG. 20 illustrates a processor's micro-architecture, according to at least one embodiment;

FIG. 21 illustrates a deep learning application processor, according to at least one embodiment;

FIG. 22 illustrates an example neuromorphic processor, according to at least one embodiment;

FIGS. 23 and 24 illustrate at least portions of a graphics processor, according to at least one embodiment;

FIG. 25 illustrates at least portions of a graphics processor core, according to at least one embodiment;

FIGS. 26A-26B illustrate at least portions of a graphics processor core, according to at least one embodiment;

FIG. 27 illustrates a parallel processing unit ("PPU"), according to at least one embodiment;

FIG. 28 illustrates a general processing cluster ("GPC"), according to at least one embodiment;

FIG. 29 illustrates a memory partition unit of a parallel processing unit ("PPU"), according to at least one embodiment;

FIG. 30 illustrates a streaming multi-processor, according to at least one embodiment;

FIG. 31 is an example data flow diagram for an advanced computing pipeline, in accordance with at least one embodiment;

FIG. 32 is a system diagram for an example system for training, adapting, instantiating and deploying machine learning models in an advanced computing pipeline, in accordance with at least one embodiment;

FIG. 33A illustrates a data flow diagram for a process to train a machine learning model, in accordance with at least one embodiment; and

FIG. 33B is an example illustration of a client-server architecture to enhance annotation tools with pre-trained annotation models, in accordance with at least one embodiment.

DETAILED DESCRIPTION

In at least one embodiment, a user **102** can view digital content via a display mechanism, as may include a display **106** internal to a virtual reality (VR) headset. In at least one embodiment, such content can be presented using an alternative display mechanism, such as a monitor, touchscreen, augmented reality (AR) display, or projection. In at least one embodiment, for VR content this VR headset may be in communication with a separate client device **110** that can provide, transmit, or render content to be presented via VR headset **104**. In at least one embodiment, VR headset **104** can include one or more orientation sensors that provide data for movement of a head of user **102**, which can be used to control a point of view (POV) of content rendered via

headset **104**, in order to provide a realistic and immersive experience for user **102**. In at least one embodiment, a VR application **112** executing on client device **110** can accept orientation data from headset **104** and use this information for rendering VR content to be presented via headset **104**. In at least one embodiment, client device **110** may be in contact with at least one content server **130** that can stream or otherwise transmit content over at least one network **120**, such as a wired or wireless communication network, to be used in rendering or presenting this VR content. In at least one embodiment, a transmission manager **132** on content server **130** can be responsible for transmitting content in an appropriate format and channel, such as in a streaming video format, over an appropriate network connection, such as an Internet or cellular connection. In at least one embodiment, VR content to be displayed can be rendered on content server **130**, as may relate to orientation data received from headset **104**, and transmitted over network **120** to client device **110** and ultimately VR headset **104**.

In at least one embodiment, VR headset **104** can include one or more cameras, such as eye tracking cameras **108**, that can capture images of one or both eyes of a user. In at least one embodiment, VR headset **104** can include a gaze-tracking subsystem that includes one or more cameras configured to capture images of one or both eyes of a user wearing headset **104**. In at least one embodiment, one or more neural networks of a gaze tracking module **114** on client device **110** can be used to determine gaze information for individual images or video frames, along with a confidence for each indication. In at least one embodiment, gaze tracking module **114** can instead collect gaze-related information from headset **104** and forward that information to a content manager **116** component or module, which can correlate that gaze-related data with corresponding images for frames of content that were being presented. In at least one embodiment, this information can be transmitted to content server **130**, which can execute a content application **134** that includes a content manager **140** component as well. In at least one embodiment, content manager **140** can receive gaze-related data and provide that data to a gaze tracking module **138** that may include one or more neural networks for estimating gaze using this received data. In at least one embodiment, content manager **140** may work with session manager **136** to determine objects or other content for a session, and use information for those objects or content to determine objects at which this user is, or was, gazing at a specific point in time. In at least one embodiment, information about a user stored in a user database **144** or content that was displayed from a content database **142** can be used in these determinations as well. In at least one embodiment, all processing may be performed on client device **110** or headset **104**, without need to contact a remote server **130** over a network **120**. In at least one embodiment, gaze determinations may be performed on client device **110**, with results of that tracking provided to server **130** for modifying or providing content based at least in part upon that gaze data.

In at least one embodiment, it can be desirable for various applications to be able to accurately estimate a center position of one or more pupils of a user, and to perform tracking with low latency and under a wide variety of lighting conditions, including for users wearing glasses. In at least one embodiment, such pupil center tracking can be useful for near-eye gaze tracking, such as may be useful for virtual reality (AR) or augmented reality (AR) applications, as well as remote gaze tracking for automotive and other such applications.

In at least one embodiment, pupil localization can be performed on a processor such as graphics processing unit (GPU), which can be optimized in such a way as to provide fast performance for localization determinations. In at least one embodiment, various approaches can also improve accuracy of determinations while still achieving such levels of performance. In at least one embodiment, such advantages can be obtained for both monocular and binocular pupil center estimation. In at least one embodiment, pupil localization for gaze estimation using deep learning can start with an uploading or providing of one or more trained neural networks to one or more GPUs of a device to perform inferencing. In at least one embodiment, at inference time, a sequence of images will be received that include views of one or more eyes of a user during a relevant session or experience. In at least one embodiment, this image data can be stored in system memory until such time as it is transferred to GPU memory for inferencing. In at least one embodiment, network inferencing can be run on this GPU to compute pupil center position in image space. In at least one embodiment, this pupil position result data can then be copied back to system memory, where this data can be accessed by an application or CPU programmed to process that data, such as to determine gaze information based on this pupil data.

In at least one embodiment, a system **200** for gaze estimation can utilize components as illustrated in FIG. 2. In at least one embodiment, a device such as a head-mounted device **202** (or device that is not head-mounted) can include one or more cameras **204** that, when worn by, or positioned with respect to, a user during a presentation of content, enables that camera to capture images or video representative of this user's eyes, which can be used for purposes such as gaze estimation or gaze tracking. In at least one embodiment, images for each of a user's eyes can be provided to a gaze estimator **206**, which may be a component or subsystem executing on a client device in communication with head-mounted device (HMD) **202**, may be a component or subsystem on HMD **202**, or may be a remote device or subsystem, as may be available over one or more wired or wireless networks. In at least one embodiment, these eye images can be analyzed to produce pupil position or gaze data, which can then be provided or otherwise made available to an application **218** or device that seeks that gaze data as input. In at least one embodiment, this application **218** may also receive information about content displayed through HMD **202** corresponding to that gaze data, in order to determine an object of interest or object otherwise drawing gaze of this user.

In at least one embodiment, a two-phase approach to pupil position determination can be utilized that takes advantage of at least two neural networks, such as convolutional neural networks (CNNs). In at least one embodiment, these networks include a coarse CNN **212** and a fine CNN **214**. In at least one embodiment, pupil center estimation accuracy is achieved by making use of these different CNNs to process incoming frames of data from an eye tracking camera. In at least one embodiment, a coarse CNN **212** can receive as input full eye images as captured by an eye-tracker camera, and can process these images to provide a coarse estimate, or first estimate, of pupil center position. In at least one embodiment, a fine CNN **214** can then analyze portions of these input images that correspond to pupil regions, as determined by coarse CNN **212**. In at least one embodiment, fine CNN **214** can provide more accurate pupil center estimates than coarse CNN **212**, which analyzes entire captured images. In at least one embodiment, fine CNN **214**

can further be trained to provide accurate pupil localization within approximate pupil regions of captured images.

In at least one embodiment, coarse CNN 212 can be trained on synthetic data and real eye images with high variety in pupil position and pupil appearance across a full eye tracking camera image. In at least one embodiment, this enables such an implementation to achieve significant robustness against lighting changes and camera noise. In at least one embodiment, however, accuracy of a pupil center estimation may be limited by complexity of a network. In at least one embodiment, a network needs to contain a high number of learned parameters to achieve high accuracy, which can significantly slow inference time determinations.

In at least one embodiment, a single network architecture can be used for both coarse CNN 212 and fine CNN 214. In at least one embodiment, this can involve a convolutional neural network with a limited number of layers, such as only seven layers. In at least one embodiment, this network can use a decreasing kernel size of 9×9 , 7×7 , 5×5 , 5×5 , 3×3 , 3×3 , and 3×3 , with output channels of 24, 36, 52, 80, 124, 256, and 512. In at least one embodiment, a size of an input image can be specified, such as for a size of 293×293 pixels. In at least one embodiment, such architecture provides for high accuracy, while smaller networks with fewer convolutions may trade off some accuracy for increases in inference speed.

In at least one embodiment, fine CNN 214 can utilize this same network architecture, but fine CNN 214 can be trained for pupil center estimation using only determined pupil regions, as may be obtained from coarse CNN 212. In at least one embodiment, fine CNN 214 would then consume an image with higher detail of a pupil, such that fine CNN 214 is enabled to produce a more accurate pupil center estimate. In at least one embodiment, input images can have a same size (e.g., 293×293 pixels) for both coarse and fine CNNs, but smaller represented regions in images for fine CNN 214 provide for higher detail for this pupil region. In at least one embodiment, a size of image to be used can be selected to be small enough to allow for fast processing, but large enough to provide sufficient detail. In at least one embodiment, input of fine CNN can be computed by extracting, from an eye tracking camera frame, a region of interest corresponding to a coarse pupil region as determined by coarse CNN 212. In at least one embodiment, output of fine CNN 214 can then be a more precise estimation of pupil center location. In at least one embodiment, a module or subsystem such as data manager 210 can direct full images of a user's eyes to coarse CNN 212, and images of determined pupil regions to fine CNN 214. In at least one embodiment, data manager 210 can also resize or otherwise process images from an eye tracker camera to meet specifications of these networks.

In at least one embodiment, estimation of pupil center can be accelerated by taking advantage of these separate coarse and fine phases. In at least one embodiment, a monocular (single eye) implementation can accelerate this phased approach by assuming that for many frames a human eye will be fixating on a single target, exhibiting behavior known as fixational eye motion. In at least one embodiment, once a user's gaze switches to a new position, that user's eyes will fixate on that object for at least a short period of time whereby a pupil position will not change significantly. In at least one embodiment, if a determination can be made, such as by data manager 210, that pupil position has not changed significantly or otherwise does not satisfy criteria for an actionable change, such as by demonstrating less than a threshold amount of movement in any given direction, then

for those frames only fine inferencing can be performed. In at least one embodiment, coarse inferencing does not need to be applied again to these frames because this approximate pupil region has already been determined, and it has been determined that a location of this region has remained substantially unchanged, such that a same pupil region could be provided to fine CNN 214 for analysis. In at least one embodiment, coarse inferencing can be applied for subsequent frames where it is determined that pupil location has changed by more than an allowable threshold amount.

In at least one embodiment, a metric such as an L2-distance can be used as a threshold to determine whether to apply a coarse step or coarse phase to an input frame, or pair of frames. In at least one embodiment, this distance metric can be applied between a sum over pixel intensities of this pupil region, when a coarse estimate has been determined by coarse CNN, and a sum over pixel intensities of this same pupil region for a current frame. In at least one embodiment, as long as this distance is below this threshold value, then no substantial movement or change has occurred and only fine inferencing is to be applied to a current frame, or pair of frames.

In at least one embodiment, for larger eye motions or saccades, a pupil would move to a different region or position, resulting in a distance (or change in corresponding intensities of a pupil region) by more than this threshold amount. In at least one embodiment, coarse processing could then be applied to a current frame, or pair of frames, in order to determine a new or updated coarse pupil center position. In at least one embodiment, such an approach can significantly reduce a number of calls for network inferencing due at least in part to a high number of fixational eye motions in natural human gaze behavior.

In at least one embodiment, binocular (two eye) pupil center estimation can be accelerated using this threshold-based approach to coarse and fine inferencing. In at least one embodiment, this is enabled due to human eyes having coupled movement. In at least one embodiment, estimation can be further accelerated by running inferencing for both eyes simultaneously, or at least partially concurrently or in parallel, rather than independently. In at least one embodiment, such an approach would not require a change to this network architecture, weights, or training setup.

In at least one embodiment, simultaneous processing can be accomplished in part by concatenating images from both eye tracking cameras into a single frame or image. In at least one embodiment, a pair of images received from one or more eye tracking cameras can be concatenated into a single image or image frame, showing both eyes of a user at an instant in time. In at least one embodiment, for a person sitting upright in a typical position this may involve concatenating these images horizontally into a single frame, or such that a "top" of each eye is towards a "top" of a resulting image frame once concatenated. In at least one embodiment, a convolution operator is translation-invariant, such that same convolution kernels can be used on this larger input. In at least one embodiment, padding can be added between these images when concatenated in order to avoid cross-talk from misaligned convolutions. In at least one embodiment, this can involve adding 0-value or black pixels to a region between these individual images. In at least one embodiment an amount of padding to be used between these concatenated images can be computed based on one or more assumptions. In at least one embodiment it can be assumed that a limited (e.g., seven) layer network architecture is used with a stride size of 2×2 . In at least one embodiment, for this network architecture a kernel of size k operating on concatenated

images with p padding pixels between them yields output with padding as may be given by:

$$p'=((k+p)/2)-1$$

In at least one embodiment, p' is then a measure of pixels of padding between newly convolved images. In at least one embodiment, initial padding value p should then be selected such that after every layer of convolution, p' correctly aligns a next convolution with newly convolved images. In at least one embodiment, p was selected to be 91 pixels of padding, ending with a final two pixels of padding between final outputs of network convolutional layers. In at least one embodiment, this extra padding is accounted for in a fully connected layer.

In at least one embodiment, a last layer of each network is a fully connected layer. In at least one embodiment, compared to monocular input for a concatenated binocular input case, an incoming matrix is longer in one dimension since there were multiple inputs. In at least one embodiment, this is managed by matching a size of this dimension in a fully connected layer, and passing that as an input parameter, enabling gaze coordinates to be extracted from a correct location in this output.

In at least one embodiment, such architecture can provide for high overall accuracy with a lower inferencing time, as well as a flexible trade-off between inference time and pupil localization estimation. In at least one embodiment, such architecture can also provide for an overall acceleration by re-using coarse pupil position data during periods of fixation. In at least one embodiment, binocular inference acceleration further allows for high inference performance with an added benefit of freeing GPU cycles for other tasks. In at least one embodiment, advantages such as low latency, robustness, and gaze accuracy enable such approaches to be used for applications such as VR, gaze-contingent displays, training scenarios, and gaze-based interaction. In at least one embodiment, such architecture can further reduce latency as needed by skipping a fine inference on at least some passes and utilizing position from a coarse inference, which can be reused for frames within a fixation period.

In at least one embodiment, such architecture can be used for gaze estimation, or identifying a line of sight for each eye of a human user at a single instant, as opposed to gaze tracking which defines a continuous process for following a user's line of sight over time. In at least one embodiment, gaze estimation will need to be performed with extremely low latency, such as on an order of milliseconds, to be useful for real-time interaction. In at least one embodiment, this can include applications such as foveated displays and accurate motion blur rendering which require a tracking system to return a result faster than a frame duration, or an image can be displayed incorrectly. In at least one embodiment, there can be at least two camera scenarios for gaze estimators: remote images captured from a monitor or dashboard-mounted camera, and images from near-eye cameras, which are often intended for use with head-mounted displays. In at least one embodiment, an architecture as presented is flexible to handle both types of images, as well as images from different types of cameras such as on-axis or off-axis cameras.

In at least one embodiment, detectable aspects of a human eye can be used to determine gaze in an accurately calibrated system, such as illustrated in view 300 of FIG. 3. In at least one embodiment, elements 310 of a human eye are illustrated with axis disparity. In at least one embodiment, varying aspects of human eye are illustrated 320, 330, that show nasal-superior shift under pupil constriction. In at least

one embodiment, a variety of other eye aspects are illustrated in view 350 of FIG. 3B. In at least one embodiment, these full images can be passed through a coarse CNN, with smaller sub-regions of these images being passed through a fine CNN where those sub-regions correspond to pupil regions determined from coarse pupil estimates.

In at least one embodiment, an architecture with a limited number of convolutional layers can compensate for significant noise in images by increasing kernel sizes of an initial number of convolutional layers. In at least one embodiment, a layer can be added to increase robustness against image noise, such as reflections and bad lighting conditions. In at least one embodiment, this network can be evaluated very quickly on a GPU or other such processor. In at least one embodiment, a 2×2 stride can be used at each convolutional layer, adding dropout layers after each convolutional layer, and applying no padding or pooling. In at least one embodiment, because a line of sight is not being estimated with this network, no per-subject post-process transformation is applied after a fully-connected final layer. In at least one embodiment, accuracy of pupil estimation can be given in form of a probability of estimating pupil location with a maximum distance of five pixels from a ground truth pupil location, otherwise known as a "5-pixel error." In at least one embodiment, a network as presented herein is able to estimate pupil center even for very challenging cases such as bad lighting conditions, dark eye lashes, partly occluded pupils, and reflections.

In at least one embodiment, a process 400 for determining gaze of a user can be utilized as illustrated in FIG. 4. In at least one embodiment, one or more eye images can be captured 402 for a user, such as by using one or more cameras. In at least one embodiment, these can be ambient or infrared images of any appropriate resolution and format. In at least one embodiment, a determination can be made 404 as to whether there are separate images obtained for both eyes of this user. If so, those images can be concatenated 406 into a single image, separated by a determined amount of padding to prevent cross-talk and other such issues. In at least one embodiment, this current image can be compared 408 against a prior image to determine whether there has been a significant change in pupil position. In at least one embodiment, this can be based on intensities or other information in those images. In at least one embodiment, if it is determined 410 there has been an actionable change, such as an amount of change meeting or exceeding a determined threshold, then that image can be analyzed 412 using a coarse CNN to obtain a coarse pupil position estimate. In at least one embodiment, this threshold may relate to an amount of lateral movement of a pupil in an image or other such metric as discussed herein. In at least one embodiment, if it is determined that there was not a significant or actionable change since a prior image or frame, then a prior coarse pupil estimate can be reused 414 without performing another coarse inference. In at least one embodiment, this pupil estimate can be used to obtain an image of a pupil region that can be provided as input to a fine CNN. In at least one embodiment, this can include taking image data from this received image that corresponds to one or more pupil regions, and generating images of an appropriate size that includes that image data for substantially only those pupil regions. In at least one embodiment, this image of an estimated pupil region can then be analyzed 416 using this fine CNN to obtain a fine pupil position estimate. In at least one embodiment, this fine pupil estimate can be used to determine 418 user gaze.

In at least one embodiment, a process **500** for determining gaze can be used as illustrated in FIG. **5**. In at least one embodiment, image data can be captured for one or more eyes of one or more users. In at least one embodiment, a coarse gaze can be determined for one or more users. In at least one embodiment, a fine gaze can then be determined for these one or more users based at least in part upon this determined coarse gaze information.

Inference and Training Logic

FIG. **6A** illustrates inference and/or training logic **615** used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIG. **6A** and/or **6B**.

In at least one embodiment, inference and/or training logic **615** may include, without limitation, code and/or data storage **601** to store forward and/or output weight and/or input/output data, and/or other parameters to configure neurons or layers of a neural network trained and/or used for inferencing in aspects of one or more embodiments. In at least one embodiment, training logic **615** may include, or be coupled to code and/or data storage **601** to store graph code or other software to control timing and/or order, in which weight and/or other parameter information is to be loaded to configure, logic, including integer and/or floating point units (collectively, arithmetic logic units (ALUs)). In at least one embodiment, code, such as graph code, loads weight or other parameter information into processor ALUs based on architecture of a neural network to which this code corresponds. In at least one embodiment, code and/or data storage **601** stores weight parameters and/or input/output data of each layer of a neural network trained or used in conjunction with one or more embodiments during forward propagation of input/output data and/or weight parameters during training and/or inferencing using aspects of one or more embodiments. In at least one embodiment, any portion of code and/or data storage **601** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory.

In at least one embodiment, any portion of code and/or data storage **601** may be internal or external to one or more processors or other hardware logic devices or circuits. In at least one embodiment, code and/or data storage **601** may be cache memory, dynamic randomly addressable memory ("DRAM"), static randomly addressable memory ("SRAM"), non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, choice of whether code and/or data storage **601** is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors.

In at least one embodiment, inference and/or training logic **615** may include, without limitation, a code and/or data storage **605** to store backward and/or output weight and/or input/output data corresponding to neurons or layers of a neural network trained and/or used for inferencing in aspects of one or more embodiments. In at least one embodiment, code and/or data storage **605** stores weight parameters and/or input/output data of each layer of a neural network trained or used in conjunction with one or more embodiments during backward propagation of input/output data and/or weight parameters during training and/or inferencing using aspects of one or more embodiments. In at least one embodiment, training logic **615** may include, or be coupled

to code and/or data storage **605** to store graph code or other software to control timing and/or order, in which weight and/or other parameter information is to be loaded to configure, logic, including integer and/or floating point units (collectively, arithmetic logic units (ALUs)). In at least one embodiment, code, such as graph code, loads weight or other parameter information into processor ALUs based on an architecture of a neural network to which this code corresponds. In at least one embodiment, any portion of code and/or data storage **605** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory. In at least one embodiment, any portion of code and/or data storage **605** may be internal or external to on one or more processors or other hardware logic devices or circuits. In at least one embodiment, code and/or data storage **605** may be cache memory, DRAM, SRAM, non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, choice of whether code and/or data storage **605** is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors.

In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be separate storage structures. In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be same storage structure. In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be partially same storage structure and partially separate storage structures. In at least one embodiment, any portion of code and/or data storage **601** and code and/or data storage **605** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory.

In at least one embodiment, inference and/or training logic **615** may include, without limitation, one or more arithmetic logic unit(s) ("ALU(s)") **610**, including integer and/or floating point units, to perform logical and/or mathematical operations based, at least in part on, or indicated by, training and/or inference code (e.g., graph code), a result of which may produce activations (e.g., output values from layers or neurons within a neural network) stored in an activation storage **620** that are functions of input/output and/or weight parameter data stored in code and/or data storage **601** and/or code and/or data storage **605**. In at least one embodiment, activations stored in activation storage **620** are generated according to linear algebraic and or matrix-based mathematics performed by ALU(s) **610** in response to performing instructions or other code, wherein weight values stored in code and/or data storage **605** and/or code and/or data storage **601** are used as operands along with other values, such as bias values, gradient information, momentum values, or other parameters or hyperparameters, any or all of which may be stored in code and/or data storage **605** or code and/or data storage **601** or another storage on or off-chip.

In at least one embodiment, ALU(s) **610** are included within one or more processors or other hardware logic devices or circuits, whereas in another embodiment, ALU(s) **610** may be external to a processor or other hardware logic device or circuit that uses them (e.g., a co-processor). In at least one embodiment, ALUs **610** may be included within a processor's execution units or otherwise within a bank of ALUs accessible by a processor's execution units either

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within same processor or distributed between different processors of different types (e.g., central processing units, graphics processing units, fixed function units, etc.). In at least one embodiment, code and/or data storage 601, code and/or data storage 605, and activation storage 620 may be on same processor or other hardware logic device or circuit, whereas in another embodiment, they may be in different processors or other hardware logic devices or circuits, or some combination of same and different processors or other hardware logic devices or circuits. In at least one embodiment, any portion of activation storage 620 may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory. Furthermore, inferencing and/or training code may be stored with other code accessible to a processor or other hardware logic or circuit and fetched and/or processed using a processor's fetch, decode, scheduling, execution, retirement and/or other logical circuits.

In at least one embodiment, activation storage 620 may be cache memory, DRAM, SRAM, non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, activation storage 620 may be completely or partially within or external to one or more processors or other logical circuits. In at least one embodiment, choice of whether activation storage 620 is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors. In at least one embodiment, inference and/or training logic 615 illustrated in FIG. 6A may be used in conjunction with an application-specific integrated circuit ("ASIC"), such as Tensorflow® Processing Unit from Google, an inference processing unit (IPU) from Graphcore™, or a Nervana® (e.g., "Lake Crest") processor from Intel Corp. In at least one embodiment, inference and/or training logic 615 illustrated in FIG. 6A may be used in conjunction with central processing unit ("CPU") hardware, graphics processing unit ("GPU") hardware or other hardware, such as field programmable gate arrays ("FPGAs").

FIG. 6B illustrates inference and/or training logic 615, according to at least one or more embodiments. In at least one embodiment, inference and/or training logic 615 may include, without limitation, hardware logic in which computational resources are dedicated or otherwise exclusively used in conjunction with weight values or other information corresponding to one or more layers of neurons within a neural network. In at least one embodiment, inference and/or training logic 615 illustrated in FIG. 6B may be used in conjunction with an application-specific integrated circuit (ASIC), such as Tensorflow® Processing Unit from Google, an inference processing unit (IPU) from Graphcore™, or a Nervana® (e.g., "Lake Crest") processor from Intel Corp. In at least one embodiment, inference and/or training logic 615 illustrated in FIG. 6B may be used in conjunction with central processing unit (CPU) hardware, graphics processing unit (GPU) hardware or other hardware, such as field programmable gate arrays (FPGAs). In at least one embodiment, inference and/or training logic 615 includes, without limitation, code and/or data storage 601 and code and/or data storage 605, which may be used to store code (e.g., graph code), weight values and/or other information, including bias values, gradient information, momentum values, and/or other parameter or hyperparameter information. In at least one embodiment illustrated in FIG. 6B, each of code and/or

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data storage 601 and code and/or data storage 605 is associated with a dedicated computational resource, such as computational hardware 602 and computational hardware 606, respectively. In at least one embodiment, each of computational hardware 602 and computational hardware 606 comprises one or more ALUs that perform mathematical functions, such as linear algebraic functions, only on information stored in code and/or data storage 601 and code and/or data storage 605, respectively, result of which is stored in activation storage 620.

In at least one embodiment, each of code and/or data storage 601 and 605 and corresponding computational hardware 602 and 606, respectively, correspond to different layers of a neural network, such that resulting activation from one "storage/computational pair 601/602" of code and/or data storage 601 and computational hardware 602 is provided as an input to "storage/computational pair 605/606" of code and/or data storage 605 and computational hardware 606, in order to mirror conceptual organization of a neural network. In at least one embodiment, each of storage/computational pairs 601/602 and 605/606 may correspond to more than one neural network layer. In at least one embodiment, additional storage/computation pairs (not shown) subsequent to or in parallel with storage computation pairs 601/602 and 605/606 may be included in inference and/or training logic 615.

Data Center

FIG. 7 illustrates an example data center 700, in which at least one embodiment may be used. In at least one embodiment, data center 700 includes a data center infrastructure layer 710, a framework layer 720, a software layer 730, and an application layer 740.

In at least one embodiment, as shown in FIG. 7, data center infrastructure layer 710 may include a resource orchestrator 712, grouped computing resources 714, and node computing resources ("node C.R.s") 716(1)-716(N), where "N" represents any whole, positive integer. In at least one embodiment, node C.R.s 716(1)-716(N) may include, but are not limited to, any number of central processing units ("CPUs") or other processors (including accelerators, field programmable gate arrays (FPGAs), graphics processors, etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output ("NW I/O") devices, network switches, virtual machines ("VMs"), power modules, and cooling modules, etc. In at least one embodiment, one or more node C.R.s from among node C.R.s 716(1)-716(N) may be a server having one or more of above-mentioned computing resources.

In at least one embodiment, grouped computing resources 714 may include separate groupings of node C.R.s housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node C.R.s within grouped computing resources 714 may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s including CPUs or processors may grouped within one or more racks to provide compute resources to support one or more workloads. In at least one embodiment, one or more racks may also include any number of power modules, cooling modules, and network switches, in any combination.

In at least one embodiment, resource orchestrator 712 may configure or otherwise control one or more node C.R.s 716(1)-716(N) and/or grouped computing resources 714. In at least one embodiment, resource orchestrator 712 may

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include a software design infrastructure (“SDI”) management entity for data center 700. In at least one embodiment, resource orchestrator may include hardware, software or some combination thereof.

In at least one embodiment, as shown in FIG. 7, framework layer 720 includes a job scheduler 722, a configuration manager 724, a resource manager 726 and a distributed file system 728. In at least one embodiment, framework layer 720 may include a framework to support software 732 of software layer 730 and/or one or more application(s) 742 of application layer 740. In at least one embodiment, software 732 or application(s) 742 may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. In at least one embodiment, framework layer 720 may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter “Spark”) that may utilize distributed file system 728 for large-scale data processing (e.g., “big data”). In at least one embodiment, job scheduler 722 may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center 700. In at least one embodiment, configuration manager 724 may be capable of configuring different layers such as software layer 730 and framework layer 720 including Spark and distributed file system 728 for supporting large-scale data processing. In at least one embodiment, resource manager 726 may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system 728 and job scheduler 722. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource 714 at data center infrastructure layer 710. In at least one embodiment, resource manager 726 may coordinate with resource orchestrator 712 to manage these mapped or allocated computing resources.

In at least one embodiment, software 732 included in software layer 730 may include software used by at least portions of node C.R.s 716(1)-716(N), grouped computing resources 714, and/or distributed file system 728 of framework layer 720. one or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

In at least one embodiment, application(s) 742 included in application layer 740 may include one or more types of applications used by at least portions of node C.R.s 716(1)-716(N), grouped computing resources 714, and/or distributed file system 728 of framework layer 720. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.) or other machine learning applications used in conjunction with one or more embodiments.

In at least one embodiment, any of configuration manager 724, resource manager 726, and resource orchestrator 712 may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. In at least one embodiment, self-modifying actions may relieve a data center operator of data center 700 from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

In at least one embodiment, data center 700 may include tools, services, software or other resources to train one or

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more machine learning models or predict or infer information using one or more machine learning models according to one or more embodiments described herein. For example, in at least one embodiment, a machine learning model may be trained by calculating weight parameters according to a neural network architecture using software and computing resources described above with respect to data center 700. In at least one embodiment, trained machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to data center 700 by using weight parameters calculated through one or more training techniques described herein.

In at least one embodiment, data center may use CPUs, application-specific integrated circuits (ASICs), GPUs, FPGAs, or other hardware to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in system FIG. 7 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

Computer Systems

FIG. 8 is a block diagram illustrating an exemplary computer system, which may be a system with interconnected devices and components, a system-on-a-chip (SOC) or some combination thereof 800 formed with a processor that may include execution units to execute an instruction, according to at least one embodiment. In at least one embodiment, computer system 800 may include, without limitation, a component, such as a processor 802 to employ execution units including logic to perform algorithms for process data, in accordance with present disclosure, such as in embodiment described herein. In at least one embodiment, computer system 800 may include processors, such as PENTIUM® Processor family, Xeon™, Itanium®, XScale™ and/or StrongARM™, Intel® Core™, or Intel® Nervana™ microprocessors available from Intel Corporation of Santa Clara, California, although other systems (including PCs having other microprocessors, engineering workstations, set-top boxes and like) may also be used. In at least one embodiment, computer system 800 may execute a version of WINDOWS® operating system available from Microsoft Corporation of Redmond, Wash., although other operating systems (UNIX and Linux for example), embedded software, and/or graphical user interfaces, may also be used.

Embodiments may be used in other devices such as handheld devices and embedded applications. Some examples of handheld devices include cellular phones, Internet Protocol devices, digital cameras, personal digital assistants (“PDAs”), and handheld PCs. In at least one embodi-

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ment, embedded applications may include a microcontroller, a digital signal processor (“DSP”), system on a chip, network computers (“NetPCs”), set-top boxes, network hubs, wide area network (“WAN”) switches, or any other system that may perform one or more instructions in accordance with at least one embodiment.

In at least one embodiment, computer system **800** may include, without limitation, processor **802** that may include, without limitation, one or more execution units **808** to perform machine learning model training and/or inferencing according to techniques described herein. In at least one embodiment, computer system **800** is a single processor desktop or server system, but in another embodiment computer system **800** may be a multiprocessor system. In at least one embodiment, processor **802** may include, without limitation, a complex instruction set computer (“CISC”) microprocessor, a reduced instruction set computing (“RISC”) microprocessor, a very long instruction word (“VLIW”) microprocessor, a processor implementing a combination of instruction sets, or any other processor device, such as a digital signal processor, for example. In at least one embodiment, processor **802** may be coupled to a processor bus **810** that may transmit data signals between processor **802** and other components in computer system **800**.

In at least one embodiment, processor **802** may include, without limitation, a Level 1 (“L1”) internal cache memory (“cache”) **804**. In at least one embodiment, processor **802** may have a single internal cache or multiple levels of internal cache. In at least one embodiment, cache memory may reside external to processor **802**. Other embodiments may also include a combination of both internal and external caches depending on particular implementation and needs. In at least one embodiment, register file **806** may store different types of data in various registers including, without limitation, integer registers, floating point registers, status registers, and instruction pointer register.

In at least one embodiment, execution unit **808**, including, without limitation, logic to perform integer and floating point operations, also resides in processor **802**. In at least one embodiment, processor **802** may also include a microcode (“ucode”) read only memory (“ROM”) that stores microcode for certain macro instructions. In at least one embodiment, execution unit **808** may include logic to handle a packed instruction set **809**. In at least one embodiment, by including packed instruction set **809** in an instruction set of a general-purpose processor **802**, along with associated circuitry to execute instructions, operations used by many multimedia applications may be performed using packed data in a general-purpose processor **802**. In one or more embodiments, many multimedia applications may be accelerated and executed more efficiently by using full width of a processor’s data bus for performing operations on packed data, which may eliminate need to transfer smaller units of data across processor’s data bus to perform one or more operations one data element at a time.

In at least one embodiment, execution unit **808** may also be used in microcontrollers, embedded processors, graphics devices, DSPs, and other types of logic circuits. In at least one embodiment, computer system **800** may include, without limitation, a memory **820**. In at least one embodiment, memory **820** may be implemented as a Dynamic Random Access Memory (“DRAM”) device, a Static Random Access Memory (“SRAM”) device, flash memory device, or other memory device. In at least one embodiment, memory **820** may store instruction(s) **819** and/or data **821** represented by data signals that may be executed by processor **802**.

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In at least one embodiment, system logic chip may be coupled to processor bus **810** and memory **820**. In at least one embodiment, system logic chip may include, without limitation, a memory controller hub (“MCH”) **816**, and processor **802** may communicate with MCH **816** via processor bus **810**. In at least one embodiment, MCH **816** may provide a high bandwidth memory path **818** to memory **820** for instruction and data storage and for storage of graphics commands, data and textures. In at least one embodiment, MCH **816** may direct data signals between processor **802**, memory **820**, and other components in computer system **800** and to bridge data signals between processor bus **810**, memory **820**, and a system I/O **822**. In at least one embodiment, system logic chip may provide a graphics port for coupling to a graphics controller. In at least one embodiment, MCH **816** may be coupled to memory **820** through a high bandwidth memory path **818** and graphics/video card **812** may be coupled to MCH **816** through an Accelerated Graphics Port (“AGP”) interconnect **814**.

In at least one embodiment, computer system **800** may use system I/O **822** that is a proprietary hub interface bus to couple MCH **816** to I/O controller hub (“ICH”) **830**. In at least one embodiment, ICH **830** may provide direct connections to some I/O devices via a local I/O bus. In at least one embodiment, local I/O bus may include, without limitation, a high-speed I/O bus for connecting peripherals to memory **820**, chipset, and processor **802**. Examples may include, without limitation, an audio controller **829**, a firmware hub (“flash BIOS”) **828**, a wireless transceiver **826**, a data storage **824**, a legacy I/O controller **823** containing user input and keyboard interfaces **825**, a serial expansion port **827**, such as Universal Serial Bus (“USB”), and a network controller **834**. data storage **824** may comprise a hard disk drive, a floppy disk drive, a CD-ROM device, a flash memory device, or other mass storage device.

In at least one embodiment, FIG. **8** illustrates a system, which includes interconnected hardware devices or “chips”, whereas in other embodiments, FIG. **8** may illustrate an exemplary System on a Chip (“SoC”). In at least one embodiment, devices illustrated in FIG. **8** may be interconnected with proprietary interconnects, standardized interconnects (e.g., PCIe) or some combination thereof. In at least one embodiment, one or more components of computer system **800** are interconnected using compute express link (CXL) interconnects.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **8** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **9** is a block diagram illustrating an electronic device **900** for utilizing a processor **910**, according to at least one embodiment. In at least one embodiment, electronic device **900** may be, for example and without limitation, a notebook, a tower server, a rack server, a blade server, a laptop, a desktop, a tablet, a mobile device, a phone, an embedded computer, or any other suitable electronic device.

In at least one embodiment, system **900** may include, without limitation, processor **910** communicatively coupled to any suitable number or kind of components, peripherals, modules, or devices. In at least one embodiment, processor **910** coupled using a bus or interface, such as a 1° C. bus, a System Management Bus (“SMBus”), a Low Pin Count (LPC) bus, a Serial Peripheral Interface (“SPI”), a High Definition Audio (“HDA”) bus, a Serial Advance Technology Attachment (“SATA”) bus, a Universal Serial Bus (“USB”) (versions 1, 2, 3), or a Universal Asynchronous Receiver/Transmitter (“UART”) bus. In at least one embodiment, FIG. **9** illustrates a system, which includes interconnected hardware devices or “chips”, whereas in other embodiments, FIG. **9** may illustrate an exemplary System on a Chip (“SoC”). In at least one embodiment, devices illustrated in FIG. **9** may be interconnected with proprietary interconnects, standardized interconnects (e.g., PCIe) or some combination thereof. In at least one embodiment, one or more components of FIG. **9** are interconnected using compute express link (CXL) interconnects.

In at least one embodiment, FIG. **9** may include a display **924**, a touch screen **925**, a touch pad **930**, a Near Field Communications unit (“NFC”) **945**, a sensor hub **940**, a thermal sensor **946**, an Express Chipset (“EC”) **935**, a Trusted Platform Module (“TPM”) **938**, BIOS/firmware/flash memory (“BIOS, FW Flash”) **922**, a DSP **960**, a drive **920** such as a Solid State Disk (“SSD”) or a Hard Disk Drive (“HDD”), a wireless local area network unit (“WLAN”) **950**, a Bluetooth unit **952**, a Wireless Wide Area Network unit (“WWAN”) **956**, a Global Positioning System (GPS) **955**, a camera (“USB 3.0 camera”) **954** such as a USB 3.0 camera, and/or a Low Power Double Data Rate (“LPDDR”) memory unit (“LPDDR3”) **915** implemented in, for example, LPDDR3 standard. These components may each be implemented in any suitable manner.

In at least one embodiment, other components may be communicatively coupled to processor **910** through components discussed above. In at least one embodiment, an accelerometer **941**, Ambient Light Sensor (“ALS”) **942**, compass **943**, and a gyroscope **944** may be communicatively coupled to sensor hub **940**. In at least one embodiment, thermal sensor **939**, a fan **937**, a keyboard **946**, and a touch pad **930** may be communicatively coupled to EC **935**. In at least one embodiment, speaker **963**, headphones **964**, and microphone (“mic”) **965** may be communicatively coupled to an audio unit (“audio codec and class d amp”) **962**, which may in turn be communicatively coupled to DSP **960**. In at least one embodiment, audio unit **964** may include, for example and without limitation, an audio coder/decoder (“codec”) and a class D amplifier. In at least one embodiment, SIM card (“SIM”) **957** may be communicatively coupled to WWAN unit **956**. In at least one embodiment, components such as WLAN unit **950** and Bluetooth unit **952**, as well as WWAN unit **956** may be implemented in a Next Generation Form Factor (“NGFF”).

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **9** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or

more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **10** illustrates a computer system **1000**, according to at least one embodiment. In at least one embodiment, computer system **1000** is configured to implement various processes and methods described throughout this disclosure.

In at least one embodiment, computer system **1000** comprises, without limitation, at least one central processing unit (“CPU”) **1002** that is connected to a communication bus **1010** implemented using any suitable protocol, such as PCI (“Peripheral Component Interconnect”), peripheral component interconnect express (“PCI-Express”), AGP (“Accelerated Graphics Port”), HyperTransport, or any other bus or point-to-point communication protocol(s). In at least one embodiment, computer system **1000** includes, without limitation, a main memory **1004** and control logic (e.g., implemented as hardware, software, or a combination thereof) and data are stored in main memory **1004** which may take form of random access memory (“RAM”). In at least one embodiment, a network interface subsystem (“network interface”) **1022** provides an interface to other computing devices and networks for receiving data from and transmitting data to other systems from computer system **1000**.

In at least one embodiment, computer system **1000**, in at least one embodiment, includes, without limitation, input devices **1008**, parallel processing system **1012**, and display devices **1006** which can be implemented using a conventional cathode ray tube (“CRT”), liquid crystal display (“LCD”), light emitting diode (“LED”), plasma display, or other suitable display technologies. In at least one embodiment, user input is received from input devices **1008** such as keyboard, mouse, touchpad, microphone, and more. In at least one embodiment, each of foregoing modules can be situated on a single semiconductor platform to form a processing system.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **10** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **11** illustrates a computer system **1100**, according to at least one embodiment. In at least one embodiment, computer system **1100** includes, without limitation, a computer **1110** and a USB stick **1120**. In at least one embodiment, computer **1110** may include, without limitation, any number and type of processor(s) (not shown) and a memory (not shown). In at least one embodiment, computer **1110** includes, without limitation, a server, a cloud instance, a laptop, and a desktop computer.

In at least one embodiment, USB stick **1120** includes, without limitation, a processing unit **1130**, a USB interface **1140**, and USB interface logic **1150**. In at least one embodiment, processing unit **1130** may be any instruction execution system, apparatus, or device capable of executing instructions. In at least one embodiment, processing unit **1130** may include, without limitation, any number and type of pro-

cessing cores (not shown). In at least one embodiment, processing core **1130** comprises an application specific integrated circuit (“ASIC”) that is optimized to perform any amount and type of operations associated with machine learning. For instance, in at least one embodiment, processing core **1130** is a tensor processing unit (“TPC”) that is optimized to perform machine learning inference operations. In at least one embodiment, processing core **1130** is a vision processing unit (“VPU”) that is optimized to perform machine vision and machine learning inference operations.

In at least one embodiment, USB interface **1140** may be any type of USB connector or USB socket. For instance, in at least one embodiment, USB interface **1140** is a USB 3.0 Type-C socket for data and power. In at least one embodiment, USB interface **1140** is a USB 3.0 Type-A connector. In at least one embodiment, USB interface logic **1150** may include any amount and type of logic that enables processing unit **1130** to interface with or devices (e.g., computer **1110**) via USB connector **1140**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **11** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **12A** illustrates an exemplary architecture in which a plurality of GPUs **1210-1213** is communicatively coupled to a plurality of multi-core processors **1205-1206** over high-speed links **1240-1243** (e.g., buses, point-to-point interconnects, etc.). In one embodiment, high-speed links **1240-1243** support a communication throughput of 4 GB/s, 30 GB/s, 80 GB/s or higher. Various interconnect protocols may be used including, but not limited to, PCIe 4.0 or 5.0 and NVLink 2.0.

In addition, and in one embodiment, two or more of GPUs **1210-1213** are interconnected over high-speed links **1229-1230**, which may be implemented using same or different protocols/links than those used for high-speed links **1240-1243**. Similarly, two or more of multi-core processors **1205-1206** may be connected over high speed link **1228** which may be symmetric multi-processor (SMP) buses operating at 20 GB/s, 30 GB/s, 120 GB/s or higher. Alternatively, all communication between various system components shown in FIG. **12A** may be accomplished using same protocols/links (e.g., over a common interconnection fabric).

In one embodiment, each multi-core processor **1205-1206** is communicatively coupled to a processor memory **1201-1202**, via memory interconnects **1226-1227**, respectively, and each GPU **1210-1213** is communicatively coupled to GPU memory **1220-1223** over GPU memory interconnects **1250-1253**, respectively. Memory interconnects **1226-1227** and **1250-1253** may utilize same or different memory access technologies. By way of example, and not limitation, processor memories **1201-1202** and GPU memories **1220-1223** may be volatile memories such as dynamic random access memories (DRAMs) (including stacked DRAMs), Graphics DDR SDRAM (GDDR) (e.g., GDDR5, GDDR6), or High Bandwidth Memory (HBM) and/or may be non-volatile

memories such as 3D XPoint or Nano-Ram. In one embodiment, some portion of processor memories **1201-1202** may be volatile memory and another portion may be non-volatile memory (e.g., using a two-level memory (2LM) hierarchy).

As described below, although various processors **1205-1206** and GPUs **1210-1213** may be physically coupled to a particular memory **1201-1202**, **1220-1223**, respectively, a unified memory architecture may be implemented in which a same virtual system address space (also referred to as “effective address” space) is distributed among various physical memories. For example, processor memories **1201-1202** may each comprise 64 GB of system memory address space and GPU memories **1220-1223** may each comprise 32 GB of system memory address space (resulting in a total of 256 GB addressable memory in this example).

FIG. **12B** illustrates additional details for an interconnection between a multi-core processor **1207** and a graphics acceleration module **1246** in accordance with one exemplary embodiment. Graphics acceleration module **1246** may include one or more GPU chips integrated on a line card which is coupled to processor **1207** via high-speed link **1240**. Alternatively, graphics acceleration module **1246** may be integrated on a same package or chip as processor **1207**.

In at least one embodiment, illustrated processor **1207** includes a plurality of cores **1260A-1260D**, each with a translation lookaside buffer **1261A-1261D** and one or more caches **1262A-1262D**. In at least one embodiment, cores **1260A-1260D** may include various other components for executing instructions and processing data which are not illustrated. Caches **1262A-1262D** may comprise level 1 (L1) and level 2 (L2) caches. In addition, one or more shared caches **1256** may be included in caches **1262A-1262D** and shared by sets of cores **1260A-1260D**. For example, one embodiment of processor **1207** includes 24 cores, each with its own L1 cache, twelve shared L2 caches, and twelve shared L3 caches. In this embodiment, one or more L2 and L3 caches are shared by two adjacent cores. Processor **1207** and graphics acceleration module **1246** connect with system memory **1214**, which may include processor memories **1201-1202** of FIG. **12A**.

Coherency is maintained for data and instructions stored in various caches **1262A-1262D**, **1256** and system memory **1214** via inter-core communication over a coherence bus **1264**. For example, each cache may have cache coherency logic/circuitry associated therewith to communicate to over coherence bus **1264** in response to detected reads or writes to particular cache lines. In one implementation, a cache snooping protocol is implemented over coherence bus **1264** to snoop cache accesses.

In one embodiment, a proxy circuit **1225** communicatively couples graphics acceleration module **1246** to coherence bus **1264**, allowing graphics acceleration module **1246** to participate in a cache coherence protocol as a peer of cores **1260A-1260D**. In particular, an interface **1235** provides connectivity to proxy circuit **1225** over high-speed link **1240** (e.g., a PCIe bus, NVLink, etc.) and an interface **1237** connects graphics acceleration module **1246** to link **1240**.

In one implementation, an accelerator integration circuit **1236** provides cache management, memory access, context management, and interrupt management services on behalf of a plurality of graphics processing engines **1231**, **1232**, N of graphics acceleration module **1246**. Graphics processing engines **1231**, **1232**, N may each comprise a separate graphics processing unit (GPU). Alternatively, graphics processing engines **1231**, **1232**, N may comprise different types of graphics processing engines within a GPU such as

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graphics execution units, media processing engines (e.g., video encoders/decoders), samplers, and blit engines. In at least one embodiment, graphics acceleration module **1246** may be a GPU with a plurality of graphics processing engines **1231-1232**, N or graphics processing engines **1231-1232**, N may be individual GPUs integrated on a common package, line card, or chip.

In one embodiment, accelerator integration circuit **1236** includes a memory management unit (MMU) **1239** for performing various memory management functions such as virtual-to-physical memory translations (also referred to as effective-to-real memory translations) and memory access protocols for accessing system memory **1214**. MMU **1239** may also include a translation lookaside buffer (TLB) (not shown) for caching virtual/effective to physical/real address translations. In one implementation, a cache **1238** stores commands and data for efficient access by graphics processing engines **1231-1232**, N. In one embodiment, data stored in cache **1238** and graphics memories **1233-1234**, M is kept coherent with core caches **1262A-1262D**, **1256**, and system memory **1214**. As mentioned above, this may be accomplished via proxy circuit **1225** on behalf of cache **1238** and memories **1233-1234**, M (e.g., sending updates to cache **1238** related to modifications/accesses of cache lines on processor caches **1262A-1262D**, **1256**, and receiving updates from cache **1238**).

A set of registers **1245** store context data for threads executed by graphics processing engines **1231-1232**, N and a context management circuit **1248** manages thread contexts. For example, context management circuit **1248** may perform save and restore operations to save and restore contexts of various threads during contexts switches (e.g., where a first thread is saved and a second thread is stored so that a second thread can be executed by a graphics processing engine). For example, on a context switch, context management circuit **1248** may store current register values to a designated region in memory (e.g., identified by a context pointer). It may then restore register values when returning to a context. In one embodiment, an interrupt management circuit **1247** receives and processes interrupts received from system devices.

In one implementation, virtual/effective addresses from a graphics processing engine **1231** are translated to real/physical addresses in system memory **1214** by MMU **1239**. One embodiment of accelerator integration circuit **1236** supports multiple (e.g., 4, 8, 16) graphics accelerator modules **1246** and/or other accelerator devices. Graphics accelerator module **1246** may be dedicated to a single application executed on processor **1207** or may be shared between multiple applications. In one embodiment, a virtualized graphics execution environment is presented in which resources of graphics processing engines **1231-1232**, N are shared with multiple applications or virtual machines (VMs). In at least one embodiment, resources may be subdivided into “slices” which are allocated to different VMs and/or applications based on processing requirements and priorities associated with VMs and/or applications.

In at least one embodiment, accelerator integration circuit **1236** performs as a bridge to a system for graphics acceleration module **1246** and provides address translation and system memory cache services. In addition, accelerator integration circuit **1236** may provide virtualization facilities for a host processor to manage virtualization of graphics processing engines **1231-1232**, N, interrupts, and memory management.

Because hardware resources of graphics processing engines **1231-1232**, N are mapped explicitly to a real

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address space seen by host processor **1207**, any host processor can address these resources directly using an effective address value. One function of accelerator integration circuit **1236**, in one embodiment, is physical separation of graphics processing engines **1231-1232**, N so that they appear to a system as independent units.

In at least one embodiment, one or more graphics memories **1233-1234**, M are coupled to each of graphics processing engines **1231-1232**, N, respectively. Graphics memories **1233-1234**, M store instructions and data being processed by each of graphics processing engines **1231-1232**, N. Graphics memories **1233-1234**, M may be volatile memories such as DRAMs (including stacked DRAMs), GDDR memory (e.g., GDDR5, GDDR6), or HBM, and/or may be non-volatile memories such as 3D XPoint or Nano-Ram.

In one embodiment, to reduce data traffic over link **1240**, biasing techniques are used to ensure that data stored in graphics memories **1233-1234**, M is data which will be used most frequently by graphics processing engines **1231-1232**, N and preferably not used by cores **1260A-1260D** (at least not frequently). Similarly, a biasing mechanism attempts to keep data needed by cores (and preferably not graphics processing engines **1231-1232**, N) within caches **1262A-1262D**, **1256** of cores and system memory **1214**.

FIG. **12C** illustrates another exemplary embodiment in which accelerator integration circuit **1236** is integrated within processor **1207**. In at least this embodiment, graphics processing engines **1231-1232**, N communicate directly over high-speed link **1240** to accelerator integration circuit **1236** via interface **1237** and interface **1235** (which, again, may be utilize any form of bus or interface protocol). Accelerator integration circuit **1236** may perform same operations as those described with respect to FIG. **12B**, but potentially at a higher throughput given its close proximity to coherence bus **1264** and caches **1262A-1262D**, **1256**. At least one embodiment supports different programming models including a dedicated-process programming model (no graphics acceleration module virtualization) and shared programming models (with virtualization), which may include programming models which are controlled by accelerator integration circuit **1236** and programming models which are controlled by graphics acceleration module **1246**.

In at least one embodiment, graphics processing engines **1231-1232**, N are dedicated to a single application or process under a single operating system. In at least one embodiment, a single application can funnel other application requests to graphics processing engines **1231-1232**, N, providing virtualization within a VM/partition.

In at least one embodiment, graphics processing engines **1231-1232**, N, may be shared by multiple VM/application partitions. In at least one embodiment, shared models may use a system hypervisor to virtualize graphics processing engines **1231-1232**, N to allow access by each operating system. For single-partition systems without a hypervisor, graphics processing engines **1231-1232**, N are owned by an operating system. In at least one embodiment, an operating system can virtualize graphics processing engines **1231-1232**, N to provide access to each process or application.

In at least one embodiment, graphics acceleration module **1246** or an individual graphics processing engine **1231-1232**, N selects a process element using a process handle. In at least one embodiment, process elements are stored in system memory **1214** and are addressable using an effective address to real address translation techniques described herein. In at least one embodiment, a process handle may be an implementation-specific value provided to a host process when registering its context with graphics processing engine

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1231-1232, N (that is, calling system software to add a process element to a process element linked list). In at least one embodiment, a lower 16-bits of a process handle may be an offset of a process element within a process element linked list.

FIG. 12D illustrates an exemplary accelerator integration slice **1290**. As used herein, a “slice” comprises a specified portion of processing resources of accelerator integration circuit **1236**. Application effective address space **1282** within system memory **1214** stores process elements **1283**. In one embodiment, process elements **1283** are stored in response to GPU invocations **1281** from applications **1280** executed on processor **1207**. A process element **1283** contains process state for corresponding application **1280**. A work descriptor (WD) **1284** contained in process element **1283** can be a single job requested by an application or may contain a pointer to a queue of jobs. In at least one embodiment, WD **1284** is a pointer to a job request queue in an application’s address space **1282**.

Graphics acceleration module **1246** and/or individual graphics processing engines **1231-1232**, N can be shared by all or a subset of processes in a system. In at least one embodiment, an infrastructure for setting up process state and sending a WD **1284** to a graphics acceleration module **1246** to start a job in a virtualized environment may be included.

In at least one embodiment, a dedicated-process programming model is implementation-specific. In this model, a single process owns graphics acceleration module **1246** or an individual graphics processing engine **1231**. Because graphics acceleration module **1246** is owned by a single process, a hypervisor initializes accelerator integration circuit **1236** for an owning partition and an operating system initializes accelerator integration circuit **1236** for an owning process when graphics acceleration module **1246** is assigned.

In operation, a WD fetch unit **1291** in accelerator integration slice **1290** fetches next WD **1284** which includes an indication of work to be done by one or more graphics processing engines of graphics acceleration module **1246**. Data from WD **1284** may be stored in registers **1245** and used by MMU **1239**, interrupt management circuit **1247**, and/or context management circuit **1248** as illustrated. For example, one embodiment of MMU **1239** includes segment/page walk circuitry for accessing segment/page tables **1286** within OS virtual address space **1285**. Interrupt management circuit **1247** may process interrupt events **1292** received from graphics acceleration module **1246**. When performing graphics operations, an effective address **1293** generated by a graphics processing engine **1231-1232**, N is translated to a real address by MMU **1239**.

In one embodiment, a same set of registers **1245** are duplicated for each graphics processing engine **1231-1232**, N and/or graphics acceleration module **1246** and may be initialized by a hypervisor or operating system. Each of these duplicated registers may be included in an accelerator integration slice **1290**. Exemplary registers that may be initialized by a hypervisor are shown in Table 1.

TABLE 1

Hypervisor Initialized Registers	
1	Slice Control Register
2	Real Address (RA) Scheduled Processes Area Pointer
3	Authority Mask Override Register
4	Interrupt Vector Table Entry Offset

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TABLE 1-continued

Hypervisor Initialized Registers	
5	Interrupt Vector Table Entry Limit
6	State Register
7	Logical Partition ID
8	Real address (RA) Hypervisor Accelerator Utilization Record Pointer
9	Storage Description Register

Exemplary registers that may be initialized by an operating system are shown in Table 2.

TABLE 2

Operating System Initialized Registers	
1	Process and Thread Identification
2	Effective Address (EA) Context Save/Restore Pointer
3	Virtual Address (VA) Accelerator Utilization Record Pointer
4	Virtual Address (VA) Storage Segment Table Pointer
5	Authority Mask
6	Work descriptor

In one embodiment, each WD **1284** is specific to a particular graphics acceleration module **1246** and/or graphics processing engines **1231-1232**, N. It contains all information required by a graphics processing engine **1231-1232**, N to do work or it can be a pointer to a memory location where an application has set up a command queue of work to be completed.

FIG. 12E illustrates additional details for one exemplary embodiment of a shared model. This embodiment includes a hypervisor real address space **1298** in which a process element list **1299** is stored. Hypervisor real address space **1298** is accessible via a hypervisor **1296** which virtualizes graphics acceleration module engines for operating system **1295**.

In at least one embodiment, shared programming models allow for all or a subset of processes from all or a subset of partitions in a system to use a graphics acceleration module **1246**. There are two programming models where graphics acceleration module **1246** is shared by multiple processes and partitions: time-sliced shared and graphics-directed shared.

In this model, system hypervisor **1296** owns graphics acceleration module **1246** and makes its function available to all operating systems **1295**. For a graphics acceleration module **1246** to support virtualization by system hypervisor **1296**, graphics acceleration module **1246** may adhere to the following: 1) An application’s job request must be autonomous (that is, state does not need to be maintained between jobs), or graphics acceleration module **1246** must provide a context save and restore mechanism. 2) An application’s job request is guaranteed by graphics acceleration module **1246** to complete in a specified amount of time, including any translation faults, or graphics acceleration module **1246** provides an ability to preempt processing of a job. 3) Graphics acceleration module **1246** must be guaranteed fairness between processes when operating in a directed shared programming model.

In at least one embodiment, application **1280** is required to make an operating system **1295** system call with a graphics acceleration module **1246** type, a work descriptor (WD), an authority mask register (AMR) value, and a context save/restore area pointer (CSRP). In at least one embodiment, graphics acceleration module **1246** type describes a targeted acceleration function for a system call. In at least one embodiment, graphics acceleration module

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1246 type may be a system-specific value. In at least one embodiment, WD is formatted specifically for graphics acceleration module 1246 and can be in a form of a graphics acceleration module 1246 command, an effective address pointer to a user-defined structure, an effective address pointer to a queue of commands, or any other data structure to describe work to be done by graphics acceleration module 1246. In one embodiment, an AMR value is an AMR state to use for a current process. In at least one embodiment, a value passed to an operating system is similar to an application setting an AMR. If accelerator integration circuit 1236 and graphics acceleration module 1246 implementations do not support a User Authority Mask Override Register (UAMOR), an operating system may apply a current UAMOR value to an AMR value before passing an AMR in a hypervisor call. Hypervisor 1296 may optionally apply a current Authority Mask Override Register (AMOR) value before placing an AMR into process element 1283. In at least one embodiment, CSRP is one of registers 1245 containing an effective address of an area in an application's effective address space 1282 for graphics acceleration module 1246 to save and restore context state. This pointer is optional if no state is required to be saved between jobs or when a job is preempted. In at least one embodiment, context save/restore area may be pinned system memory.

Upon receiving a system call, operating system 1295 may verify that application 1280 has registered and been given authority to use graphics acceleration module 1246. Operating system 1295 then calls hypervisor 1296 with information shown in Table 3.

TABLE 3

OS to Hypervisor Call Parameters

- 1 A work descriptor (WD)
- 2 An Authority Mask Register (AMR) value (potentially masked)
- 3 An effective address (EA) Context Save/
Restore Area Pointer (CSRP)
- 4 A process ID (PID) and optional thread ID (TID)
- 5 A virtual address (VA) accelerator utilization record pointer (AURP)
- 6 Virtual address of storage segment table pointer (SSTP)
- 7 A logical interrupt service number (LISN)

Upon receiving a hypervisor call, hypervisor 1296 verifies that operating system 1295 has registered and been given authority to use graphics acceleration module 1246. Hypervisor 1296 then puts process element 1283 into a process element linked list for a corresponding graphics acceleration module 1246 type. A process element may include information shown in Table 4.

TABLE 4

Process Element Information

- 1 A work descriptor (WD)
- 2 An Authority Mask Register (AMR) value (potentially masked).
- 3 An effective address (EA) Context Save/
Restore Area Pointer (CSRP)
- 4 A process ID (PID) and optional thread ID (TID)
- 5 A virtual address (VA) accelerator utilization record pointer (AURP)
- 6 Virtual address of storage segment table pointer (SSTP)
- 7 A logical interrupt service number (LISN)
- 8 Interrupt vector table, derived from hypervisor call parameters
- 9 A state register (SR) value
- 10 A logical partition ID (LPID)
- 11 A real address (RA) hypervisor accelerator utilization record pointer
- 12 Storage Descriptor Register (SDR)

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In at least one embodiment, hypervisor initializes a plurality of accelerator integration slice 1290 registers 1245.

As illustrated in FIG. 12F, in at least one embodiment, a unified memory is used, addressable via a common virtual memory address space used to access physical processor memories 1201-1202 and GPU memories 1220-1223. In this implementation, operations executed on GPUs 1210-1213 utilize a same virtual/effective memory address space to access processor memories 1201-1202 and vice versa, thereby simplifying programmability. In one embodiment, a first portion of a virtual/effective address space is allocated to processor memory 1201, a second portion to second processor memory 1202, a third portion to GPU memory 1220, and so on. In at least one embodiment, an entire virtual/effective memory space (sometimes referred to as an effective address space) is thereby distributed across each of processor memories 1201-1202 and GPU memories 1220-1223, allowing any processor or GPU to access any physical memory with a virtual address mapped to that memory.

In one embodiment, bias/coherence management circuitry 1294A-1294E within one or more of MMUs 1239A-1239E ensures cache coherence between caches of one or more host processors (e.g., 1205) and GPUs 1210-1213 and implements biasing techniques indicating physical memories in which certain types of data should be stored. While multiple instances of bias/coherence management circuitry 1294A-1294E are illustrated in FIG. 12F, bias/coherence circuitry may be implemented within an MMU of one or more host processors 1205 and/or within accelerator integration circuit 1236.

One embodiment allows GPU-attached memory 1220-1223 to be mapped as part of system memory, and accessed using shared virtual memory (SVM) technology, but without suffering performance drawbacks associated with full system cache coherence. In at least one embodiment, an ability for GPU-attached memory 1220-1223 to be accessed as system memory without onerous cache coherence overhead provides a beneficial operating environment for GPU offload. This arrangement allows host processor 1205 software to setup operands and access computation results, without overhead of tradition I/O DMA data copies. Such traditional copies involve driver calls, interrupts and memory mapped I/O (MMIO) accesses that are all inefficient relative to simple memory accesses. In at least one embodiment, an ability to access GPU attached memory 1220-1223 without cache coherence overheads can be critical to execution time of an offloaded computation. In cases with substantial streaming write memory traffic, for example, cache coherence overhead can significantly reduce an effective write bandwidth seen by a GPU 1210-1213. In at least one embodiment, efficiency of operand setup, efficiency of results access, and efficiency of GPU computation may play a role in determining effectiveness of a GPU offload.

In at least one embodiment, selection of GPU bias and host processor bias is driven by a bias tracker data structure. A bias table may be used, for example, which may be a page-granular structure (i.e., controlled at a granularity of a memory page) that includes 1 or 2 bits per GPU-attached memory page. In at least one embodiment, a bias table may be implemented in a stolen memory range of one or more GPU-attached memories 1220-1223, with or without a bias cache in GPU 1210-1213 (e.g., to cache frequently/recently used entries of a bias table). Alternatively, an entire bias table may be maintained within a GPU.

In at least one embodiment, a bias table entry associated with each access to GPU-attached memory 1220-1223 is accessed prior to actual access to a GPU memory, causing

the following operations. First, local requests from GPU **1210-1213** that find their page in GPU bias are forwarded directly to a corresponding GPU memory **1220-1223**. Local requests from a GPU that find their page in host bias are forwarded to processor **1205** (e.g., over a high-speed link as discussed above). In one embodiment, requests from processor **1205** that find a requested page in host processor bias complete a request like a normal memory read. Alternatively, requests directed to a GPU-biased page may be forwarded to GPU **1210-1213**. In at least one embodiment, a GPU may then transition a page to a host processor bias if it is not currently using a page. In at least one embodiment, bias state of a page can be changed either by a software-based mechanism, a hardware-assisted software-based mechanism, or, for a limited set of cases, a purely hardware-based mechanism.

One mechanism for changing bias state employs an API call (e.g., OpenCL), which, in turn, calls a GPU's device driver which, in turn, sends a message (or enqueues a command descriptor) to a GPU directing it to change a bias state and, for some transitions, perform a cache flushing operation in a host. In at least one embodiment, cache flushing operation is used for a transition from host processor **1205** bias to GPU bias, but is not for an opposite transition.

In one embodiment, cache coherency is maintained by temporarily rendering GPU-biased pages uncacheable by host processor **1205**. To access these pages, processor **1205** may request access from GPU **1210** which may or may not grant access right away. Thus, to reduce communication between processor **1205** and GPU **1210** it is beneficial to ensure that GPU-biased pages are those which are required by a GPU but not host processor **1205** and vice versa.

Inference and/or training logic **615** are used to perform one or more embodiments. Details regarding the inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **13** illustrates exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores, according to various embodiments described herein. In addition to what is illustrated, other logic and circuits may be included in at least one embodiment, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores.

FIG. **13** is a block diagram illustrating an exemplary system on a chip integrated circuit **1300** that may be fabricated using one or more IP cores, according to at least one embodiment. In at least one embodiment, integrated circuit **1300** includes one or more application processor(s) **1305** (e.g., CPUs), at least one graphics processor **1310**, and may additionally include an image processor **1315** and/or a video processor **1320**, any of which may be a modular IP core. In at least one embodiment, integrated circuit **1300** includes peripheral or bus logic including a USB controller **1325**, UART controller **1330**, an SPI/SDIO controller **1335**, and an I²S/I²C controller **1340**. In at least one embodiment, integrated circuit **1300** can include a display device **1345** coupled to one or more of a high-definition multimedia interface (HDMI) controller **1350** and a mobile industry processor interface (MIPI) display interface **1355**. In at least one embodiment, storage may be provided by a flash

memory subsystem **1360** including flash memory and a flash memory controller. In at least one embodiment, memory interface may be provided via a memory controller **1365** for access to SDRAM or SRAM memory devices. In at least one embodiment, some integrated circuits additionally include an embedded security engine **1370**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in integrated circuit **1300** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIGS. **14A-14B** illustrate exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores, according to various embodiments described herein. In addition to what is illustrated, other logic and circuits may be included in at least one embodiment, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores.

FIGS. **14A-14B** are block diagrams illustrating exemplary graphics processors for use within an SoC, according to embodiments described herein. FIG. **14A** illustrates an exemplary graphics processor **1410** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to at least one embodiment. FIG. **14B** illustrates an additional exemplary graphics processor **1440** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to at least one embodiment. In at least one embodiment, graphics processor **1410** of FIG. **14A** is a low power graphics processor core. In at least one embodiment, graphics processor **1440** of FIG. **14B** is a higher performance graphics processor core. In at least one embodiment, each of graphics processors **1410**, **1440** can be variants of graphics processor **1310** of FIG. **13**.

In at least one embodiment, graphics processor **1410** includes a vertex processor **1405** and one or more fragment processor(s) **1415A-1415N** (e.g., **1415A**, **1415B**, **1415C**, **1415D**, through **1415N-1**, and **1415N**). In at least one embodiment, graphics processor **1410** can execute different shader programs via separate logic, such that vertex processor **1405** is optimized to execute operations for vertex shader programs, while one or more fragment processor(s) **1415A-1415N** execute fragment (e.g., pixel) shading operations for fragment or pixel shader programs. In at least one embodiment, vertex processor **1405** performs a vertex processing stage of a 3D graphics pipeline and generates primitives and vertex data. In at least one embodiment, fragment processor(s) **1415A-1415N** use primitive and vertex data generated by vertex processor **1405** to produce a framebuffer that is displayed on a display device. In at least one embodiment, fragment processor(s) **1415A-1415N** are optimized to execute fragment shader programs as provided for in an OpenGL API, which may be used to perform similar operations as a pixel shader program as provided for in a Direct 3D API.

In at least one embodiment, graphics processor **1410** additionally includes one or more memory management

units (MMUs) **1420A-1420B**, cache(s) **1425A-1425B**, and circuit interconnect(s) **1430A-1430B**. In at least one embodiment, one or more MMU(s) **1420A-1420B** provide for virtual to physical address mapping for graphics processor **1410**, including for vertex processor **1405** and/or fragment processor(s) **1415A-1415N**, which may reference vertex or image/texture data stored in memory, in addition to vertex or image/texture data stored in one or more cache(s) **1425A-1425B**. In at least one embodiment, one or more MMU(s) **1420A-1420B** may be synchronized with other MMUs within system, including one or more MMUs associated with one or more application processor(s) **1305**, image processors **1315**, and/or video processors **1320** of FIG. **13**, such that each processor **1305-1320** can participate in a shared or unified virtual memory system. In at least one embodiment, one or more circuit interconnect(s) **1430A-1430B** enable graphics processor **1410** to interface with other IP cores within SoC, either via an internal bus of SoC or via a direct connection.

In at least one embodiment, graphics processor **1440** includes one or more MMU(s) **1420A-1420B**, cache(s) **1425A-1425B**, and circuit interconnect(s) **1430A-1430B** of graphics processor **1410** of FIG. **14A**. In at least one embodiment, graphics processor **1440** includes one or more shader core(s) **1455A-1455N** (e.g., **1455A**, **1455B**, **1455C**, **1455D**, **1455E**, **1455F**, through **1455N-1**, and **1455N**), which provides for a unified shader core architecture in which a single core or type or core can execute all types of programmable shader code, including shader program code to implement vertex shaders, fragment shaders, and/or compute shaders. In at least one embodiment, a number of shader cores can vary. In at least one embodiment, graphics processor **1440** includes an inter-core task manager **1445**, which acts as a thread dispatcher to dispatch execution threads to one or more shader cores **1455A-1455N** and a tiling unit **1458** to accelerate tiling operations for tile-based rendering, in which rendering operations for a scene are subdivided in image space, for example to exploit local spatial coherence within a scene or to optimize use of internal caches.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in integrated circuit **14A** and/or **14B** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein. Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIGS. **15A-15B** illustrate additional exemplary graphics processor logic according to embodiments described herein. FIG. **15A** illustrates a graphics core **1500** that may be included within graphics processor **1310** of FIG. **13**, in at least one embodiment, and may be a unified shader core **1455A-1455N** as in FIG. **14B** in at least one embodiment. FIG. **15B** illustrates a highly-parallel general-purpose graphics processing unit **1530** suitable for deployment on a multi-chip module in at least one embodiment.

In at least one embodiment, graphics core **1500** includes a shared instruction cache **1502**, a texture unit **1518**, and a cache/shared memory **1520** that are common to execution

resources within graphics core **1500**. In at least one embodiment, graphics core **1500** can include multiple slices **1501A-1501N** or partition for each core, and a graphics processor can include multiple instances of graphics core **1500**. Slices **1501A-1501N** can include support logic including a local instruction cache **1504A-1504N**, a thread scheduler **1506A-1506N**, a thread dispatcher **1508A-1508N**, and a set of registers **1510A-1510N**. In at least one embodiment, slices **1501A-1501N** can include a set of additional function units (AFUs **1512A-1512N**), floating-point units (FPU **1514A-1514N**), integer arithmetic logic units (ALUs **1516-1516N**), address computational units (ACU **1513A-1513N**), double-precision floating-point units (DPFPU **1515A-1515N**), and matrix processing units (MPU **1517A-1517N**).

In at least one embodiment, FPUs **1514A-1514N** can perform single-precision (32-bit) and half-precision (16-bit) floating point operations, while DPFPU **1515A-1515N** perform double precision (64-bit) floating point operations. In at least one embodiment, ALUs **1516A-1516N** can perform variable precision integer operations at 8-bit, 16-bit, and 32-bit precision, and can be configured for mixed precision operations. In at least one embodiment, MPUs **1517A-1517N** can also be configured for mixed precision matrix operations, including half-precision floating point and 8-bit integer operations. In at least one embodiment, MPUs **1517A-1517N** can perform a variety of matrix operations to accelerate machine learning application frameworks, including enabling support for accelerated general matrix to matrix multiplication (GEMM). In at least one embodiment, AFUs **1512A-1512N** can perform additional logic operations not supported by floating-point or integer units, including trigonometric operations (e.g., Sine, Cosine, etc.).

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics core **1500** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **15B** illustrates a general-purpose processing unit (GPGPU) **1530** that can be configured to enable highly-parallel compute operations to be performed by an array of graphics processing units, in at least one embodiment. In at least one embodiment, GPGPU **1530** can be linked directly to other instances of GPGPU **1530** to create a multi-GPU cluster to improve training speed for deep neural networks. In at least one embodiment, GPGPU **1530** includes a host interface **1532** to enable a connection with a host processor. In at least one embodiment, host interface **1532** is a PCI Express interface. In at least one embodiment, host interface **1532** can be a vendor specific communications interface or communications fabric. In at least one embodiment, GPGPU **1530** receives commands from a host processor and uses a global scheduler **1534** to distribute execution threads associated with those commands to a set of compute clusters **1536A-1536H**. In at least one embodiment, compute clusters **1536A-1536H** share a cache memory **1538**. In at least one

embodiment, cache memory **1538** can serve as a higher-level cache for cache memories within compute clusters **1536A-1536H**.

In at least one embodiment, GPGPU **1530** includes memory **1544A-1544B** coupled with compute clusters **1536A-1536H** via a set of memory controllers **1542A-1542B**. In at least one embodiment, memory **1544A-1544B** can include various types of memory devices including dynamic random access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory.

In at least one embodiment, compute clusters **1536A-1536H** each include a set of graphics cores, such as graphics core **1500** of FIG. **15A**, which can include multiple types of integer and floating point logic units that can perform computational operations at a range of precisions including suited for machine learning computations. For example, in at least one embodiment, at least a subset of floating point units in each of compute clusters **1536A-1536H** can be configured to perform 16-bit or 32-bit floating point operations, while a different subset of floating point units can be configured to perform 64-bit floating point operations.

In at least one embodiment, multiple instances of GPGPU **1530** can be configured to operate as a compute cluster. In at least one embodiment, communication used by compute clusters **1536A-1536H** for synchronization and data exchange varies across embodiments. In at least one embodiment, multiple instances of GPGPU **1530** communicate over host interface **1532**. In at least one embodiment, GPGPU **1530** includes an I/O hub **1539** that couples GPGPU **1530** with a GPU link **1540** that enables a direct connection to other instances of GPGPU **1530**. In at least one embodiment, GPU link **1540** is coupled to a dedicated GPU-to-GPU bridge that enables communication and synchronization between multiple instances of GPGPU **1530**. In at least one embodiment, GPU link **1540** couples with a high speed interconnect to transmit and receive data to other GPGPUs or parallel processors. In at least one embodiment, multiple instances of GPGPU **1530** are located in separate data processing systems and communicate via a network device that is accessible via host interface **1532**. In at least one embodiment GPU, link **1540** can be configured to enable a connection to a host processor in addition to or as an alternative to host interface **1532**.

In at least one embodiment, GPGPU **1530** can be configured to train neural networks. In at least one embodiment, GPGPU **1530** can be used within a inferencing platform. In at least one embodiment, in which GPGPU **1530** is used for inferencing, GPGPU may include fewer compute clusters **1536A-1536H** relative to when GPGPU is used for training a neural network. In at least one embodiment, memory technology associated with memory **1544A-1544B** may differ between inferencing and training configurations, with higher bandwidth memory technologies devoted to training configurations. In at least one embodiment, inferencing configuration of GPGPU **1530** can support inferencing specific instructions. For example, in at least one embodiment, an inferencing configuration can provide support for one or more 8-bit integer dot product instructions, which may be used during inferencing operations for deployed neural networks.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or

training logic **615** may be used in GPGPU **1530** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **16** is a block diagram illustrating a computing system **1600** according to at least one embodiment. In at least one embodiment, computing system **1600** includes a processing subsystem **1601** having one or more processor(s) **1602** and a system memory **1604** communicating via an interconnection path that may include a memory hub **1605**. In at least one embodiment, memory hub **1605** may be a separate component within a chipset component or may be integrated within one or more processor(s) **1602**. In at least one embodiment, memory hub **1605** couples with an I/O subsystem **1611** via a communication link **1606**. In at least one embodiment, I/O subsystem **1611** includes an I/O hub **1607** that can enable computing system **1600** to receive input from one or more input device(s) **1608**. In at least one embodiment, I/O hub **1607** can enable a display controller, which may be included in one or more processor(s) **1602**, to provide outputs to one or more display device(s) **1610A**. In at least one embodiment, one or more display device(s) **1610A** coupled with I/O hub **1607** can include a local, internal, or embedded display device.

In at least one embodiment, processing subsystem **1601** includes one or more parallel processor(s) **1612** coupled to memory hub **1605** via a bus or other communication link **1613**. In at least one embodiment, communication link **1613** may be one of any number of standards based communication link technologies or protocols, such as, but not limited to PCI Express, or may be a vendor specific communications interface or communications fabric. In at least one embodiment, one or more parallel processor(s) **1612** form a computationally focused parallel or vector processing system that can include a large number of processing cores and/or processing clusters, such as a many integrated core (MIC) processor. In at least one embodiment, one or more parallel processor(s) **1612** form a graphics processing subsystem that can output pixels to one of one or more display device(s) **1610A** coupled via I/O Hub **1607**. In at least one embodiment, one or more parallel processor(s) **1612** can also include a display controller and display interface (not shown) to enable a direct connection to one or more display device(s) **1610B**.

In at least one embodiment, a system storage unit **1614** can connect to I/O hub **1607** to provide a storage mechanism for computing system **1600**. In at least one embodiment, an I/O switch **1616** can be used to provide an interface mechanism to enable connections between I/O hub **1607** and other components, such as a network adapter **1618** and/or wireless network adapter **1619** that may be integrated into a platform(s), and various other devices that can be added via one or more add-in device(s) **1620**. In at least one embodiment, network adapter **1618** can be an Ethernet adapter or another wired network adapter. In at least one embodiment, wireless network adapter **1619** can include one or more of a Wi-Fi, Bluetooth, near field communication (NFC), or other network device that includes one or more wireless radios.

In at least one embodiment, computing system **1600** can include other components not explicitly shown, including USB or other port connections, optical storage drives, video

capture devices, and like, may also be connected to I/O hub 1607. In at least one embodiment, communication paths interconnecting various components in FIG. 16 may be implemented using any suitable protocols, such as PCI (Peripheral Component Interconnect) based protocols (e.g., PCI-Express), or other bus or point-to-point communication interfaces and/or protocol(s), such as NV-Link high-speed interconnect, or interconnect protocols.

In at least one embodiment, one or more parallel processor(s) 1612 incorporate circuitry optimized for graphics and video processing, including, for example, video output circuitry, and constitutes a graphics processing unit (GPU). In at least one embodiment, one or more parallel processor(s) 1612 incorporate circuitry optimized for general purpose processing. In at least one embodiment, components of computing system 1600 may be integrated with one or more other system elements on a single integrated circuit. For example, in at least one embodiment, one or more parallel processor(s) 1612, memory hub 1605, processor(s) 1602, and I/O hub 1607 can be integrated into a system on chip (SoC) integrated circuit. In at least one embodiment, components of computing system 1600 can be integrated into a single package to form a system in package (SIP) configuration. In at least one embodiment, at least a portion of components of computing system 1600 can be integrated into a multi-chip module (MCM), which can be interconnected with other multi-chip modules into a modular computing system.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in system FIG. 1600 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

Processors

FIG. 17A illustrates a parallel processor 1700 according to at least one embodiment. In at least one embodiment, various components of parallel processor 1700 may be implemented using one or more integrated circuit devices, such as programmable processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGA). In at least one embodiment, illustrated parallel processor 1700 is a variant of one or more parallel processor(s) 1612 shown in FIG. 16 according to an exemplary embodiment.

In at least one embodiment, parallel processor 1700 includes a parallel processing unit 1702. In at least one embodiment, parallel processing unit 1702 includes an I/O unit 1704 that enables communication with other devices, including other instances of parallel processing unit 1702. In at least one embodiment, I/O unit 1704 may be directly connected to other devices. In at least one embodiment, I/O unit 1704 connects with other devices via use of a hub or switch interface, such as memory hub 1605. In at least one embodiment, connections between memory hub 1605 and I/O unit 1704 form a communication link 1613. In at least one embodiment, I/O unit 1704 connects with a host interface 1706 and a memory crossbar 1716, where host interface

1706 receives commands directed to performing processing operations and memory crossbar 1716 receives commands directed to performing memory operations.

In at least one embodiment, when host interface 1706 receives a command buffer via I/O unit 1704, host interface 1706 can direct work operations to perform those commands to a front end 1708. In at least one embodiment, front end 1708 couples with a scheduler 1710, which is configured to distribute commands or other work items to a processing cluster array 1712. In at least one embodiment, scheduler 1710 ensures that processing cluster array 1712 is properly configured and in a valid state before tasks are distributed to processing cluster array 1712. In at least one embodiment, scheduler 1710 is implemented via firmware logic executing on a microcontroller. In at least one embodiment, microcontroller implemented scheduler 1710 is configurable to perform complex scheduling and work distribution operations at coarse and fine granularity, enabling rapid preemption and context switching of threads executing on processing array 1712. In at least one embodiment, host software can prove workloads for scheduling on processing array 1712 via one of multiple graphics processing doorbells. In at least one embodiment, workloads can then be automatically distributed across processing array 1712 by scheduler 1710 logic within a microcontroller including scheduler 1710.

In at least one embodiment, processing cluster array 1712 can include up to "N" processing clusters (e.g., cluster 1714A, cluster 1714B, through cluster 1714N). In at least one embodiment, each cluster 1714A-1714N of processing cluster array 1712 can execute a large number of concurrent threads. In at least one embodiment, scheduler 1710 can allocate work to clusters 1714A-1714N of processing cluster array 1712 using various scheduling and/or work distribution algorithms, which may vary depending on workload arising for each type of program or computation. In at least one embodiment, scheduling can be handled dynamically by scheduler 1710, or can be assisted in part by compiler logic during compilation of program logic configured for execution by processing cluster array 1712. In at least one embodiment, different clusters 1714A-1714N of processing cluster array 1712 can be allocated for processing different types of programs or for performing different types of computations.

In at least one embodiment, processing cluster array 1712 can be configured to perform various types of parallel processing operations. In at least one embodiment, processing cluster array 1712 is configured to perform general-purpose parallel compute operations. For example, in at least one embodiment, processing cluster array 1712 can include logic to execute processing tasks including filtering of video and/or audio data, performing modeling operations, including physics operations, and performing data transformations.

In at least one embodiment, processing cluster array 1712 is configured to perform parallel graphics processing operations. In at least one embodiment, processing cluster array 1712 can include additional logic to support execution of such graphics processing operations, including, but not limited to texture sampling logic to perform texture operations, as well as tessellation logic and other vertex processing logic. In at least one embodiment, processing cluster array 1712 can be configured to execute graphics processing related shader programs such as, but not limited to vertex shaders, tessellation shaders, geometry shaders, and pixel shaders. In at least one embodiment, parallel processing unit 1702 can transfer data from system memory via I/O unit 1704 for processing. In at least one embodiment, during

processing, transferred data can be stored to on-chip memory (e.g., parallel processor memory 1722) during processing, then written back to system memory.

In at least one embodiment, when parallel processing unit 1702 is used to perform graphics processing, scheduler 1710 can be configured to divide a processing workload into approximately equal sized tasks, to better enable distribution of graphics processing operations to multiple clusters 1714A-1714N of processing cluster array 1712. In at least one embodiment, portions of processing cluster array 1712 can be configured to perform different types of processing. For example, in at least one embodiment, a first portion may be configured to perform vertex shading and topology generation, a second portion may be configured to perform tessellation and geometry shading, and a third portion may be configured to perform pixel shading or other screen space operations, to produce a rendered image for display. In at least one embodiment, intermediate data produced by one or more of clusters 1714A-1714N may be stored in buffers to allow intermediate data to be transmitted between clusters 1714A-1714N for further processing.

In at least one embodiment, processing cluster array 1712 can receive processing tasks to be executed via scheduler 1710, which receives commands defining processing tasks from front end 1708. In at least one embodiment, processing tasks can include indices of data to be processed, e.g., surface (patch) data, primitive data, vertex data, and/or pixel data, as well as state parameters and commands defining how data is to be processed (e.g., what program is to be executed). In at least one embodiment, scheduler 1710 may be configured to fetch indices corresponding to tasks or may receive indices from front end 1708. In at least one embodiment, front end 1708 can be configured to ensure processing cluster array 1712 is configured to a valid state before a workload specified by incoming command buffers (e.g., batch-buffers, push buffers, etc.) is initiated.

In at least one embodiment, each of one or more instances of parallel processing unit 1702 can couple with parallel processor memory 1722. In at least one embodiment, parallel processor memory 1722 can be accessed via memory crossbar 1716, which can receive memory requests from processing cluster array 1712 as well as I/O unit 1704. In at least one embodiment, memory crossbar 1716 can access parallel processor memory 1722 via a memory interface 1718. In at least one embodiment, memory interface 1718 can include multiple partition units (e.g., partition unit 1720A, partition unit 1720B, through partition unit 1720N) that can each couple to a portion (e.g., memory unit) of parallel processor memory 1722. In at least one embodiment, a number of partition units 1720A-1720N is configured to be equal to a number of memory units, such that a first partition unit 1720A has a corresponding first memory unit 1724A, a second partition unit 1720B has a corresponding memory unit 1724B, and a Nth partition unit 1720N has a corresponding Nth memory unit 1724N. In at least one embodiment, a number of partition units 1720A-1720N may not be equal to a number of memory devices.

In at least one embodiment, memory units 1724A-1724N can include various types of memory devices, including dynamic random access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. In at least one embodiment, memory units 1724A-1724N may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM). In at least one embodiment, render targets, such as frame buffers or texture maps may be stored across memory

units 1724A-1724N, allowing partition units 1720A-1720N to write portions of each render target in parallel to efficiently use available bandwidth of parallel processor memory 1722. In at least one embodiment, a local instance of parallel processor memory 1722 may be excluded in favor of a unified memory design that utilizes system memory in conjunction with local cache memory.

In at least one embodiment, any one of clusters 1714A-1714N of processing cluster array 1712 can process data that will be written to any of memory units 1724A-1724N within parallel processor memory 1722. In at least one embodiment, memory crossbar 1716 can be configured to transfer an output of each cluster 1714A-1714N to any partition unit 1720A-1720N or to another cluster 1714A-1714N, which can perform additional processing operations on an output. In at least one embodiment, each cluster 1714A-1714N can communicate with memory interface 1718 through memory crossbar 1716 to read from or write to various external memory devices. In at least one embodiment, memory crossbar 1716 has a connection to memory interface 1718 to communicate with I/O unit 1704, as well as a connection to a local instance of parallel processor memory 1722, enabling processing units within different processing clusters 1714A-1714N to communicate with system memory or other memory that is not local to parallel processing unit 1702. In at least one embodiment, memory crossbar 1716 can use virtual channels to separate traffic streams between clusters 1714A-1714N and partition units 1720A-1720N.

In at least one embodiment, multiple instances of parallel processing unit 1702 can be provided on a single add-in card, or multiple add-in cards can be interconnected. In at least one embodiment, different instances of parallel processing unit 1702 can be configured to inter-operate even if different instances have different numbers of processing cores, different amounts of local parallel processor memory, and/or other configuration differences. For example, in at least one embodiment, some instances of parallel processing unit 1702 can include higher precision floating point units relative to other instances. In at least one embodiment, systems incorporating one or more instances of parallel processing unit 1702 or parallel processor 1700 can be implemented in a variety of configurations and form factors, including but not limited to desktop, laptop, or handheld personal computers, servers, workstations, game consoles, and/or embedded systems.

FIG. 17B is a block diagram of a partition unit 1720 according to at least one embodiment. In at least one embodiment, partition unit 1720 is an instance of one of partition units 1720A-1720N of FIG. 17A. In at least one embodiment, partition unit 1720 includes an L2 cache 1721, a frame buffer interface 1725, and a raster operations unit ("ROP") 1726. L2 cache 1721 is a read/write cache that is configured to perform load and store operations received from memory crossbar 1716 and ROP 1726. In at least one embodiment, read misses and urgent write-back requests are output by L2 cache 1721 to frame buffer interface 1725 for processing. In at least one embodiment, updates can also be sent to a frame buffer via frame buffer interface 1725 for processing. In at least one embodiment, frame buffer interface 1725 interfaces with one of memory units in parallel processor memory, such as memory units 1724A-1724N of FIG. 17 (e.g., within parallel processor memory 1722).

In at least one embodiment, ROP 1726 is a processing unit that performs raster operations such as stencil, z test, blending, and so forth. In at least one embodiment, ROP 1726 then outputs processed graphics data that is stored in graphics memory. In at least one embodiment, ROP 1726 includes

compression logic to compress depth or color data that is written to memory and decompress depth or color data that is read from memory. In at least one embodiment, compression logic can be lossless compression logic that makes use of one or more of multiple compression algorithms. Compression logic that is performed by ROP 1726 can vary based on statistical characteristics of data to be compressed. For example, in at least one embodiment, delta color compression is performed on depth and color data on a per-tile basis.

In at least one embodiment, ROP 1726 is included within each processing cluster (e.g., cluster 1714A-1714N of FIG. 17A) instead of within partition unit 1720. In at least one embodiment, read and write requests for pixel data are transmitted over memory crossbar 1716 instead of pixel fragment data. In at least one embodiment, processed graphics data may be displayed on a display device, such as one of one or more display device(s) 1610 of FIG. 16, routed for further processing by processor(s) 1602, or routed for further processing by one of processing entities within parallel processor 1700 of FIG. 17A.

FIG. 17C is a block diagram of a processing cluster 1714 within a parallel processing unit according to at least one embodiment. In at least one embodiment, a processing cluster is an instance of one of processing clusters 1714A-1714N of FIG. 17A. In at least one embodiment, one of more of processing cluster(s) 1714 can be configured to execute many threads in parallel, where “thread” refers to an instance of a particular program executing on a particular set of input data. In at least one embodiment, single-instruction, multiple-data (SIMD) instruction issue techniques are used to support parallel execution of a large number of threads without providing multiple independent instruction units. In at least one embodiment, single-instruction, multiple-thread (SIMT) techniques are used to support parallel execution of a large number of generally synchronized threads, using a common instruction unit configured to issue instructions to a set of processing engines within each one of processing clusters.

In at least one embodiment, operation of processing cluster 1714 can be controlled via a pipeline manager 1732 that distributes processing tasks to SIMT parallel processors. In at least one embodiment, pipeline manager 1732 receives instructions from scheduler 1710 of FIG. 17A and manages execution of those instructions via a graphics multiprocessor 1734 and/or a texture unit 1736. In at least one embodiment, graphics multiprocessor 1734 is an exemplary instance of a SIMT parallel processor. However, in at least one embodiment, various types of SIMT parallel processors of differing architectures may be included within processing cluster 1714. In at least one embodiment, one or more instances of graphics multiprocessor 1734 can be included within a processing cluster 1714. In at least one embodiment, graphics multiprocessor 1734 can process data and a data crossbar 1740 can be used to distribute processed data to one of multiple possible destinations, including other shader units. In at least one embodiment, pipeline manager 1732 can facilitate distribution of processed data by specifying destinations for processed data to be distributed via data crossbar 1740.

In at least one embodiment, each graphics multiprocessor 1734 within processing cluster 1714 can include an identical set of functional execution logic (e.g., arithmetic logic units, load-store units, etc.). In at least one embodiment, functional execution logic can be configured in a pipelined manner in which new instructions can be issued before previous instructions are complete. In at least one embodiment,

functional execution logic supports a variety of operations including integer and floating point arithmetic, comparison operations, Boolean operations, bit-shifting, and computation of various algebraic functions. In at least one embodiment, same functional-unit hardware can be leveraged to perform different operations and any combination of functional units may be present.

In at least one embodiment, instructions transmitted to processing cluster 1714 constitute a thread. In at least one embodiment, a set of threads executing across a set of parallel processing engines is a thread group. In at least one embodiment, thread group executes a program on different input data. In at least one embodiment, each thread within a thread group can be assigned to a different processing engine within a graphics multiprocessor 1734. In at least one embodiment, a thread group may include fewer threads than a number of processing engines within graphics multiprocessor 1734. In at least one embodiment, when a thread group includes fewer threads than a number of processing engines, one or more processing engines may be idle during cycles in which that thread group is being processed. In at least one embodiment, a thread group may also include more threads than a number of processing engines within graphics multiprocessor 1734. In at least one embodiment, when a thread group includes more threads than processing engines within graphics multiprocessor 1734, processing can be performed over consecutive clock cycles. In at least one embodiment, multiple thread groups can be executed concurrently on a graphics multiprocessor 1734.

In at least one embodiment, graphics multiprocessor 1734 includes an internal cache memory to perform load and store operations. In at least one embodiment, graphics multiprocessor 1734 can forego an internal cache and use a cache memory (e.g., L1 cache 1748) within processing cluster 1714. In at least one embodiment, each graphics multiprocessor 1734 also has access to L2 caches within partition units (e.g., partition units 1720A-1720N of FIG. 17A) that are shared among all processing clusters 1714 and may be used to transfer data between threads. In at least one embodiment, graphics multiprocessor 1734 may also access off-chip global memory, which can include one or more of local parallel processor memory and/or system memory. In at least one embodiment, any memory external to parallel processing unit 1702 may be used as global memory. In at least one embodiment, processing cluster 1714 includes multiple instances of graphics multiprocessor 1734 can share common instructions and data, which may be stored in L1 cache 1748.

In at least one embodiment, each processing cluster 1714 may include a memory management unit (“MMU”) 1745 that is configured to map virtual addresses into physical addresses. In at least one embodiment, one or more instances of MMU 1745 may reside within memory interface 1718 of FIG. 17A. In at least one embodiment, MMU 1745 includes a set of page table entries (PTEs) used to map a virtual address to a physical address of a tile and optionally a cache line index. In at least one embodiment, MMU 1745 may include address translation lookaside buffers (TLB) or caches that may reside within graphics multiprocessor 1734 or L1 cache or processing cluster 1714. In at least one embodiment, physical address is processed to distribute surface data access locality to allow efficient request interleaving among partition units. In at least one embodiment, cache line index may be used to determine whether a request for a cache line is a hit or miss.

In at least one embodiment, a processing cluster 1714 may be configured such that each graphics multiprocessor 1734

is coupled to a texture unit 1736 for performing texture mapping operations, e.g., determining texture sample positions, reading texture data, and filtering texture data. In at least one embodiment, texture data is read from an internal texture L1 cache (not shown) or from an L1 cache within graphics multiprocessor 1734 and is fetched from an L2 cache, local parallel processor memory, or system memory, as needed. In at least one embodiment, each graphics multiprocessor 1734 outputs processed tasks to data crossbar 1740 to provide processed task(s) to another processing cluster 1714 for further processing or to store processed task(s) in an L2 cache, local parallel processor memory, or system memory via memory crossbar 1716. In at least one embodiment, preROP 1742 (pre-raster operations unit) is configured to receive data from graphics multiprocessor 1734, direct data to ROP units, which may be located with partition units as described herein (e.g., partition units 1720A-1720N of FIG. 17A). In at least one embodiment, PreROP 1742 unit can perform optimizations for color blending, organize pixel color data, and perform address translations.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in graphics processing cluster 1714 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. 17D shows a graphics multiprocessor 1734 according to at least one embodiment. In at least one embodiment, graphics multiprocessor 1734 couples with pipeline manager 1732 of processing cluster 1714. In at least one embodiment, graphics multiprocessor 1734 has an execution pipeline including but not limited to an instruction cache 1752, an instruction unit 1754, an address mapping unit 1756, a register file 1758, one or more general purpose graphics processing unit (GPGPU) cores 1762, and one or more load/store units 1766. GPGPU core(s) 1762 and load/store unit(s) 1766 are coupled with cache memory 1772 and shared memory 1770 via a memory and cache interconnect 1768.

In at least one embodiment, instruction cache 1752 receives a stream of instructions to execute from pipeline manager 1732. In at least one embodiment, instructions are cached in instruction cache 1752 and dispatched for execution by instruction unit 1754. In at least one embodiment, instruction unit 1754 can dispatch instructions as thread groups (e.g., warps), with each thread group assigned to a different execution unit within GPGPU core(s) 1762. In at least one embodiment, an instruction can access any of a local, shared, or global address space by specifying an address within a unified address space. In at least one embodiment, address mapping unit 1756 can be used to translate addresses in a unified address space into a distinct memory address that can be accessed by load/store unit(s) 1766.

In at least one embodiment, register file 1758 provides a set of registers for functional units of graphics multiprocessor 1734.

In at least one embodiment, register file 1758 provides temporary storage for operands connected to data paths of functional units (e.g., GPGPU cores 1762, load/store units 1766) of graphics multiprocessor 1734. In at least one embodiment, register file 1758 is divided between each of functional units such that each functional unit is allocated a dedicated portion of register file 1758. In at least one embodiment, register file 1758 is divided between different warps being executed by graphics multiprocessor 1734.

In at least one embodiment, GPGPU cores 1762 can each include floating point units (FPUs) and/or integer arithmetic logic units (ALUs) that are used to execute instructions of graphics multiprocessor 1734. GPGPU cores 1762 can be similar in architecture or can differ in architecture. In at least one embodiment, a first portion of GPGPU cores 1762 include a single precision FPU and an integer ALU while a second portion of GPGPU cores include a double precision FPU. In at least one embodiment, FPUs can implement IEEE 754-2008 standard for floating point arithmetic or enable variable precision floating point arithmetic. In at least one embodiment, graphics multiprocessor 1734 can additionally include one or more fixed function or special function units to perform specific functions such as copy rectangle or pixel blending operations. In at least one embodiment one or more of GPGPU cores can also include fixed or special function logic.

In at least one embodiment, GPGPU cores 1762 include SIMD logic capable of performing a single instruction on multiple sets of data. In at least one embodiment GPGPU cores 1762 can physically execute SIMD4, SIMD8, and SIMD16 instructions and logically execute SIMD1, SIMD2, and SIMD32 instructions. In at least one embodiment, SIMD instructions for GPGPU cores can be generated at compile time by a shader compiler or automatically generated when executing programs written and compiled for single program multiple data (SPMD) or SIMT architectures. In at least one embodiment, multiple threads of a program configured for an SIMT execution model can be executed via a single SIMD instruction. For example, in at least one embodiment, eight SIMT threads that perform same or similar operations can be executed in parallel via a single SIMD8 logic unit.

In at least one embodiment, memory and cache interconnect 1768 is an interconnect network that connects each functional unit of graphics multiprocessor 1734 to register file 1758 and to shared memory 1770. In at least one embodiment, memory and cache interconnect 1768 is a crossbar interconnect that allows load/store unit 1766 to implement load and store operations between shared memory 1770 and register file 1758. In at least one embodiment, register file 1758 can operate at a same frequency as GPGPU cores 1762, thus data transfer between GPGPU cores 1762 and register file 1758 is very low latency. In at least one embodiment, shared memory 1770 can be used to enable communication between threads that execute on functional units within graphics multiprocessor 1734. In at least one embodiment, cache memory 1772 can be used as a data cache for example, to cache texture data communicated between functional units and texture unit 1736. In at least one embodiment, shared memory 1770 can also be used as a program managed cache. In at least one embodiment, threads executing on GPGPU cores 1762 can programmatically store data within shared memory in addition to automatically cached data that is stored within cache memory 1772.

In at least one embodiment, a parallel processor or GPGPU as described herein is communicatively coupled to

host/processor cores to accelerate graphics operations, machine-learning operations, pattern analysis operations, and various general purpose GPU (GPGPU) functions. In at least one embodiment, GPU may be communicatively coupled to host processor/cores over a bus or other interconnect (e.g., a high speed interconnect such as PCIe or NVLink). In at least one embodiment, GPU may be integrated on same package or chip as cores and communicatively coupled to cores over an internal processor bus/interconnect (i.e., internal to package or chip). In at least one embodiment, regardless of manner in which GPU is connected, processor cores may allocate work to GPU in form of sequences of commands/instructions contained in a work descriptor. In at least one embodiment, GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics multiprocessor **1734** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **18** illustrates a multi-GPU computing system **1800**, according to at least one embodiment. In at least one embodiment, multi-GPU computing system **1800** can include a processor **1802** coupled to multiple general purpose graphics processing units (GPGPUs) **1806A-D** via a host interface switch **1804**. In at least one embodiment, host interface switch **1804** is a PCI express switch device that couples processor **1802** to a PCI express bus over which processor **1802** can communicate with GPGPUs **1806A-D**. GPGPUs **1806A-D** can interconnect via a set of high-speed point to point GPU to GPU links **1816**. In at least one embodiment, GPU to GPU links **1816** connect to each of GPGPUs **1806A-D** via a dedicated GPU link. In at least one embodiment, P2P GPU links **1816** enable direct communication between each of GPGPUs **1806A-D** without requiring communication over host interface bus **1804** to which processor **1802** is connected. In at least one embodiment, with GPU-to-GPU traffic directed to P2P GPU links **1816**, host interface bus **1804** remains available for system memory access or to communicate with other instances of multi-GPU computing system **1800**, for example, via one or more network devices. While in at least one embodiment GPGPUs **1806A-D** connect to processor **1802** via host interface switch **1804**, in at least one embodiment processor **1802** includes direct support for P2P GPU links **1816** and can connect directly to GPGPUs **1806A-D**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in multi-GPU computing system **1800** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural

network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **19** is a block diagram of a graphics processor **1900**, according to at least one embodiment. In at least one embodiment, graphics processor **1900** includes a ring interconnect **1902**, a pipeline front-end **1904**, a media engine **1937**, and graphics cores **1980A-1980N**. In at least one embodiment, ring interconnect **1902** couples graphics processor **1900** to other processing units, including other graphics processors or one or more general-purpose processor cores. In at least one embodiment, graphics processor **1900** is one of many processors integrated within a multi-core processing system.

In at least one embodiment, graphics processor **1900** receives batches of commands via ring interconnect **1902**. In at least one embodiment, incoming commands are interpreted by a command streamer **1903** in pipeline front-end **1904**. In at least one embodiment, graphics processor **1900** includes scalable execution logic to perform 3D geometry processing and media processing via graphics core(s) **1980A-1980N**. In at least one embodiment, for 3D geometry processing commands, command streamer **1903** supplies commands to geometry pipeline **1936**. In at least one embodiment, for at least some media processing commands, command streamer **1903** supplies commands to a video front end **1934**, which couples with a media engine **1937**. In at least one embodiment, media engine **1937** includes a Video Quality Engine (VQE) **1930** for video and image post-processing and a multi-format encode/decode (MFX) **1933** engine to provide hardware-accelerated media data encode and decode. In at least one embodiment, geometry pipeline **1936** and media engine **1937** each generate execution threads for thread execution resources provided by at least one graphics core **1980A**.

In at least one embodiment, graphics processor **1900** includes scalable thread execution resources featuring modular cores **1980A-1980N** (sometimes referred to as core slices), each having multiple sub-cores **1950A-1950N**, **1960A-1960N** (sometimes referred to as core sub-slices). In at least one embodiment, graphics processor **1900** can have any number of graphics cores **1980A** through **1980N**. In at least one embodiment, graphics processor **1900** includes a graphics core **1980A** having at least a first sub-core **1950A** and a second sub-core **1960A**. In at least one embodiment, graphics processor **1900** is a low power processor with a single sub-core (e.g., **1950A**). In at least one embodiment, graphics processor **1900** includes multiple graphics cores **1980A-1980N**, each including a set of first sub-cores **1950A-1950N** and a set of second sub-cores **1960A-1960N**. In at least one embodiment, each sub-core in first sub-cores **1950A-1950N** includes at least a first set of execution units **1952A-1952N** and media/texture samplers **1954A-1954N**. In at least one embodiment, each sub-core in second sub-cores **1960A-1960N** includes at least a second set of execution units **1962A-1962N** and samplers **1964A-1964N**. In at least one embodiment, each sub-core **1950A-1950N**, **1960A-1960N** shares a set of shared resources **1970A-1970N**. In at least one embodiment, shared resources include shared cache memory and pixel operation logic.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or

more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics processor **1900** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **20** is a block diagram illustrating micro-architecture for a processor **2000** that may include logic circuits to perform instructions, according to at least one embodiment. In at least one embodiment, processor **2000** may perform instructions, including x86 instructions, ARM instructions, specialized instructions for application-specific integrated circuits (ASICs), etc. In at least one embodiment, processor **2000** may include registers to store packed data, such as 64-bit wide MMX™ registers in microprocessors enabled with MMX technology from Intel Corporation of Santa Clara, Calif. In at least one embodiment, MMX registers, available in both integer and floating point forms, may operate with packed data elements that accompany single instruction, multiple data (“SIMD”) and streaming SIMD extensions (“SSE”) instructions. In at least one embodiment, 128-bit wide XMM registers relating to SSE2, SSE3, SSE4, AVX, or beyond (referred to generically as “SSEx”) technology may hold such packed data operands. In at least one embodiment, processor **2000** may perform instructions to accelerate machine learning or deep learning algorithms, training, or inferencing.

In at least one embodiment, processor **2000** includes an in-order front end (“front end”) **2001** to fetch instructions to be executed and prepare instructions to be used later in processor pipeline. In at least one embodiment, front end **2001** may include several units. In at least one embodiment, an instruction prefetcher **2026** fetches instructions from memory and feeds instructions to an instruction decoder **2028** which in turn decodes or interprets instructions. For example, in at least one embodiment, instruction decoder **2028** decodes a received instruction into one or more operations called “micro-instructions” or “micro-operations” (also called “micro ops” or “uops”) that machine may execute. In at least one embodiment, instruction decoder **2028** parses instruction into an opcode and corresponding data and control fields that may be used by micro-architecture to perform operations in accordance with at least one embodiment. In at least one embodiment, a trace cache **2030** may assemble decoded uops into program ordered sequences or traces in a uop queue **2034** for execution. In at least one embodiment, when trace cache **2030** encounters a complex instruction, a microcode ROM **2032** provides uops needed to complete operation.

In at least one embodiment, some instructions may be converted into a single micro-op, whereas others need several micro-ops to complete full operation. In at least one embodiment, if more than four micro-ops are needed to complete an instruction, instruction decoder **2028** may access microcode ROM **2032** to perform instruction. In at least one embodiment, an instruction may be decoded into a small number of micro-ops for processing at instruction decoder **2028**. In at least one embodiment, an instruction may be stored within microcode ROM **2032** should a number of micro-ops be needed to accomplish operation. In

at least one embodiment, trace cache **2030** refers to an entry point programmable logic array (“PLA”) to determine a correct micro-instruction pointer for reading microcode sequences to complete one or more instructions from microcode ROM **2032** in accordance with at least one embodiment. In at least one embodiment, after microcode ROM **2032** finishes sequencing micro-ops for an instruction, front end **2001** of machine may resume fetching micro-ops from trace cache **2030**.

In at least one embodiment, out-of-order execution engine (“out of order engine”) **2003** may prepare instructions for execution. In at least one embodiment, out-of-order execution logic has a number of buffers to smooth out and re-order flow of instructions to optimize performance as they go down pipeline and get scheduled for execution. In at least one embodiment, out-of-order execution engine **2003** includes, without limitation, an allocator/register renamer **2040**, a memory uop queue **2042**, an integer/floating point uop queue **2044**, a memory scheduler **2046**, a fast scheduler **2002**, a slow/general floating point scheduler (“slow/general FP scheduler”) **2004**, and a simple floating point scheduler (“simple FP scheduler”) **2006**. In at least one embodiment, fast scheduler **2002**, slow/general floating point scheduler **2004**, and simple floating point scheduler **2006** are also collectively referred to herein as “uop schedulers **2002**, **2004**, **2006**.” In at least one embodiment, allocator/register renamer **2040** allocates machine buffers and resources that each uop needs in order to execute. In at least one embodiment, allocator/register renamer **2040** renames logic registers onto entries in a register file. In at least one embodiment, allocator/register renamer **2040** also allocates an entry for each uop in one of two uop queues, memory uop queue **2042** for memory operations and integer/floating point uop queue **2044** for non-memory operations, in front of memory scheduler **2046** and uop schedulers **2002**, **2004**, **2006**. In at least one embodiment, uop schedulers **2002**, **2004**, **2006** determine when a uop is ready to execute based on readiness of their dependent input register operand sources and availability of execution resources uops need to complete their operation. In at least one embodiment, fast scheduler **2002** of at least one embodiment may schedule on each half of main clock cycle while slow/general floating point scheduler **2004** and simple floating point scheduler **2006** may schedule once per main processor clock cycle. In at least one embodiment, uop schedulers **2002**, **2004**, **2006** arbitrate for dispatch ports to schedule uops for execution.

In at least one embodiment, execution block **2011** includes, without limitation, an integer register file/bypass network **2008**, a floating point register file/bypass network (“FP register file/bypass network”) **2010**, address generation units (“AGUs”) **2012** and **2014**, fast Arithmetic Logic Units (ALUs) (“fast ALUs”) **2016** and **2018**, a slow Arithmetic Logic Unit (“slow ALU”) **2020**, a floating point ALU (“FP”) **2022**, and a floating point move unit (“FP move”) **2024**. In at least one embodiment, integer register file/bypass network **2008** and floating point register file/bypass network **2010** are also referred to herein as “register files **2008**, **2010**.” In at least one embodiment, AGUs **2012** and **2014**, fast ALUs **2016** and **2018**, slow ALU **2020**, floating point ALU **2022**, and floating point move unit **2024** are also referred to herein as “execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, and **2024**.” In at least one embodiment, execution block **2011** may include, without limitation, any number (including zero) and type of register files, bypass networks, address generation units, and execution units, in any combination.

In at least one embodiment, register files **2008**, **2010** may be arranged between uop schedulers **2002**, **2004**, **2006**, and

execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, and **2024**. In at least one embodiment, integer register file/bypass network **2008** performs integer operations. In at least one embodiment, floating point register file/bypass network **2010** performs floating point operations. In at least one embodiment, each of register files **2008**, **2010** may include, without limitation, a bypass network that may bypass or forward just completed results that have not yet been written into register file to new dependent uops. In at least one embodiment, register files **2008**, **2010** may communicate data with each other. In at least one embodiment, integer register file/bypass network **2008** may include, without limitation, two separate register files, one register file for low-order thirty-two bits of data and a second register file for high order thirty-two bits of data. In at least one embodiment, floating point register file/bypass network **2010** may include, without limitation, 128-bit wide entries because floating point instructions typically have operands from 64 to 128 bits in width.

In at least one embodiment, execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, **2024** may execute instructions. In at least one embodiment, register files **2008**, **2010** store integer and floating point data operand values that micro-instructions need to execute. In at least one embodiment, processor **2000** may include, without limitation, any number and combination of execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, **2024**. In at least one embodiment, floating point ALU **2022** and floating point move unit **2024**, may execute floating point, MMX, SIMD, AVX and SSE, or other operations, including specialized machine learning instructions. In at least one embodiment, floating point ALU **2022** may include, without limitation, a 64-bit by 64-bit floating point divider to execute divide, square root, and remainder micro ops. In at least one embodiment, instructions involving a floating point value may be handled with floating point hardware. In at least one embodiment, ALU operations may be passed to fast ALUs **2016**, **2018**. In at least one embodiment, fast ALUS **2016**, **2018** may execute fast operations with an effective latency of half a clock cycle. In at least one embodiment, most complex integer operations go to slow ALU **2020** as slow ALU **2020** may include, without limitation, integer execution hardware for long-latency type of operations, such as a multiplier, shifts, flag logic, and branch processing. In at least one embodiment, memory load/store operations may be executed by AGUS **2012**, **2014**. In at least one embodiment, fast ALU **2016**, fast ALU **2018**, and slow ALU **2020** may perform integer operations on 64-bit data operands. In at least one embodiment, fast ALU **2016**, fast ALU **2018**, and slow ALU **2020** may be implemented to support a variety of data bit sizes including sixteen, thirty-two, 128, 256, etc. In at least one embodiment, floating point ALU **2022** and floating point move unit **2024** may be implemented to support a range of operands having bits of various widths. In at least one embodiment, floating point ALU **2022** and floating point move unit **2024** may operate on 128-bit wide packed data operands in conjunction with SIMD and multimedia instructions.

In at least one embodiment, uop schedulers **2002**, **2004**, **2006**, dispatch dependent operations before parent load has finished executing. In at least one embodiment, as uops may be speculatively scheduled and executed in processor **2000**, processor **2000** may also include logic to handle memory misses. In at least one embodiment, if a data load misses in data cache, there may be dependent operations in flight in pipeline that have left scheduler with temporarily incorrect data. In at least one embodiment, a replay mechanism tracks and re-executes instructions that use incorrect data. In at

least one embodiment, dependent operations might need to be replayed and independent ones may be allowed to complete. In at least one embodiment, schedulers and replay mechanism of at least one embodiment of a processor may also be designed to catch instruction sequences for text string comparison operations.

In at least one embodiment, term “registers” may refer to on-board processor storage locations that may be used as part of instructions to identify operands. In at least one embodiment, registers may be those that may be usable from outside of processor (from a programmer’s perspective). In at least one embodiment, registers might not be limited to a particular type of circuit. Rather, in at least one embodiment, a register may store data, provide data, and perform functions described herein. In at least one embodiment, registers described herein may be implemented by circuitry within a processor using any number of different techniques, such as dedicated physical registers, dynamically allocated physical registers using register renaming, combinations of dedicated and dynamically allocated physical registers, etc. In at least one embodiment, integer registers store 32-bit integer data. A register file of at least one embodiment also contains eight multimedia SIMD registers for packed data.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into execution block **2011** and other memory or registers shown or not shown. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs illustrated in execution block **2011**. Moreover, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of execution block **2011** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **21** illustrates a deep learning application processor **2100**, according to at least one embodiment. In at least one embodiment, deep learning application processor **2100** uses instructions that, if executed by deep learning application processor **2100**, cause deep learning application processor **2100** to perform some or all of processes and techniques described throughout this disclosure. In at least one embodiment, deep learning application processor **2100** is an application-specific integrated circuit (ASIC). In at least one embodiment, application processor **2100** performs matrix multiply operations either “hard-wired” into hardware as a result of performing one or more instructions or both. In at least one embodiment, deep learning application processor **2100** includes, without limitation, processing clusters **2110** (1)-**2110**(12), Inter-Chip Links (“ICLs”) **2120**(1)-**2120**(12), Inter-Chip Controllers (“ICCs”) **2130**(1)-**2130**(2), memory controllers (“Mem Ctrlrs”) **2142**(1)-**2142**(4), high bandwidth memory physical layer (“HBM PHY”) **2144**(1)-**2144**(4), a management-controller central processing unit (“management-controller CPU”) **2150**, a peripheral component interconnect express controller and direct memory access block (“PCIe Controller and DMA”) **2170**, and a sixteen-lane peripheral component interconnect express port (“PCI Express x 16”) **2180**.

In at least one embodiment, processing clusters **2110** may perform deep learning operations, including inference or prediction operations based on weight parameters calculated one or more training techniques, including those described herein. In at least one embodiment, each processing cluster **2110** may include, without limitation, any number and type of processors. In at least one embodiment, deep learning application processor **2100** may include any number and type of processing clusters **2100**. In at least one embodiment, Inter-Chip Links **2120** are bi-directional. In at least one embodiment, Inter-Chip Links **2120** and Inter-Chip Controllers **2130** enable multiple deep learning application processors **2100** to exchange information, including activation information resulting from performing one or more machine learning algorithms embodied in one or more neural networks. In at least one embodiment, deep learning application processor **2100** may include any number (including zero) and type of ICLs **2120** and ICCs **2130**.

In at least one embodiment, HBM2s **2140** provide a total of 32 Gigabytes (GB) of memory. HBM2 **2140(i)** is associated with both memory controller **2142(i)** and HBM PHY **2144(i)**. In at least one embodiment, any number of HBM2s **2140** may provide any type and total amount of high bandwidth memory and may be associated with any number (including zero) and type of memory controllers **2142** and HBM PHYs **2144**. In at least one embodiment, SPI, I2C, GPIO **2160**, PCIe Controller and DMA **2170**, and/or PCIe **2180** may be replaced with any number and type of blocks that enable any number and type of communication standards in any technically feasible fashion.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, deep learning application processor **2100** is used to train a machine learning model, such as a neural network, to predict or infer information provided to deep learning application processor **2100**. In at least one embodiment, deep learning application processor **2100** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by deep learning application processor **2100**. In at least one embodiment, processor **2100** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **22** is a block diagram of a neuromorphic processor **2200**, according to at least one embodiment. In at least one embodiment, neuromorphic processor **2200** may receive one or more inputs from sources external to neuromorphic processor **2200**. In at least one embodiment, these inputs may be transmitted to one or more neurons **2202** within neuromorphic processor **2200**. In at least one embodiment, neurons **2202** and components thereof may be implemented using circuitry or logic, including one or more arithmetic logic units (ALUs). In at least one embodiment, neuromorphic processor **2200** may include, without limitation, thousands or millions of instances of neurons **2202**, but any suitable number of neurons **2202** may be used. In at least one embodiment, each instance of neuron **2202** may include a neuron input **2204** and a neuron output **2206**. In at least one embodiment, neurons **2202** may generate outputs that may be transmitted to inputs of other instances of neurons **2202**.

For example, in at least one embodiment, neuron inputs **2204** and neuron outputs **2206** may be interconnected via synapses **2208**.

In at least one embodiment, neurons **2202** and synapses **2208** may be interconnected such that neuromorphic processor **2200** operates to process or analyze information received by neuromorphic processor **2200**. In at least one embodiment, neurons **2202** may transmit an output pulse (or “fire” or “spike”) when inputs received through neuron input **2204** exceed a threshold. In at least one embodiment, neurons **2202** may sum or integrate signals received at neuron inputs **2204**. For example, in at least one embodiment, neurons **2202** may be implemented as leaky integrate-and-fire neurons, wherein if a sum (referred to as a “membrane potential”) exceeds a threshold value, neuron **2202** may generate an output (or “fire”) using a transfer function such as a sigmoid or threshold function. In at least one embodiment, a leaky integrate-and-fire neuron may sum signals received at neuron inputs **2204** into a membrane potential and may also apply a decay factor (or leak) to reduce a membrane potential. In at least one embodiment, a leaky integrate-and-fire neuron may fire if multiple input signals are received at neuron inputs **2204** rapidly enough to exceed a threshold value (i.e., before a membrane potential decays too low to fire). In at least one embodiment, neurons **2202** may be implemented using circuits or logic that receive inputs, integrate inputs into a membrane potential, and decay a membrane potential. In at least one embodiment, inputs may be averaged, or any other suitable transfer function may be used. Furthermore, in at least one embodiment, neurons **2202** may include, without limitation, comparator circuits or logic that generate an output spike at neuron output **2206** when result of applying a transfer function to neuron input **2204** exceeds a threshold. In at least one embodiment, once neuron **2202** fires, it may disregard previously received input information by, for example, resetting a membrane potential to 0 or another suitable default value. In at least one embodiment, once membrane potential is reset to 0, neuron **2202** may resume normal operation after a suitable period of time (or refractory period).

In at least one embodiment, neurons **2202** may be interconnected through synapses **2208**. In at least one embodiment, synapses **2208** may operate to transmit signals from an output of a first neuron **2202** to an input of a second neuron **2202**. In at least one embodiment, neurons **2202** may transmit information over more than one instance of synapse **2208**. In at least one embodiment, one or more instances of neuron output **2206** may be connected, via an instance of synapse **2208**, to an instance of neuron input **2204** in same neuron **2202**. In at least one embodiment, an instance of neuron **2202** generating an output to be transmitted over an instance of synapse **2208** may be referred to as a “pre-synaptic neuron” with respect to that instance of synapse **2208**. In at least one embodiment, an instance of neuron **2202** receiving an input transmitted over an instance of synapse **2208** may be referred to as a “post-synaptic neuron” with respect to that instance of synapse **2208**. Because an instance of neuron **2202** may receive inputs from one or more instances of synapse **2208**, and may also transmit outputs over one or more instances of synapse **2208**, a single instance of neuron **2202** may therefore be both a “pre-synaptic neuron” and “post-synaptic neuron,” with respect to various instances of synapses **2208**, in at least one embodiment.

In at least one embodiment, neurons **2202** may be organized into one or more layers. Each instance of neuron **2202** may have one neuron output **2206** that may fan out through

one or more synapses **2208** to one or more neuron inputs **2204**. In at least one embodiment, neuron outputs **2206** of neurons **2202** in a first layer **2210** may be connected to neuron inputs **2204** of neurons **2202** in a second layer **2212**. In at least one embodiment, layer **2210** may be referred to as a “feed-forward layer.” In at least one embodiment, each instance of neuron **2202** in an instance of first layer **2210** may fan out to each instance of neuron **2202** in second layer **2212**. In at least one embodiment, first layer **2210** may be referred to as a “fully connected feed-forward layer.” In at least one embodiment, each instance of neuron **2202** in an instance of second layer **2212** may fan out to fewer than all instances of neuron **2202** in a third layer **2214**. In at least one embodiment, second layer **2212** may be referred to as a “sparsely connected feed-forward layer.” In at least one embodiment, neurons **2202** in second layer **2212** may fan out to neurons **2202** in multiple other layers, including to neurons **2202** in (same) second layer **2212**. In at least one embodiment, second layer **2212** may be referred to as a “recurrent layer.” In at least one embodiment, neuromorphic processor **2200** may include, without limitation, any suitable combination of recurrent layers and feed-forward layers, including, without limitation, both sparsely connected feed-forward layers and fully connected feed-forward layers.

In at least one embodiment, neuromorphic processor **2200** may include, without limitation, a reconfigurable interconnect architecture or dedicated hard wired interconnects to connect synapse **2208** to neurons **2202**. In at least one embodiment, neuromorphic processor **2200** may include, without limitation, circuitry or logic that allows synapses to be allocated to different neurons **2202** as needed based on neural network topology and neuron fan-in/out. For example, in at least one embodiment, synapses **2208** may be connected to neurons **2202** using an interconnect fabric, such as network-on-chip, or with dedicated connections. In at least one embodiment, synapse interconnections and components thereof may be implemented using circuitry or logic.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **23** is a block diagram of a processing system, according to at least one embodiment. In at least one embodiment, system **2300** includes one or more processors **2302** and one or more graphics processors **2308**, and may be a single processor desktop system, a multiprocessor workstation system, or a server system having a large number of processors **2302** or processor cores **2307**. In at least one embodiment, system **2300** is a processing platform incorporated within a system-on-a-chip (SoC) integrated circuit for use in mobile, handheld, or embedded devices.

In at least one embodiment, system **2300** can include, or be incorporated within a server-based gaming platform, a game console, including a game and media console, a mobile gaming console, a handheld game console, or an online game console. In at least one embodiment, system **2300** is a mobile phone, smart phone, tablet computing device or mobile Internet device. In at least one embodiment, processing system **2300** can also include, couple with, or be integrated within a wearable device, such as a smart watch wearable device, smart eyewear device, augmented reality device, or virtual reality device. In at least one embodiment, processing system **2300** is a television or set

top box device having one or more processors **2302** and a graphical interface generated by one or more graphics processors **2308**.

In at least one embodiment, one or more processors **2302** each include one or more processor cores **2307** to process instructions which, when executed, perform operations for system and user software. In at least one embodiment, each of one or more processor cores **2307** is configured to process a specific instruction set **2309**. In at least one embodiment, instruction set **2309** may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). In at least one embodiment, processor cores **2307** may each process a different instruction set **2309**, which may include instructions to facilitate emulation of other instruction sets. In at least one embodiment, processor core **2307** may also include other processing devices, such as a Digital Signal Processor (DSP).

In at least one embodiment, processor **2302** includes cache memory **2304**. In at least one embodiment, processor **2302** can have a single internal cache or multiple levels of internal cache. In at least one embodiment, cache memory is shared among various components of processor **2302**. In at least one embodiment, processor **2302** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **2307** using known cache coherency techniques. In at least one embodiment, register file **2306** is additionally included in processor **2302** which may include different types of registers for storing different types of data (e.g., integer registers, floating point registers, status registers, and an instruction pointer register). In at least one embodiment, register file **2306** may include general-purpose registers or other registers.

In at least one embodiment, one or more processor(s) **2302** are coupled with one or more interface bus(es) **2310** to transmit communication signals such as address, data, or control signals between processor **2302** and other components in system **2300**. In at least one embodiment, interface bus **2310**, in one embodiment, can be a processor bus, such as a version of a Direct Media Interface (DMI) bus. In at least one embodiment, interface **2310** is not limited to a DMI bus, and may include one or more Peripheral Component Interconnect buses (e.g., PCI, PCI Express), memory busses, or other types of interface busses. In at least one embodiment, processor(s) **2302** include an integrated memory controller **2316** and a platform controller hub **2330**. In at least one embodiment, memory controller **2316** facilitates communication between a memory device and other components of system **2300**, while platform controller hub (PCH) **2330** provides connections to I/O devices via a local I/O bus.

In at least one embodiment, memory device **2320** can be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. In at least one embodiment memory device **2320** can operate as system memory for system **2300**, to store data **2322** and instructions **2321** for use when one or more processors **2302** executes an application or process. In at least one embodiment, memory controller **2316** also couples with an optional external graphics processor **2312**, which may communicate with one or more graphics processors **2308** in processors **2302** to perform graphics and media operations. In at least one embodiment, a display device **2311** can connect to processor(s) **2302**. In at least one embodiment display device **2311** can include one or more of

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an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). In at least one embodiment, display device **2311** can include a head mounted display (HMD) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

In at least one embodiment, platform controller hub **2330** enables peripherals to connect to memory device **2320** and processor **2302** via a high-speed I/O bus. In at least one embodiment, I/O peripherals include, but are not limited to, an audio controller **2346**, a network controller **2334**, a firmware interface **2328**, a wireless transceiver **2326**, touch sensors **2325**, a data storage device **2324** (e.g., hard disk drive, flash memory, etc.). In at least one embodiment, data storage device **2324** can connect via a storage interface (e.g., SATA) or via a peripheral bus, such as a Peripheral Component Interconnect bus (e.g., PCI, PCI Express). In at least one embodiment, touch sensors **2325** can include touch screen sensors, pressure sensors, or fingerprint sensors. In at least one embodiment, wireless transceiver **2326** can be a Wi-Fi transceiver, a Bluetooth transceiver, or a mobile network transceiver such as a 3G, 4G, or Long Term Evolution (LTE) transceiver. In at least one embodiment, firmware interface **2328** enables communication with system firmware, and can be, for example, a unified extensible firmware interface (UEFI). In at least one embodiment, network controller **2334** can enable a network connection to a wired network. In at least one embodiment, a high-performance network controller (not shown) couples with interface bus **2310**. In at least one embodiment, audio controller **2346** is a multi-channel high definition audio controller. In at least one embodiment, system **2300** includes an optional legacy I/O controller **2340** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to system. In at least one embodiment, platform controller hub **2330** can also connect to one or more Universal Serial Bus (USB) controllers **2342** connect input devices, such as keyboard and mouse **2343** combinations, a camera **2344**, or other USB input devices.

In at least one embodiment, an instance of memory controller **2316** and platform controller hub **2330** may be integrated into a discreet external graphics processor, such as external graphics processor **2312**. In at least one embodiment, platform controller hub **2330** and/or memory controller **2316** may be external to one or more processor(s) **2302**. For example, in at least one embodiment, system **2300** can include an external memory controller **2316** and platform controller hub **2330**, which may be configured as a memory controller hub and peripheral controller hub within a system chip that is in communication with processor(s) **2302**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into graphics processor **2300**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2300** to perform one or more machine

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learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **24** is a block diagram of a processor **2400** having one or more processor cores **2402A-2402N**, an integrated memory controller **2414**, and an integrated graphics processor **2408**, according to at least one embodiment. In at least one embodiment, processor **2400** can include additional cores up to and including additional core **2402N** represented by dashed lined boxes. In at least one embodiment, each of processor cores **2402A-2402N** includes one or more internal cache units **2404A-2404N**. In at least one embodiment, each processor core also has access to one or more shared cached units **2406**.

In at least one embodiment, internal cache units **2404A-2404N** and shared cache units **2406** represent a cache memory hierarchy within processor **2400**. In at least one embodiment, cache memory units **2404A-2404N** may include at least one level of instruction and data cache within each processor core and one or more levels of shared mid-level cache, such as a Level 2 (L2), Level 3 (L3), Level 4 (L4), or other levels of cache, where a highest level of cache before external memory is classified as an LLC. In at least one embodiment, cache coherency logic maintains coherency between various cache units **2406** and **2404A-2404N**.

In at least one embodiment, processor **2400** may also include a set of one or more bus controller units **2416** and a system agent core **2410**. In at least one embodiment, one or more bus controller units **2416** manage a set of peripheral buses, such as one or more PCI or PCI express busses. In at least one embodiment, system agent core **2410** provides management functionality for various processor components. In at least one embodiment, system agent core **2410** includes one or more integrated memory controllers **2414** to manage access to various external memory devices (not shown).

In at least one embodiment, one or more of processor cores **2402A-2402N** include support for simultaneous multi-threading. In at least one embodiment, system agent core **2410** includes components for coordinating and operating cores **2402A-2402N** during multi-threaded processing. In at least one embodiment, system agent core **2410** may additionally include a power control unit (PCU), which includes logic and components to regulate one or more power states of processor cores **2402A-2402N** and graphics processor **2408**.

In at least one embodiment, processor **2400** additionally includes graphics processor **2408** to execute graphics processing operations. In at least one embodiment, graphics processor **2408** couples with shared cache units **2406**, and system agent core **2410**, including one or more integrated memory controllers **2414**. In at least one embodiment, system agent core **2410** also includes a display controller **2411** to drive graphics processor output to one or more coupled displays. In at least one embodiment, display controller **2411** may also be a separate module coupled with graphics processor **2408** via at least one interconnect, or may be integrated within graphics processor **2408**.

In at least one embodiment, a ring based interconnect unit **2412** is used to couple internal components of processor **2400**. In at least one embodiment, an alternative interconnect unit may be used, such as a point-to-point interconnect,

a switched interconnect, or other techniques. In at least one embodiment, graphics processor **2408** couples with ring interconnect **2412** via an I/O link **2413**.

In at least one embodiment, I/O link **2413** represents at least one of multiple varieties of I/O interconnects, including an on package I/O interconnect which facilitates communication between various processor components and a high-performance embedded memory module **2418**, such as an eDRAM module. In at least one embodiment, each of processor cores **2402A-2402N** and graphics processor **2408** use embedded memory modules **2418** as a shared Last Level Cache.

In at least one embodiment, processor cores **2402A-2402N** are homogenous cores executing a common instruction set architecture. In at least one embodiment, processor cores **2402A-2402N** are heterogeneous in terms of instruction set architecture (ISA), where one or more of processor cores **2402A-2402N** execute a common instruction set, while one or more other cores of processor cores **2402A-2402N** executes a subset of a common instruction set or a different instruction set. In at least one embodiment, processor cores **2402A-2402N** are heterogeneous in terms of microarchitecture, where one or more cores having a relatively higher power consumption couple with one or more power cores having a lower power consumption. In at least one embodiment, processor **2400** can be implemented on one or more chips or as an SoC integrated circuit.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into processor **2400**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**, graphics core(s) **2402A-2402N**, or other components in FIG. **24**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2400** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **25** is a block diagram of hardware logic of a graphics processor core **2500**, according to at least one embodiment described herein. In at least one embodiment, graphics processor core **2500** is included within a graphics core array. In at least one embodiment, graphics processor core **2500**, sometimes referred to as a core slice, can be one or multiple graphics cores within a modular graphics processor. In at least one embodiment, graphics processor core **2500** is exemplary of one graphics core slice, and a graphics processor as described herein may include multiple graphics core slices based on target power and performance envelopes. In at least one embodiment, each graphics core **2500** can include a fixed function block **2530** coupled with multiple sub-cores **2501A-2501F**, also referred to as sub-slices, that include modular blocks of general-purpose and fixed function logic.

In at least one embodiment, fixed function block **2530** includes a geometry/fixed function pipeline **2536** that can be shared by all sub-cores in graphics processor **2500**, for example, in lower performance and/or lower power graphics processor implementations. In at least one embodiment, geometry/fixed function pipeline **2536** includes a 3D fixed function pipeline, a video front-end unit, a thread spawner and thread dispatcher, and a unified return buffer manager, which manages unified return buffers.

In at least one embodiment fixed, function block **2530** also includes a graphics SoC interface **2537**, a graphics microcontroller **2538**, and a media pipeline **2539**. In at least one embodiment fixed, graphics SoC interface **2537** provides an interface between graphics core **2500** and other processor cores within a system on a chip integrated circuit. In at least one embodiment, graphics microcontroller **2538** is a programmable sub-processor that is configurable to manage various functions of graphics processor **2500**, including thread dispatch, scheduling, and pre-emption. In at least one embodiment, media pipeline **2539** includes logic to facilitate decoding, encoding, pre-processing, and/or post-processing of multimedia data, including image and video data. In at least one embodiment, media pipeline **2539** implements media operations via requests to compute or sampling logic within sub-cores **2501-2501F**.

In at least one embodiment, SoC interface **2537** enables graphics core **2500** to communicate with general-purpose application processor cores (e.g., CPUs) and/or other components within an SoC, including memory hierarchy elements such as a shared last level cache memory, system RAM, and/or embedded on-chip or on-package DRAM. In at least one embodiment, SoC interface **2537** can also enable communication with fixed function devices within an SoC, such as camera imaging pipelines, and enables use of and/or implements global memory atomics that may be shared between graphics core **2500** and CPUs within an SoC. In at least one embodiment, SoC interface **2537** can also implement power management controls for graphics core **2500** and enable an interface between a clock domain of graphic core **2500** and other clock domains within an SoC. In at least one embodiment, SoC interface **2537** enables receipt of command buffers from a command streamer and global thread dispatcher that are configured to provide commands and instructions to each of one or more graphics cores within a graphics processor. In at least one embodiment, commands and instructions can be dispatched to media pipeline **2539**, when media operations are to be performed, or a geometry and fixed function pipeline (e.g., geometry and fixed function pipeline **2536**, geometry and fixed function pipeline **2514**) when graphics processing operations are to be performed.

In at least one embodiment, graphics microcontroller **2538** can be configured to perform various scheduling and management tasks for graphics core **2500**. In at least one embodiment, graphics microcontroller **2538** can perform graphics and/or compute workload scheduling on various graphics parallel engines within execution unit (EU) arrays **2502A-2502F**, **2504A-2504F** within sub-cores **2501A-2501F**. In at least one embodiment, host software executing on a CPU core of an SoC including graphics core **2500** can submit workloads one of multiple graphic processor doorbells, which invokes a scheduling operation on an appropriate graphics engine. In at least one embodiment, scheduling operations include determining which workload to run next, submitting a workload to a command streamer, preempting existing workloads running on an engine, monitoring progress of a workload, and notifying host software

when a workload is complete. In at least one embodiment, graphics microcontroller **2538** can also facilitate low-power or idle states for graphics core **2500**, providing graphics core **2500** with an ability to save and restore registers within graphics core **2500** across low-power state transitions independently from an operating system and/or graphics driver software on a system.

In at least one embodiment, graphics core **2500** may have greater than or fewer than illustrated sub-cores **2501A-2501F**, up to N modular sub-cores. For each set of N sub-cores, in at least one embodiment, graphics core **2500** can also include shared function logic **2510**, shared and/or cache memory **2512**, a geometry/fixed function pipeline **2514**, as well as additional fixed function logic **2516** to accelerate various graphics and compute processing operations. In at least one embodiment, shared function logic **2510** can include logic units (e.g., sampler, math, and/or inter-thread communication logic) that can be shared by each N sub-cores within graphics core **2500**. In at least one embodiment fixed, shared and/or cache memory **2512** can be a last-level cache for N sub-cores **2501A-2501F** within graphics core **2500** and can also serve as shared memory that is accessible by multiple sub-cores. In at least one embodiment, geometry/fixed function pipeline **2514** can be included instead of geometry/fixed function pipeline **2536** within fixed function block **2530** and can include same or similar logic units.

In at least one embodiment, graphics core **2500** includes additional fixed function logic **2516** that can include various fixed function acceleration logic for use by graphics core **2500**. In at least one embodiment, additional fixed function logic **2516** includes an additional geometry pipeline for use in position only shading. In position-only shading, at least two geometry pipelines exist, whereas in a full geometry pipeline within geometry/fixed function pipeline **2516**, **2536**, and a cull pipeline, which is an additional geometry pipeline which may be included within additional fixed function logic **2516**. In at least one embodiment, cull pipeline is a trimmed down version of a full geometry pipeline. In at least one embodiment, a full pipeline and a cull pipeline can execute different instances of an application, each instance having a separate context. In at least one embodiment, position only shading can hide long cull runs of discarded triangles, enabling shading to be completed earlier in some instances. For example, in at least one embodiment, cull pipeline logic within additional fixed function logic **2516** can execute position shaders in parallel with a main application and generally generates critical results faster than a full pipeline, as cull pipeline fetches and shades position attribute of vertices, without performing rasterization and rendering of pixels to a frame buffer. In at least one embodiment, cull pipeline can use generated critical results to compute visibility information for all triangles without regard to whether those triangles are culled. In at least one embodiment, full pipeline (which in this instance may be referred to as a replay pipeline) can consume visibility information to skip culled triangles to shade only visible triangles that are finally passed to a rasterization phase.

In at least one embodiment, additional fixed function logic **2516** can also include machine-learning acceleration logic, such as fixed function matrix multiplication logic, for implementations including optimizations for machine learning training or inferencing.

In at least one embodiment, within each graphics sub-core **2501A-2501F** includes a set of execution resources that may be used to perform graphics, media, and compute operations in response to requests by graphics pipeline, media pipeline,

or shader programs. In at least one embodiment, graphics sub-cores **2501A-2501F** include multiple EU arrays **2502A-2502F**, **2504A-2504F**, thread dispatch and inter-thread communication (TD/IC) logic **2503A-2503F**, a 3D (e.g., texture) sampler **2505A-2505F**, a media sampler **2506A-2506F**, a shader processor **2507A-2507F**, and shared local memory (SLM) **2508A-2508F**. EU arrays **2502A-2502F**, **2504A-2504F** each include multiple execution units, which are general-purpose graphics processing units capable of performing floating-point and integer/fixed-point logic operations in service of a graphics, media, or compute operation, including graphics, media, or compute shader programs. In at least one embodiment, TD/IC logic **2503A-2503F** performs local thread dispatch and thread control operations for execution units within a sub-core and facilitate communication between threads executing on execution units of a sub-core. In at least one embodiment, 3D sampler **2505A-2505F** can read texture or other 3D graphics related data into memory. In at least one embodiment, 3D sampler can read texture data differently based on a configured sample state and texture format associated with a given texture. In at least one embodiment, media sampler **2506A-2506F** can perform similar read operations based on a type and format associated with media data. In at least one embodiment, each graphics sub-core **2501A-2501F** can alternately include a unified 3D and media sampler. In at least one embodiment, threads executing on execution units within each of sub-cores **2501A-2501F** can make use of shared local memory **2508A-2508F** within each sub-core, to enable threads executing within a thread group to execute using a common pool of on-chip memory.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, portions or all of inference and/or training logic **615** may be incorporated into graphics processor **2510**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**, graphics microcontroller **2538**, geometry & fixed function pipeline **2514** and **2536**, or other logic in FIG. **24**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2500** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIGS. **26A-26B** illustrate thread execution logic **2600** including an array of processing elements of a graphics processor core according to at least one embodiment. FIG. **26A** illustrates at least one embodiment, in which thread execution logic **2600** is used. FIG. **26B** illustrates exemplary internal details of an execution unit, according to at least one embodiment.

As illustrated in FIG. **26A**, in at least one embodiment, thread execution logic **2600** includes a shader processor **2602**, a thread dispatcher **2604**, instruction cache **2606**, a scalable execution unit array including a plurality of execu-

tion units **2608A-2608N**, sampler(s) **2610**, a data cache **2612**, and a data port **2614**. In at least one embodiment a scalable execution unit array can dynamically scale by enabling or disabling one or more execution units (e.g., any of execution unit **2608A**, **2608B**, **2608C**, **2608D**, through **2608N-1** and **2608N**) based on computational requirements of a workload, for example. In at least one embodiment, scalable execution units are interconnected via an interconnect fabric that links to each of execution unit. In at least one embodiment, thread execution logic **2600** includes one or more connections to memory, such as system memory or cache memory, through one or more of instruction cache **2606**, data port **2614**, sampler **2610**, and execution units **2608A-2608N**. In at least one embodiment, each execution unit (e.g., **2608A**) is a stand-alone programmable general-purpose computational unit that is capable of executing multiple simultaneous hardware threads while processing multiple data elements in parallel for each thread. In at least one embodiment, array of execution units **2608A-2608N** is scalable to include any number individual execution units.

In at least one embodiment, execution units **2608A-2608N** are primarily used to execute shader programs. In at least one embodiment, shader processor **2602** can process various shader programs and dispatch execution threads associated with shader programs via a thread dispatcher **2604**. In at least one embodiment, thread dispatcher **2604** includes logic to arbitrate thread initiation requests from graphics and media pipelines and instantiate requested threads on one or more execution units in execution units **2608A-2608N**. For example, in at least one embodiment, a geometry pipeline can dispatch vertex, tessellation, or geometry shaders to thread execution logic for processing. In at least one embodiment, thread dispatcher **2604** can also process runtime thread spawning requests from executing shader programs.

In at least one embodiment, execution units **2608A-2608N** support an instruction set that includes native support for many standard 3D graphics shader instructions, such that shader programs from graphics libraries (e.g., Direct 3D and OpenGL) are executed with a minimal translation. In at least one embodiment, execution units support vertex and geometry processing (e.g., vertex programs, geometry programs, vertex shaders), pixel processing (e.g., pixel shaders, fragment shaders) and general-purpose processing (e.g., compute and media shaders). In at least one embodiment, each of execution units **2608A-2608N**, which include one or more arithmetic logic units (ALUs), is capable of multi-issue single instruction multiple data (SIMD) execution and multi-threaded operation enables an efficient execution environment despite higher latency memory accesses. In at least one embodiment, each hardware thread within each execution unit has a dedicated high-bandwidth register file and associated independent thread-state. In at least one embodiment, execution is multi-issue per clock to pipelines capable of integer, single and double precision floating point operations, SIMD branch capability, logical operations, transcendental operations, and other miscellaneous operations. In at least one embodiment, while waiting for data from memory or one of shared functions, dependency logic within execution units **2608A-2608N** causes a waiting thread to sleep until requested data has been returned. In at least one embodiment, while a waiting thread is sleeping, hardware resources may be devoted to processing other threads. For example, in at least one embodiment, during a delay associated with a vertex shader operation, an execution unit can

perform operations for a pixel shader, fragment shader, or another type of shader program, including a different vertex shader.

In at least one embodiment, each execution unit in execution units **2608A-2608N** operates on arrays of data elements. In at least one embodiment, a number of data elements is "execution size," or number of channels for an instruction. In at least one embodiment, an execution channel is a logical unit of execution for data element access, masking, and flow control within instructions. In at least one embodiment, a number of channels may be independent of a number of physical Arithmetic Logic Units (ALUs) or Floating Point Units (FPUs) for a particular graphics processor. In at least one embodiment, execution units **2608A-2608N** support integer and floating-point data types.

In at least one embodiment, an execution unit instruction set includes SIMD instructions. In at least one embodiment, various data elements can be stored as a packed data type in a register and execution unit will process various elements based on data size of elements. For example, in at least one embodiment, when operating on a 256-bit wide vector, 256 bits of a vector are stored in a register and an execution unit operates on a vector as four separate 64-bit packed data elements (Quad-Word (QW) size data elements), eight separate 32-bit packed data elements (Double Word (DW) size data elements), sixteen separate 16-bit packed data elements (Word (W) size data elements), or thirty-two separate 8-bit data elements (byte (B) size data elements). However, in at least one embodiment, different vector widths and register sizes are possible.

In at least one embodiment, one or more execution units can be combined into a fused execution unit **2609A-2609N** having thread control logic (**2607A-2607N**) that is common to fused EUs. In at least one embodiment, multiple EUs can be fused into an EU group. In at least one embodiment, each EU in fused EU group can be configured to execute a separate SIMD hardware thread. Number of EUs in a fused EU group can vary according to various embodiments. In at least one embodiment, various SIMD widths can be performed per-EU, including but not limited to SIMD8, SIMD16, and SIMD32. In at least one embodiment, each fused graphics execution unit **2609A-2609N** includes at least two execution units. For example, in at least one embodiment, fused execution unit **2609A** includes a first EU **2608A**, second EU **2608B**, and thread control logic **2607A** that is common to first EU **2608A** and second EU **2608B**. In at least one embodiment, thread control logic **2607A** controls threads executed on fused graphics execution unit **2609A**, allowing each EU within fused execution units **2609A-2609N** to execute using a common instruction pointer register.

In at least one embodiment, one or more internal instruction caches (e.g., **2606**) are included in thread execution logic **2600** to cache thread instructions for execution units. In at least one embodiment, one or more data caches (e.g., **2612**) are included to cache thread data during thread execution. In at least one embodiment, a sampler **2610** is included to provide texture sampling for 3D operations and media sampling for media operations. In at least one embodiment, sampler **2610** includes specialized texture or media sampling functionality to process texture or media data during a sampling process before providing sampled data to an execution unit.

During execution, in at least one embodiment, graphics and media pipelines send thread initiation requests to thread execution logic **2600** via thread spawning and dispatch logic. In at least one embodiment, once a group of geometric

objects has been processed and rasterized into pixel data, pixel processor logic (e.g., pixel shader logic, fragment shader logic, etc.) within shader processor **2602** is invoked to further compute output information and cause results to be written to output surfaces (e.g., color buffers, depth buffers, stencil buffers, etc.). In at least one embodiment, a pixel shader or fragment shader calculates values of various vertex attributes that are to be interpolated across a rasterized object. In at least one embodiment, pixel processor logic within shader processor **2602** then executes an application programming interface (API)-supplied pixel or fragment shader program. In at least one embodiment, to execute a shader program, shader processor **2602** dispatches threads to an execution unit (e.g., **2608A**) via thread dispatcher **2604**. In at least one embodiment, shader processor **2602** uses texture sampling logic in sampler **2610** to access texture data in texture maps stored in memory. In at least one embodiment, arithmetic operations on texture data and input geometry data compute pixel color data for each geometric fragment, or discards one or more pixels from further processing.

In at least one embodiment, data port **2614** provides a memory access mechanism for thread execution logic **2600** to output processed data to memory for further processing on a graphics processor output pipeline. In at least one embodiment, data port **2614** includes or couples to one or more cache memories (e.g., data cache **2612**) to cache data for memory access via a data port.

As illustrated in FIG. **26B**, in at least one embodiment, a graphics execution unit **2608** can include an instruction fetch unit **2637**, a general register file array (GRF) **2624**, an architectural register file array (ARF) **2626**, a thread arbiter **2622**, a send unit **2630**, a branch unit **2632**, a set of SIMD floating point units (FPU(s)) **2634**, and, in at least one embodiment, a set of dedicated integer SIMD ALUs **2635**. In at least one embodiment, GRF **2624** and ARF **2626** includes a set of general register files and architecture register files associated with each simultaneous hardware thread that may be active in graphics execution unit **2608**. In at least one embodiment, per thread architectural state is maintained in ARF **2626**, while data used during thread execution is stored in GRF **2624**. In at least one embodiment, execution state of each thread, including instruction pointers for each thread, can be held in thread-specific registers in ARF **2626**.

In at least one embodiment, graphics execution unit **2608** has an architecture that is a combination of Simultaneous Multi-Threading (SMT) and fine-grained Interleaved Multi-Threading (IMT). In at least one embodiment, architecture has a modular configuration that can be fine-tuned at design time based on a target number of simultaneous threads and number of registers per execution unit, where execution unit resources are divided across logic used to execute multiple simultaneous threads.

In at least one embodiment, graphics execution unit **2608** can co-issue multiple instructions, which may each be different instructions. In at least one embodiment, thread arbiter **2622** of graphics execution unit thread **2608** can dispatch instructions to one of send unit **2630**, branch unit **2642**, or SIMD FPU(s) **2634** for execution. In at least one embodiment, each execution thread can access 128 general-purpose registers within GRF **2624**, where each register can store 32 bytes, accessible as a SIMD 8-element vector of 32-bit data elements. In at least one embodiment, each execution unit thread has access to 4 Kbytes within GRF **2624**, although embodiments are not so limited, and greater or fewer register resources may be provided in other

embodiments. In at least one embodiment, up to seven threads can execute simultaneously, although a number of threads per execution unit can also vary according to embodiments. In at least one embodiment, in which seven threads may access 4 Kbytes, GRF **2624** can store a total of 28 Kbytes. In at least one embodiment, flexible addressing modes can permit registers to be addressed together to build effectively wider registers or to represent strided rectangular block data structures.

In at least one embodiment, memory operations, sampler operations, and other longer-latency system communications are dispatched via “send” instructions that are executed by message passing send unit **2630**. In at least one embodiment, branch instructions are dispatched to a dedicated branch unit **2632** to facilitate SIMD divergence and eventual convergence.

In at least one embodiment graphics execution unit **2608** includes one or more SIMD floating point units (FPU(s)) **2634** to perform floating-point operations. In at least one embodiment, FPU(s) **2634** also support integer computation. In at least one embodiment FPU(s) **2634** can SIMD execute up to M number of 32-bit floating-point (or integer) operations, or SIMD execute up to 2M 16-bit integer or 16-bit floating-point operations. In at least one embodiment, at least one of FPU(s) provides extended math capability to support high-throughput transcendental math functions and double precision 64-bit floating-point. In at least one embodiment, a set of 8-bit integer SIMD ALUs **2635** are also present, and may be specifically optimized to perform operations associated with machine learning computations.

In at least one embodiment, arrays of multiple instances of graphics execution unit **2608** can be instantiated in a graphics sub-core grouping (e.g., a sub-slice). In at least one embodiment, execution unit **2608** can execute instructions across a plurality of execution channels. In at least one embodiment, each thread executed on graphics execution unit **2608** is executed on a different channel.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, portions or all of inference and/or training logic **615** may be incorporated into execution logic **2600**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of execution logic **2600** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. **27** illustrates a parallel processing unit (“PPU”) **2700**, according to at least one embodiment. In at least one embodiment, PPU **2700** is configured with machine-readable code that, if executed by PPU **2700**, causes PPU **2700** to perform some or all of processes and techniques described throughout this disclosure. In at least one embodiment, PPU **2700** is a multi-threaded processor that is implemented on one or more integrated circuit devices and that utilizes multithreading as a latency-hiding technique designed to process computer-readable instructions (also referred to as

machine-readable instructions or simply instructions) on multiple threads in parallel. In at least one embodiment, a thread refers to a thread of execution and is an instantiation of a set of instructions configured to be executed by PPU 2700. In at least one embodiment, PPU 2700 is a graphics processing unit (“GPU”) configured to implement a graphics rendering pipeline for processing three-dimensional (“3D”) graphics data in order to generate two-dimensional (“2D”) image data for display on a display device such as a liquid crystal display (“LCD”) device. In at least one embodiment, PPU 2700 is utilized to perform computations such as linear algebra operations and machine-learning operations. FIG. 27 illustrates an example parallel processor for illustrative purposes only and should be construed as a non-limiting example of processor architectures contemplated within scope of this disclosure and that any suitable processor may be employed to supplement and/or substitute for same.

In at least one embodiment, one or more PPUs 2700 are configured to accelerate High Performance Computing (“HPC”), data center, and machine learning applications. In at least one embodiment, PPU 2700 is configured to accelerate deep learning systems and applications including following non-limiting examples: autonomous vehicle platforms, deep learning, high-accuracy speech, image, text recognition systems, intelligent video analytics, molecular simulations, drug discovery, disease diagnosis, weather forecasting, big data analytics, astronomy, molecular dynamics simulation, financial modeling, robotics, factory automation, real-time language translation, online search optimizations, and personalized user recommendations, and more.

In at least one embodiment, PPU 2700 includes, without limitation, an Input/Output (“I/O”) unit 2706, a front-end unit 2710, a scheduler unit 2712, a work distribution unit 2714, a hub 2716, a crossbar (“Xbar”) 2720, one or more general processing clusters (“GPCs”) 2718, and one or more partition units (“memory partition units”) 2722. In at least one embodiment, PPU 2700 is connected to a host processor or other PPUs 2700 via one or more high-speed GPU interconnects (“GPU interconnects”) 2708. In at least one embodiment, PPU 2700 is connected to a host processor or other peripheral devices via an interconnect 2702. In at least one embodiment, PPU 2700 is connected to a local memory comprising one or more memory devices (“memory”) 2704. In at least one embodiment, memory devices 2704 include, without limitation, one or more dynamic random access memory (“DRAM”) devices. In at least one embodiment, one or more DRAM devices are configured and/or configurable as high-bandwidth memory (“HBM”) subsystems, with multiple DRAM dies stacked within each device.

In at least one embodiment, high-speed GPU interconnect 2708 may refer to a wire-based multi-lane communications link that is used by systems to scale and include one or more PPUs 2700 combined with one or more central processing units (“CPUs”), supports cache coherence between PPUs 2700 and CPUs, and CPU mastering. In at least one embodiment, data and/or commands are transmitted by high-speed GPU interconnect 2708 through hub 2716 to/from other units of PPU 2700 such as one or more copy engines, video encoders, video decoders, power management units, and other components which may not be explicitly illustrated in FIG. 27.

In at least one embodiment, I/O unit 2706 is configured to transmit and receive communications (e.g., commands, data) from a host processor (not illustrated in FIG. 27) over system bus 2702. In at least one embodiment, I/O unit 2706 communicates with host processor directly via system bus 2702 or through one or more intermediate devices such as a

memory bridge. In at least one embodiment, I/O unit 2706 may communicate with one or more other processors, such as one or more of PPUs 2700 via system bus 2702. In at least one embodiment, I/O unit 2706 implements a Peripheral Component Interconnect Express (“PCIe”) interface for communications over a PCIe bus. In at least one embodiment, I/O unit 2706 implements interfaces for communicating with external devices.

In at least one embodiment, I/O unit 2706 decodes packets received via system bus 2702. In at least one embodiment, at least some packets represent commands configured to cause PPU 2700 to perform various operations. In at least one embodiment, I/O unit 2706 transmits decoded commands to various other units of PPU 2700 as specified by commands. In at least one embodiment, commands are transmitted to front-end unit 2710 and/or transmitted to hub 2716 or other units of PPU 2700 such as one or more copy engines, a video encoder, a video decoder, a power management unit, etc. (not explicitly illustrated in FIG. 27). In at least one embodiment, I/O unit 2706 is configured to route communications between and among various logical units of PPU 2700.

In at least one embodiment, a program executed by host processor encodes a command stream in a buffer that provides workloads to PPU 2700 for processing. In at least one embodiment, a workload comprises instructions and data to be processed by those instructions. In at least one embodiment, buffer is a region in a memory that is accessible (e.g., read/write) by both host processor and PPU 2700—a host interface unit may be configured to access buffer in a system memory connected to system bus 2702 via memory requests transmitted over system bus 2702 by I/O unit 2706. In at least one embodiment, host processor writes command stream to buffer and then transmits a pointer to start of command stream to PPU 2700 such that front-end unit 2710 receives pointers to one or more command streams and manages one or more command streams, reading commands from command streams and forwarding commands to various units of PPU 2700.

In at least one embodiment, front-end unit 2710 is coupled to scheduler unit 2712 that configures various GPCs 2718 to process tasks defined by one or more command streams. In at least one embodiment, scheduler unit 2712 is configured to track state information related to various tasks managed by scheduler unit 2712 where state information may indicate which of GPCs 2718 a task is assigned to, whether task is active or inactive, a priority level associated with task, and so forth. In at least one embodiment, scheduler unit 2712 manages execution of a plurality of tasks on one or more of GPCs 2718.

In at least one embodiment, scheduler unit 2712 is coupled to work distribution unit 2714 that is configured to dispatch tasks for execution on GPCs 2718. In at least one embodiment, work distribution unit 2714 tracks a number of scheduled tasks received from scheduler unit 2712 and work distribution unit 2714 manages a pending task pool and an active task pool for each of GPCs 2718. In at least one embodiment, pending task pool comprises a number of slots (e.g., 32 slots) that contain tasks assigned to be processed by a particular GPC 2718; active task pool may comprise a number of slots (e.g., 4 slots) for tasks that are actively being processed by GPCs 2718 such that as one of GPCs 2718 completes execution of a task, that task is evicted from active task pool for GPC 2718 and one of other tasks from pending task pool is selected and scheduled for execution on GPC 2718. In at least one embodiment, if an active task is idle on GPC 2718, such as while waiting for a data depen-

dency to be resolved, then active task is evicted from GPC 2718 and returned to pending task pool while another task in pending task pool is selected and scheduled for execution on GPC 2718.

In at least one embodiment, work distribution unit 2714 communicates with one or more GPCs 2718 via XBar 2720. In at least one embodiment, XBar 2720 is an interconnect network that couples many of units of PPU 2700 to other units of PPU 2700 and can be configured to couple work distribution unit 2714 to a particular GPC 2718. In at least one embodiment, one or more other units of PPU 2700 may also be connected to XBar 2720 via hub 2716.

In at least one embodiment, tasks are managed by scheduler unit 2712 and dispatched to one of GPCs 2718 by work distribution unit 2714. GPC 2718 is configured to process task and generate results. In at least one embodiment, results may be consumed by other tasks within GPC 2718, routed to a different GPC 2718 via XBar 2720, or stored in memory 2704. In at least one embodiment, results can be written to memory 2704 via partition units 2722, which implement a memory interface for reading and writing data to/from memory 2704. In at least one embodiment, results can be transmitted to another PPU 2704 or CPU via high-speed GPU interconnect 2708. In at least one embodiment, PPU 2700 includes, without limitation, a number U of partition units 2722 that is equal to number of separate and distinct memory devices 2704 coupled to PPU 2700. In at least one embodiment, partition unit 2722 will be described in more detail below in conjunction with FIG. 29.

In at least one embodiment, a host processor executes a driver kernel that implements an application programming interface ("API") that enables one or more applications executing on host processor to schedule operations for execution on PPU 2700. In at least one embodiment, multiple compute applications are simultaneously executed by PPU 2700 and PPU 2700 provides isolation, quality of service ("QoS"), and independent address spaces for multiple compute applications. In at least one embodiment, an application generates instructions (e.g., in form of API calls) that cause driver kernel to generate one or more tasks for execution by PPU 2700 and driver kernel outputs tasks to one or more streams being processed by PPU 2700. In at least one embodiment, each task comprises one or more groups of related threads, which may be referred to as a warp. In at least one embodiment, a warp comprises a plurality of related threads (e.g., 32 threads) that can be executed in parallel. In at least one embodiment, cooperating threads can refer to a plurality of threads including instructions to perform task and that exchange data through shared memory. In at least one embodiment, threads and cooperating threads are described in more detail, in accordance with at least one embodiment, in conjunction with FIG. 29.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to PPU 2700. In at least one embodiment, PPU 2700 is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by PPU 2700. In at least one embodiment, PPU 2700 may be used to perform one or more neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. 28 illustrates a general processing cluster ("GPC") 2800, according to at least one embodiment. In at least one embodiment, GPC 2800 is GPC 2718 of FIG. 27. In at least one embodiment, each GPC 2800 includes, without limitation, a number of hardware units for processing tasks and each GPC 2800 includes, without limitation, a pipeline manager 2802, a pre-raster operations unit ("PROP") 2804, a raster engine 2808, a work distribution crossbar ("WDX") 2816, a memory management unit ("MMU") 2818, one or more Data Processing Clusters ("DPCs") 2806, and any suitable combination of parts.

In at least one embodiment, operation of GPC 2800 is controlled by pipeline manager 2802. In at least one embodiment, pipeline manager 2802 manages configuration of one or more DPCs 2806 for processing tasks allocated to GPC 2800. In at least one embodiment, pipeline manager 2802 configures at least one of one or more DPCs 2806 to implement at least a portion of a graphics rendering pipeline. In at least one embodiment, DPC 2806 is configured to execute a vertex shader program on a programmable streaming multi-processor ("SM") 2814. In at least one embodiment, pipeline manager 2802 is configured to route packets received from a work distribution unit to appropriate logical units within GPC 2800, in at least one embodiment, and some packets may be routed to fixed function hardware units in PROP 2804 and/or raster engine 2808 while other packets may be routed to DPCs 2806 for processing by a primitive engine 2812 or SM 2814. In at least one embodiment, pipeline manager 2802 configures at least one of DPCs 2806 to implement a neural network model and/or a computing pipeline.

In at least one embodiment, PROP unit 2804 is configured, in at least one embodiment, to route data generated by raster engine 2808 and DPCs 2806 to a Raster Operations ("ROP") unit in partition unit 2722, described in more detail above in conjunction with FIG. 27. In at least one embodiment, PROP unit 2804 is configured to perform optimizations for color blending, organize pixel data, perform address translations, and more. In at least one embodiment, raster engine 2808 includes, without limitation, a number of fixed function hardware units configured to perform various raster operations, in at least one embodiment, and raster engine 2808 includes, without limitation, a setup engine, a coarse raster engine, a culling engine, a clipping engine, a fine raster engine, a tile coalescing engine, and any suitable combination thereof. In at least one embodiment, setup engine receives transformed vertices and generates plane equations associated with geometric primitive defined by vertices; plane equations are transmitted to coarse raster engine to generate coverage information (e.g., an x, y coverage mask for a tile) for primitive; output of coarse raster engine is transmitted to culling engine where fragments associated with primitive that fail a z-test are culled, and transmitted to a clipping engine where fragments lying outside a viewing frustum are clipped. In at least one embodiment, fragments that survive clipping and culling are passed to fine raster engine to generate attributes for pixel fragments based on plane equations generated by setup engine. In at least one embodiment, output of raster engine 2808 comprises fragments to be processed by any suitable entity such as by a fragment shader implemented within DPC 2806.

In at least one embodiment, each DPC **2806** included in GPC **2800** comprise, without limitation, an M-Pipe Controller (“MPC”) **2810**; primitive engine **2812**; one or more SMs **2814**; and any suitable combination thereof. In at least one embodiment, MPC **2810** controls operation of DPC **2806**, routing packets received from pipeline manager **2802** to appropriate units in DPC **2806**. In at least one embodiment, packets associated with a vertex are routed to primitive engine **2812**, which is configured to fetch vertex attributes associated with vertex from memory; in contrast, packets associated with a shader program may be transmitted to SM **2814**.

In at least one embodiment, SM **2814** comprises, without limitation, a programmable streaming processor that is configured to process tasks represented by a number of threads. In at least one embodiment, SM **2814** is multi-threaded and configured to execute a plurality of threads (e.g., 32 threads) from a particular group of threads concurrently and implements a Single-Instruction, Multiple-Data (“SIMD”) architecture where each thread in a group of threads (e.g., a warp) is configured to process a different set of data based on same set of instructions. In at least one embodiment, all threads in group of threads execute same instructions. In at least one embodiment, SM **2814** implements a Single-Instruction, Multiple Thread (“SMT”) architecture wherein each thread in a group of threads is configured to process a different set of data based on same set of instructions, but where individual threads in group of threads are allowed to diverge during execution. In at least one embodiment, a program counter, call stack, and execution state is maintained for each warp, enabling concurrency between warps and serial execution within warps when threads within warp diverge. In another embodiment, a program counter, call stack, and execution state is maintained for each individual thread, enabling equal concurrency between all threads, within and between warps. In at least one embodiment, execution state is maintained for each individual thread and threads executing same instructions may be converged and executed in parallel for better efficiency. At least one embodiment of SM **2814** are described in more detail below.

In at least one embodiment, MMU **2818** provides an interface between GPC **2800** and memory partition unit (e.g., partition unit **2722** of FIG. 27) and MMU **2818** provides translation of virtual addresses into physical addresses, memory protection, and arbitration of memory requests. In at least one embodiment, MMU **2818** provides one or more translation lookaside buffers (“TLBs”) for performing translation of virtual addresses into physical addresses in memory.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to GPC **2800**. In at least one embodiment, GPC **2800** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by GPC **2800**. In at least one embodiment, GPC **2800** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic

can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

FIG. 29 illustrates a memory partition unit **2900** of a parallel processing unit (“PPU”), in accordance with at least one embodiment. In at least one embodiment, memory partition unit **2900** includes, without limitation, a Raster Operations (“ROP”) unit **2902**; a level two (“L2”) cache **2904**; a memory interface **2906**; and any suitable combination thereof. In at least one embodiment, memory interface **2906** is coupled to memory. In at least one embodiment, memory interface **2906** may implement 32, 64, 128, 1024-bit data buses, or similar implementations, for high-speed data transfer. In at least one embodiment, PPU incorporates U memory interfaces **2906**, one memory interface **2906** per pair of partition units **2900**, where each pair of partition units **2900** is connected to a corresponding memory device. For example, in at least one embodiment, PPU may be connected to up to Y memory devices, such as high bandwidth memory stacks or graphics double-data-rate, version 5, synchronous dynamic random access memory (“GDDR5 SDRAM”).

In at least one embodiment, memory interface **2906** implements a high bandwidth memory second generation (“HBM2”) memory interface and Y equals half U. In at least one embodiment, HBM2 memory stacks are located on same physical package as PPU, providing substantial power and area savings compared with conventional GDDR5 SDRAM systems. In at least one embodiment, each HBM2 stack includes, without limitation, four memory dies and Y equals 4, with each HBM2 stack including two 128-bit channels per die for a total of 8 channels and a data bus width of 1024 bits. In at least one embodiment, memory supports Single-Error Correcting Double-Error Detecting (“SECCDED”) Error Correction Code (“ECC”) to protect data. In at least one embodiment, ECC provides higher reliability for compute applications that are sensitive to data corruption.

In at least one embodiment, PPU implements a multi-level memory hierarchy. In at least one embodiment, memory partition unit **2900** supports a unified memory to provide a single unified virtual address space for central processing unit (“CPU”) and PPU memory, enabling data sharing between virtual memory systems. In at least one embodiment, frequency of accesses by a PPU to memory located on other processors is traced to ensure that memory pages are moved to physical memory of PPU that is accessing pages more frequently. In at least one embodiment, high-speed GPU interconnect **2708** supports address translation services allowing PPU to directly access a CPU’s page tables and providing full access to CPU memory by PPU.

In at least one embodiment, copy engines transfer data between multiple PPUs or between PPUs and CPUs. In at least one embodiment, copy engines can generate page faults for addresses that are not mapped into page tables and memory partition unit **2900** then services page faults, mapping addresses into page table, after which copy engine performs transfer. In at least one embodiment, memory is pinned (i.e., non-pageable) for multiple copy engine operations between multiple processors, substantially reducing available memory. In at least one embodiment, with hardware page faulting, addresses can be passed to copy engines without regard as to whether memory pages are resident, and copy process is transparent.

Data from memory **2704** of FIG. 27 or other system memory is fetched by memory partition unit **2900** and stored in L2 cache **2904**, which is located on-chip and is shared

between various GPCs, in accordance with at least one embodiment. Each memory partition unit **2900**, in at least one embodiment, includes, without limitation, at least a portion of L2 cache associated with a corresponding memory device. In at least one embodiment, lower level caches are implemented in various units within GPCs. In at least one embodiment, each of SMs **2814** may implement a level one (“L1”) cache wherein L1 cache is private memory that is dedicated to a particular SM **2814** and data from L2 cache **2904** is fetched and stored in each of L1 caches for processing in functional units of SMs **2814**. In at least one embodiment, L2 cache **2904** is coupled to memory interface **2906** and XBar **2720**.

ROP unit **2902** performs graphics raster operations related to pixel color, such as color compression, pixel blending, and more, in at least one embodiment. ROP unit **2902**, in at least one embodiment, implements depth testing in conjunction with raster engine **2808**, receiving a depth for a sample location associated with a pixel fragment from culling engine of raster engine **2808**. In at least one embodiment, depth is tested against a corresponding depth in a depth buffer for a sample location associated with fragment. In at least one embodiment, if fragment passes depth test for sample location, then ROP unit **2902** updates depth buffer and transmits a result of depth test to raster engine **2808**. It will be appreciated that number of partition units **2900** may be different than number of GPCs and, therefore, each ROP unit **2902** can, in at least one embodiment, be coupled to each of GPCs. In at least one embodiment, ROP unit **2902** tracks packets received from different GPCs and determines which that a result generated by ROP unit **2902** is routed to through XBar **2720**.

FIG. **30** illustrates a streaming multi-processor (“SM”) **3000**, according to at least one embodiment. In at least one embodiment, SM **3000** is SM **2814** of FIG. **28**. In at least one embodiment, SM **3000** includes, without limitation, an instruction cache **3002**; one or more scheduler units **3004**; a register file **3008**; one or more processing cores (“cores”) **3010**; one or more special function units (“SFUs”) **3012**; one or more load/store units (“LSUs”) **3014**; an interconnect network **3016**; a shared memory/level one (“L1”) cache **3018**; and any suitable combination thereof. In at least one embodiment, a work distribution unit dispatches tasks for execution on general processing clusters (“GPCs”) of parallel processing units (“PPUs”) and each task is allocated to a particular Data Processing Cluster (“DPC”) within a GPC and, if task is associated with a shader program, task is allocated to one of SMs **3000**. In at least one embodiment, scheduler unit **3004** receives tasks from work distribution unit and manages instruction scheduling for one or more thread blocks assigned to SM **3000**. In at least one embodiment, scheduler unit **3004** schedules thread blocks for execution as warps of parallel threads, wherein each thread block is allocated at least one warp. In at least one embodiment, each warp executes threads. In at least one embodiment, scheduler unit **3004** manages a plurality of different thread blocks, allocating warps to different thread blocks and then dispatching instructions from plurality of different cooperative groups to various functional units (e.g., processing cores **3010**, SFUs **3012**, and LSUs **3014**) during each clock cycle.

In at least one embodiment, Cooperative Groups may refer to a programming model for organizing groups of communicating threads that allows developers to express granularity at which threads are communicating, enabling expression of richer, more efficient parallel decompositions. In at least one embodiment, cooperative launch APIs support

synchronization amongst thread blocks for execution of parallel algorithms. In at least one embodiment, applications of conventional programming models provide a single, simple construct for synchronizing cooperating threads: a barrier across all threads of a thread block (e.g., syncthreads () function). However, In at least one embodiment, programmers may define groups of threads at smaller than thread block granularities and synchronize within defined groups to enable greater performance, design flexibility, and software reuse in form of collective group-wide function interfaces. In at least one embodiment, Cooperative Groups enables programmers to define groups of threads explicitly at sub-block (i.e., as small as a single thread) and multi-block granularities, and to perform collective operations such as synchronization on threads in a cooperative group. In at least one embodiment, programming model supports clean composition across software boundaries, so that libraries and utility functions can synchronize safely within their local context without having to make assumptions about convergence. In at least one embodiment, Cooperative Groups primitives enable new patterns of cooperative parallelism, including, without limitation, producer-consumer parallelism, opportunistic parallelism, and global synchronization across an entire grid of thread blocks.

In at least one embodiment, a dispatch unit **3006** is configured to transmit instructions to one or more of functional units and scheduler unit **3004** includes, without limitation, two dispatch units **3006** that enable two different instructions from same warp to be dispatched during each clock cycle. In at least one embodiment, each scheduler unit **3004** includes a single dispatch unit **3006** or additional dispatch units **3006**.

In at least one embodiment, each SM **3000**, in at least one embodiment, includes, without limitation, register file **3008** that provides a set of registers for functional units of SM **3000**. In at least one embodiment, register file **3008** is divided between each of functional units such that each functional unit is allocated a dedicated portion of register file **3008**. In at least one embodiment, register file **3008** is divided between different warps being executed by SM **3000** and register file **3008** provides temporary storage for operands connected to data paths of functional units. In at least one embodiment, each SM **3000** comprises, without limitation, a plurality of L processing cores **3010**. In at least one embodiment, SM **3000** includes, without limitation, a large number (e.g., 128 or more) of distinct processing cores **3010**. In at least one embodiment, each processing core **3010**, in at least one embodiment, includes, without limitation, a fully-pipelined, single-precision, double-precision, and/or mixed precision processing unit that includes, without limitation, a floating point arithmetic logic unit and an integer arithmetic logic unit. In at least one embodiment, floating point arithmetic logic units implement IEEE 754-2008 standard for floating point arithmetic. In at least one embodiment, processing cores **3010** include, without limitation, 64 single-precision (32-bit) floating point cores, 64 integer cores, 32 double-precision (64-bit) floating point cores, and 8 tensor cores.

Tensor cores are configured to perform matrix operations in accordance with at least one embodiment. In at least one embodiment, one or more tensor cores are included in processing cores **3010**. In at least one embodiment, tensor cores are configured to perform deep learning matrix arithmetic, such as convolution operations for neural network training and inferencing. In at least one embodiment, each

tensor core operates on a 4×4 matrix and performs a matrix multiply and accumulate operation $D=A \times B + C$, where A, B, C, and D are 4×4 matrices.

In at least one embodiment, matrix multiply inputs A and B are 16-bit floating point matrices and accumulation matrices C and D are 16-bit floating point or 32-bit floating point matrices. In at least one embodiment, tensor cores operate on 16-bit floating point input data with 32-bit floating point accumulation. In at least one embodiment, 16-bit floating point multiply uses 64 operations and results in a full precision product that is then accumulated using 32-bit floating point addition with other intermediate products for a 4×4 matrix multiply. Tensor cores are used to perform much larger two-dimensional or higher dimensional matrix operations, built up from these smaller elements, in at least one embodiment. In at least one embodiment, an API, such as CUDA 9 C++ API, exposes specialized matrix load, matrix multiply and accumulate, and matrix store operations to efficiently use tensor cores from a CUDA-C++ program. In at least one embodiment, at CUDA level, warp-level interface assumes 16×16 size matrices spanning all 32 threads of warp.

In at least one embodiment, each SM 3000 comprises, without limitation, M SFUs 3012 that perform special functions (e.g., attribute evaluation, reciprocal square root, etc.). In at least one embodiment, SFUs 3012 include, without limitation, a tree traversal unit configured to traverse a hierarchical tree data structure. In at least one embodiment, SFUs 3012 include, without limitation, a texture unit configured to perform texture map filtering operations. In at least one embodiment, texture units are configured to load texture maps (e.g., a 2D array of texels) from memory and sample texture maps to produce sampled texture values for use in shader programs executed by SM 3000. In at least one embodiment, texture maps are stored in shared memory/L1 cache 3018. In at least one embodiment, texture units implement texture operations such as filtering operations using mip-maps (e.g., texture maps of varying levels of detail), in accordance with at least one embodiment. In at least one embodiment, each SM 3000 includes, without limitation, two texture units.

Each SM 3000 comprises, without limitation, N LSUs 3014 that implement load and store operations between shared memory/L1 cache 3018 and register file 3008, in at least one embodiment. Each SM 3000 includes, without limitation, interconnect network 3016 that connects each of functional units to register file 3008 and LSU 3014 to register file 3008 and shared memory/L1 cache 3018 in at least one embodiment. In at least one embodiment, interconnect network 3016 is a crossbar that can be configured to connect any of functional units to any of registers in register file 3008 and connect LSUs 3014 to register file 3008 and memory locations in shared memory/L1 cache 3018.

In at least one embodiment, shared memory/L1 cache 3018 is an array of on-chip memory that allows for data storage and communication between SM 3000 and primitive engine and between threads in SM 3000, in at least one embodiment. In at least one embodiment, shared memory/L1 cache 3018 comprises, without limitation, 128 KB of storage capacity and is in path from SM 3000 to partition unit. In at least one embodiment, shared memory/L1 cache 3018, in at least one embodiment, is used to cache reads and writes. In at least one embodiment, one or more of shared memory/L1 cache 3018, L2 cache, and memory are backing stores.

Combining data cache and shared memory functionality into a single memory block provides improved performance

for both types of memory accesses, in at least one embodiment. In at least one embodiment, capacity is used or is usable as a cache by programs that do not use shared memory, such as if shared memory is configured to use half of capacity, texture and load/store operations can use remaining capacity. Integration within shared memory/L1 cache 3018 enables shared memory/L1 cache 3018 to function as a high-throughput conduit for streaming data while simultaneously providing high-bandwidth and low-latency access to frequently reused data, in accordance with at least one embodiment. In at least one embodiment, when configured for general purpose parallel computation, a simpler configuration can be used compared with graphics processing. In at least one embodiment, fixed function graphics processing units are bypassed, creating a much simpler programming model. In general purpose parallel computation configuration, work distribution unit assigns and distributes blocks of threads directly to DPCs, in at least one embodiment. In at least one embodiment, threads in a block execute same program, using a unique thread ID in calculation to ensure each thread generates unique results, using SM 3000 to execute program and perform calculations, shared memory/L1 cache 3018 to communicate between threads, and LSU 3014 to read and write global memory through shared memory/L1 cache 3018 and memory partition unit. In at least one embodiment, when configured for general purpose parallel computation, SM 3000 writes commands that scheduler unit 3004 can use to launch new work on DPCs.

In at least one embodiment, PPU is included in or coupled to a desktop computer, a laptop computer, a tablet computer, servers, supercomputers, a smart-phone (e.g., a wireless, hand-held device), personal digital assistant ("PDA"), a digital camera, a vehicle, a head mounted display, a hand-held electronic device, and more. In at least one embodiment, PPU is embodied on a single semiconductor substrate. In at least one embodiment, PPU is included in a system-on-a-chip ("SoC") along with one or more other devices such as additional PPUs, memory, a reduced instruction set computer ("RISC") CPU, a memory management unit ("MMU"), a digital-to-analog converter ("DAC"), and like.

In at least one embodiment, PPU may be included on a graphics card that includes one or more memory devices. A graphics card may be configured to interface with a PCIe slot on a motherboard of a desktop computer. In at least one embodiment, PPU may be an integrated graphics processing unit ("iGPU") included in chipset of motherboard.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to SM 3000. In at least one embodiment, SM 3000 is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by SM 3000. In at least one embodiment, SM 3000 may be used to perform one or more neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

In at least one embodiment, a single semiconductor platform may refer to a sole unitary semiconductor-based integrated circuit or chip. In at least one embodiment, multi-chip modules may be used with increased connectivity which simulate on-chip operation, and make substantial improvements over utilizing a conventional central processing unit (“CPU”) and bus implementation. In at least one embodiment, various modules may also be situated separately or in various combinations of semiconductor platforms per desires of user.

In at least one embodiment, computer programs in form of machine-readable executable code or computer control logic algorithms are stored in main memory **1004** and/or secondary storage. Computer programs, if executed by one or more processors, enable system **1000** to perform various functions in accordance with at least one embodiment. In at least one embodiment, memory **1004**, storage, and/or any other storage are possible examples of computer-readable media. In at least one embodiment, secondary storage may refer to any suitable storage device or system such as a hard disk drive and/or a removable storage drive, representing a floppy disk drive, a magnetic tape drive, a compact disk drive, digital versatile disk (“DVD”) drive, recording device, universal serial bus (“USB”) flash memory, etc. In at least one embodiment, architecture and/or functionality of various previous figures are implemented in context of CPU **1002**; parallel processing system **1012**; an integrated circuit capable of at least a portion of capabilities of both CPU **1002**; parallel processing system **1012**; a chipset (e.g., a group of integrated circuits designed to work and sold as a unit for performing related functions, etc.); and any suitable combination of integrated circuit(s).

In at least one embodiment, architecture and/or functionality of various previous figures are implemented in context of a general computer system, a circuit board system, a game console system dedicated for entertainment purposes, an application-specific system, and more. In at least one embodiment, computer system **1000** may take form of a desktop computer, a laptop computer, a tablet computer, servers, supercomputers, a smart-phone (e.g., a wireless, hand-held device), personal digital assistant (“PDA”), a digital camera, a vehicle, a head mounted display, a hand-held electronic device, a mobile phone device, a television, workstation, game consoles, embedded system, and/or any other type of logic.

In at least one embodiment, parallel processing system **1012** includes, without limitation, a plurality of parallel processing units (“PPUs”) **1014** and associated memories **1016**. In at least one embodiment, PPUs **1014** are connected to a host processor or other peripheral devices via an interconnect **1018** and a switch **1020** or multiplexer. In at least one embodiment, parallel processing system **1012** distributes computational tasks across PPUs **1014** which can be parallelizable—for example, as part of distribution of computational tasks across multiple graphics processing unit (“GPU”) thread blocks. In at least one embodiment, memory is shared and accessible (e.g., for read and/or write access) across some or all of PPUs **1014**, although such shared memory may incur performance penalties relative to use of local memory and registers resident to a PPU **1014**. In at least one embodiment, operation of PPUs **1014** is synchronized through use of a command such as `_syncthreads()`, wherein all threads in a block (e.g., executed across multiple PPUs **1014**) to reach a certain point of execution of code before proceeding.

Virtualized Computing Platform

Embodiments are disclosed related a virtualized computing platform for advanced computing, such as image inferencing and image processing. With reference to FIG. **31** is an example data flow diagram for a process **3100** of generating and deploying an image processing and inferencing pipeline, in accordance with at least one embodiment. In at least one embodiment, process **3100** may be deployed for use with imaging devices, processing devices, genomics devices, gene sequencing devices, radiology devices, and/or other device types at one or more facilities **3102**, such as medical facilities, hospitals, healthcare institutes, clinics, research or diagnostic labs, etc. In at least one embodiment, process **3100** may be deployed to perform genomics analysis and inferencing on sequencing data. Examples of genomic analyses that may be performed using systems and processes described herein include, without limitation, variant calling, mutation detection, and gene expression quantification. Process **3100** may be executed within a training system **3104** and/or a deployment system **3106**. In at least one embodiment, training system **3104** may be used to perform training, deployment, and implementation of machine learning models (e.g., neural networks, object detection algorithms, computer vision algorithms, etc.) for use in deployment system **3106**. In at least one embodiment, deployment system **3106** may be configured to offload processing and compute resources among a distributed computing environment to reduce infrastructure requirements at facility **3102**. In at least one embodiment, deployment system **3106** may provide a streamlined platform for selecting, customizing, and implementing virtual instruments for use with imaging devices (e.g., MRI, CT Scan, X-Ray, Ultrasound, etc.) or sequencing devices at facility **3102**. In at least one embodiment, virtual instruments may include software-defined applications for performing one or more processing operations with respect to imaging data generated by imaging devices, sequencing devices, radiology devices, and/or other device types. In at least one embodiment, one or more applications in a pipeline may use or call upon services (e.g., inference, visualization, compute, AI, etc.) of deployment system **3106** during execution of applications.

In at least one embodiment, some of applications used in advanced processing and inferencing pipelines may use machine learning models or other AI to perform one or more processing steps. In at least one embodiment, machine learning models may be trained at facility **3102** using data **3108** (such as imaging data) generated at facility **3102** (and stored on one or more picture archiving and communication system (PACS) servers at facility **3102**), may be trained using imaging or sequencing data **3108** from another facility(ies) (e.g., a different hospital, lab, clinic, etc.), or a combination thereof. In at least one embodiment, training system **3104** may be used to provide applications, services, and/or other resources for generating working, deployable machine learning models for deployment system **3106**.

In at least one embodiment, model registry **3124** may be backed by object storage that may support versioning and object metadata. In at least one embodiment, object storage may be accessible through, for example, a cloud storage (e.g., cloud **3226** of FIG. **32**) compatible application programming interface (API) from within a cloud platform. In at least one embodiment, machine learning models within model registry **3124** may be uploaded, listed, modified, or deleted by developers or partners of a system interacting with an API. In at least one embodiment, an API may provide access to methods that allow users with appropriate credentials to associate models with applications, such that

models may be executed as part of execution of containerized instantiations of applications.

In at least one embodiment, training pipeline **3204** (FIG. **32**) may include a scenario where facility **3102** is training their own machine learning model, or has an existing machine learning model that needs to be optimized or updated. In at least one embodiment, imaging data **3108** generated by imaging device(s), sequencing devices, and/or other device types may be received. In at least one embodiment, once imaging data **3108** is received, AI-assisted annotation **3110** may be used to aid in generating annotations corresponding to imaging data **3108** to be used as ground truth data for a machine learning model. In at least one embodiment, AI-assisted annotation **3110** may include one or more machine learning models (e.g., convolutional neural networks (CNNs)) that may be trained to generate annotations corresponding to certain types of imaging data **3108** (e.g., from certain devices) and/or certain types of anomalies in imaging data **3108**. In at least one embodiment, AI-assisted annotations **3110** may then be used directly, or may be adjusted or fine-tuned using an annotation tool (e.g., by a researcher, a clinician, a doctor, a scientist, etc.), to generate ground truth data. In at least one embodiment, in some examples, labeled clinic data **3112** (e.g., annotations provided by a clinician, doctor, scientist, technician, etc.) may be used as ground truth data for training a machine learning model. In at least one embodiment, AI-assisted annotations **3110**, labeled clinic data **3112**, or a combination thereof may be used as ground truth data for training a machine learning model. In at least one embodiment, a trained machine learning model may be referred to as output model **3116**, and may be used by deployment system **3106**, as described herein.

In at least one embodiment, training pipeline **3204** (FIG. **32**) may include a scenario where facility **3102** needs a machine learning model for use in performing one or more processing tasks for one or more applications in deployment system **3106**, but facility **3102** may not currently have such a machine learning model (or may not have a model that is optimized, efficient, or effective for such purposes). In at least one embodiment, an existing machine learning model may be selected from a model registry **3124**. In at least one embodiment, model registry **3124** may include machine learning models trained to perform a variety of different inference tasks on imaging data. In at least one embodiment, machine learning models in model registry **3124** may have been trained on imaging data from different facilities than facility **3102** (e.g., facilities remotely located). In at least one embodiment, machine learning models may have been trained on imaging data from one location, two locations, or any number of locations. In at least one embodiment, when being trained on imaging data from a specific location, training may take place at that location, or at least in a manner that protects confidentiality of imaging data or restricts imaging data from being transferred off-premises (e.g., to comply with HIPAA regulations, privacy regulations, etc.). In at least one embodiment, once a model is trained—or partially trained—at one location, a machine learning model may be added to model registry **3124**. In at least one embodiment, a machine learning model may then be retrained, or updated, at any number of other facilities, and a retrained or updated model may be made available in model registry **3124**. In at least one embodiment, a machine learning model may then be selected from model registry **3124**—and referred to as output model **3116**—and may be

used in deployment system **3106** to perform one or more processing tasks for one or more applications of a deployment system.

In at least one embodiment, training pipeline **3204** (FIG. **32**), a scenario may include facility **3102** requiring a machine learning model for use in performing one or more processing tasks for one or more applications in deployment system **3106**, but facility **3102** may not currently have such a machine learning model (or may not have a model that is optimized, efficient, or effective for such purposes). In at least one embodiment, a machine learning model selected from model registry **3124** may not be fine-tuned or optimized for imaging data **3108** generated at facility **3102** because of differences in populations, genetic variations, robustness of training data used to train a machine learning model, diversity in anomalies of training data, and/or other issues with training data. In at least one embodiment, AI-assisted annotation **3110** may be used to aid in generating annotations corresponding to imaging data **3108** to be used as ground truth data for retraining or updating a machine learning model. In at least one embodiment, labeled clinic data **3112** (e.g., annotations provided by a clinician, doctor, scientist, etc.) may be used as ground truth data for training a machine learning model. In at least one embodiment, retraining or updating a machine learning model may be referred to as model training **3114**. In at least one embodiment, model training **3114**—e.g., AI-assisted annotations **3110**, labeled clinic data **3112**, or a combination thereof—may be used as ground truth data for retraining or updating a machine learning model. In at least one embodiment, a trained machine learning model may be referred to as output model **3116**, and may be used by deployment system **3106**, as described herein.

In at least one embodiment, deployment system **3106** may include software **3118**, services **3120**, hardware **3122**, and/or other components, features, and functionality. In at least one embodiment, deployment system **3106** may include a software “stack,” such that software **3118** may be built on top of services **3120** and may use services **3120** to perform some or all of processing tasks, and services **3120** and software **3118** may be built on top of hardware **3122** and use hardware **3122** to execute processing, storage, and/or other compute tasks of deployment system **3106**. In at least one embodiment, software **3118** may include any number of different containers, where each container may execute an instantiation of an application. In at least one embodiment, each application may perform one or more processing tasks in an advanced processing and inferencing pipeline (e.g., inferencing, object detection, feature detection, segmentation, image enhancement, calibration, etc.). In at least one embodiment, for each type of imaging device (e.g., CT, MRI, X-Ray, ultrasound, sonography, echocardiography, etc.), sequencing device, radiology device, genomics device, etc., there may be any number of containers that may perform a data processing task with respect to imaging data **3108** (or other data types, such as those described herein) generated by a device. In at least one embodiment, an advanced processing and inferencing pipeline may be defined based on selections of different containers that are desired or required for processing imaging data **3108**, in addition to containers that receive and configure imaging data for use by each container and/or for use by facility **3102** after processing through a pipeline (e.g., to convert outputs back to a usable data type, such as digital imaging and communications in medicine (DICOM) data, radiology information system (RIS) data, clinical information system (CIS) data, remote procedure call (RPC) data, data substan-

tially compliant with a representation state transfer (REST) interface, data substantially compliant with a file-based interface, and/or raw data, for storage and display at facility 3102). In at least one embodiment, a combination of containers within software 3118 (e.g., that make up a pipeline) may be referred to as a virtual instrument (as described in more detail herein), and a virtual instrument may leverage services 3120 and hardware 3122 to execute some or all processing tasks of applications instantiated in containers.

In at least one embodiment, a data processing pipeline may receive input data (e.g., imaging data 3108) in a DICOM, RIS, CIS, REST compliant, RPC, raw, and/or other format in response to an inference request (e.g., a request from a user of deployment system 3106, such as a clinician, a doctor, a radiologist, etc.). In at least one embodiment, input data may be representative of one or more images, video, and/or other data representations generated by one or more imaging devices, sequencing devices, radiology devices, genomics devices, and/or other device types. In at least one embodiment, data may undergo pre-processing as part of data processing pipeline to prepare data for processing by one or more applications. In at least one embodiment, post-processing may be performed on an output of one or more inferencing tasks or other processing tasks of a pipeline to prepare an output data for a next application and/or to prepare output data for transmission and/or use by a user (e.g., as a response to an inference request). In at least one embodiment, inferencing tasks may be performed by one or more machine learning models, such as trained or deployed neural networks, which may include output models 3116 of training system 3104.

In at least one embodiment, tasks of data processing pipeline may be encapsulated in a container(s) that each represent a discrete, fully functional instantiation of an application and virtualized computing environment that is able to reference machine learning models. In at least one embodiment, containers or applications may be published into a private (e.g., limited access) area of a container registry (described in more detail herein), and trained or deployed models may be stored in model registry 3124 and associated with one or more applications. In at least one embodiment, images of applications (e.g., container images) may be available in a container registry, and once selected by a user from a container registry for deployment in a pipeline, an image may be used to generate a container for an instantiation of an application for use by a user's system.

In at least one embodiment, developers (e.g., software developers, clinicians, doctors, etc.) may develop, publish, and store applications (e.g., as containers) for performing image processing and/or inferencing on supplied data. In at least one embodiment, development, publishing, and/or storing may be performed using a software development kit (SDK) associated with a system (e.g., to ensure that an application and/or container developed is compliant with or compatible with a system). In at least one embodiment, an application that is developed may be tested locally (e.g., at a first facility, on data from a first facility) with an SDK which may support at least some of services 3120 as a system (e.g., system 3200 of FIG. 32). In at least one embodiment, because DICOM objects may contain anywhere from one to hundreds of images or other data types, and due to a variation in data, a developer may be responsible for managing (e.g., setting constructs for, building pre-processing into an application, etc.) extraction and preparation of incoming DICOM data. In at least one embodiment, once validated by system 3200 (e.g., for accuracy, safety, patient privacy, etc.), an application may be

available in a container registry for selection and/or implementation by a user (e.g., a hospital, clinic, lab, healthcare provider, etc.) to perform one or more processing tasks with respect to data at a facility (e.g., a second facility) of a user.

In at least one embodiment, developers may then share applications or containers through a network for access and use by users of a system (e.g., system 3200 of FIG. 32). In at least one embodiment, completed and validated applications or containers may be stored in a container registry and associated machine learning models may be stored in model registry 3124. In at least one embodiment, a requesting entity (e.g., a user at a medical facility)—who provides an inference or image processing request—may browse a container registry and/or model registry 3124 for an application, container, dataset, machine learning model, etc., select a desired combination of elements for inclusion in data processing pipeline, and submit an imaging processing request. In at least one embodiment, a request may include input data (and associated patient data, in some examples) that is necessary to perform a request, and/or may include a selection of application(s) and/or machine learning models to be executed in processing a request. In at least one embodiment, a request may then be passed to one or more components of deployment system 3106 (e.g., a cloud) to perform processing of data processing pipeline. In at least one embodiment, processing by deployment system 3106 may include referencing selected elements (e.g., applications, containers, models, etc.) from a container registry and/or model registry 3124. In at least one embodiment, once results are generated by a pipeline, results may be returned to a user for reference (e.g., for viewing in a viewing application suite executing on a local, on-premises workstation or terminal). In at least one embodiment, a radiologist may receive results from an data processing pipeline including any number of application and/or containers, where results may include anomaly detection in X-rays, CT scans, MRIs, etc.

In at least one embodiment, to aid in processing or execution of applications or containers in pipelines, services 3120 may be leveraged. In at least one embodiment, services 3120 may include compute services, artificial intelligence (AI) services, visualization services, and/or other service types. In at least one embodiment, services 3120 may provide functionality that is common to one or more applications in software 3118, so functionality may be abstracted to a service that may be called upon or leveraged by applications. In at least one embodiment, functionality provided by services 3120 may run dynamically and more efficiently, while also scaling well by allowing applications to process data in parallel (e.g., using a parallel computing platform 3230 (FIG. 32)). In at least one embodiment, rather than each application that shares a same functionality offered by a service 3120 being required to have a respective instance of service 3120, service 3120 may be shared between and among various applications. In at least one embodiment, services may include an inference server or engine that may be used for executing detection or segmentation tasks, as non-limiting examples. In at least one embodiment, a model training service may be included that may provide machine learning model training and/or retraining capabilities. In at least one embodiment, a data augmentation service may further be included that may provide GPU accelerated data (e.g., DICOM, RIS, CIS, REST compliant, RPC, raw, etc.) extraction, resizing, scaling, and/or other augmentation. In at least one embodiment, a visualization service may be used that may add image rendering effects—such as ray-tracing, rasterization, denoise-

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ing, sharpening, etc.—to add realism to two-dimensional (2D) and/or three-dimensional (3D) models. In at least one embodiment, virtual instrument services may be included that provide for beam-forming, segmentation, inferencing, imaging, and/or support for other applications within pipelines of virtual instruments.

In at least one embodiment, where a service 3120 includes an AI service (e.g., an inference service), one or more machine learning models associated with an application for anomaly detection (e.g., tumors, growth abnormalities, scarring, etc.) may be executed by calling upon (e.g., as an API call) an inference service (e.g., an inference server) to execute machine learning model(s), or processing thereof, as part of application execution. In at least one embodiment, where another application includes one or more machine learning models for segmentation tasks, an application may call upon an inference service to execute machine learning models for performing one or more of processing operations associated with segmentation tasks. In at least one embodiment, software 3118 implementing advanced processing and inferencing pipeline that includes segmentation application and anomaly detection application may be streamlined because each application may call upon a same inference service to perform one or more inferencing tasks.

In at least one embodiment, hardware 3122 may include GPUs, CPUs, graphics cards, an AI/deep learning system (e.g., an AI supercomputer, such as NVIDIA's DGX), a cloud platform, or a combination thereof. In at least one embodiment, different types of hardware 3122 may be used to provide efficient, purpose-built support for software 3118 and services 3120 in deployment system 3106. In at least one embodiment, use of GPU processing may be implemented for processing locally (e.g., at facility 3102), within an AI/deep learning system, in a cloud system, and/or in other processing components of deployment system 3106 to improve efficiency, accuracy, and efficacy of image processing, image reconstruction, segmentation, Mill exams, stroke or heart attack detection (e.g., in real-time), image quality in rendering, etc. In at least one embodiment, a facility may include imaging devices, genomics devices, sequencing devices, and/or other device types on-premises that may leverage GPUs to generate imaging data representative of a subject's anatomy. In at least one embodiment, software 3118 and/or services 3120 may be optimized for GPU processing with respect to deep learning, machine learning, and/or high-performance computing, as non-limiting examples. In at least one embodiment, at least some of computing environment of deployment system 3106 and/or training system 3104 may be executed in a datacenter one or more supercomputers or high performance computing systems, with GPU optimized software (e.g., hardware and software combination of NVIDIA's DGX System). In at least one embodiment, datacenters may be compliant with provisions of HIPAA, such that receipt, processing, and transmission of imaging data and/or other patient data is securely handled with respect to privacy of patient data. In at least one embodiment, hardware 3122 may include any number of GPUs that may be called upon to perform processing of data in parallel, as described herein. In at least one embodiment, cloud platform may further include GPU processing for GPU-optimized execution of deep learning tasks, machine learning tasks, or other computing tasks. In at least one embodiment, cloud platform (e.g., NVIDIA's NGC) may be executed using an AI/deep learning supercomputer(s) and/or GPU-optimized software (e.g., as provided on NVIDIA's DGX Systems) as a hardware abstraction and scaling platform. In at least one embodiment, cloud

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platform may integrate an application container clustering system or orchestration system (e.g., KUBERNETES) on multiple GPUs to enable seamless scaling and load balancing.

FIG. 32 is a system diagram for an example system 3200 for generating and deploying an imaging deployment pipeline, in accordance with at least one embodiment. In at least one embodiment, system 3200 may be used to implement process 3100 of FIG. 31 and/or other processes including advanced processing and inferencing pipelines. In at least one embodiment, system 3200 may include training system 3104 and deployment system 3106. In at least one embodiment, training system 3104 and deployment system 3106 may be implemented using software 3118, services 3120, and/or hardware 3122, as described herein.

In at least one embodiment, system 3200 (e.g., training system 3104 and/or deployment system 3106) may implemented in a cloud computing environment (e.g., using cloud 3226). In at least one embodiment, system 3200 may be implemented locally with respect to a healthcare services facility, or as a combination of both cloud and local computing resources. In at least one embodiment, in embodiments where cloud computing is implemented, patient data may be separated from, or unprocessed by, by one or more components of system 3200 that would render processing non-compliant with HIPAA and/or other data handling and privacy regulations or laws. In at least one embodiment, access to APIs in cloud 3226 may be restricted to authorized users through enacted security measures or protocols. In at least one embodiment, a security protocol may include web tokens that may be signed by an authentication (e.g., AuthN, AuthZ, Gluecon, etc.) service and may carry appropriate authorization. In at least one embodiment, APIs of virtual instruments (described herein), or other instantiations of system 3200, may be restricted to a set of public IPs that have been vetted or authorized for interaction.

In at least one embodiment, various components of system 3200 may communicate between and among one another using any of a variety of different network types, including but not limited to local area networks (LANs) and/or wide area networks (WANs) via wired and/or wireless communication protocols. In at least one embodiment, communication between facilities and components of system 3200 (e.g., for transmitting inference requests, for receiving results of inference requests, etc.) may be communicated over data bus(es), wireless data protocols (Wi-Fi), wired data protocols (e.g., Ethernet), etc.

In at least one embodiment, training system 3104 may execute training pipelines 3204, similar to those described herein with respect to FIG. 31. In at least one embodiment, where one or more machine learning models are to be used in deployment pipelines 3210 by deployment system 3106, training pipelines 3204 may be used to train or retrain one or more (e.g. pre-trained) models, and/or implement one or more of pre-trained models 3206 (e.g., without a need for retraining or updating). In at least one embodiment, as a result of training pipelines 3204, output model(s) 3116 may be generated. In at least one embodiment, training pipelines 3204 may include any number of processing steps, such as but not limited to imaging data (or other input data) conversion or adaption (e.g., using DICOM adapter 3202A to convert DICOM images to another format suitable for processing by respective machine learning models, such as Neuroimaging Informatics Technology Initiative (NIfTI) format), AI-assisted annotation 3110, labeling or annotating of imaging data 3108 to generate labeled clinic data 3112, model selection from a model registry, model training 3114,

training, retraining, or updating models, and/or other processing steps. In at least one embodiment, for different machine learning models used by deployment system 3106, different training pipelines 3204 may be used. In at least one embodiment, training pipeline 3204 similar to a first example described with respect to FIG. 31 may be used for a first machine learning model, training pipeline 3204 similar to a second example described with respect to FIG. 31 may be used for a second machine learning model, and training pipeline 3204 similar to a third example described with respect to FIG. 31 may be used for a third machine learning model. In at least one embodiment, any combination of tasks within training system 3104 may be used depending on what is required for each respective machine learning model. In at least one embodiment, one or more of machine learning models may already be trained and ready for deployment so machine learning models may not undergo any processing by training system 3104, and may be implemented by deployment system 3106.

In at least one embodiment, output model(s) 3116 and/or pre-trained model(s) 3206 may include any types of machine learning models depending on implementation or embodiment. In at least one embodiment, and without limitation, machine learning models used by system 3200 may include machine learning model(s) using linear regression, logistic regression, decision trees, support vector machines (SVM), Naïve Bayes, k-nearest neighbor (Knn), K means clustering, random forest, dimensionality reduction algorithms, gradient boosting algorithms, neural networks (e.g., auto-encoders, convolutional, recurrent, perceptrons, Long/Short Term Memory (LSTM), Hopfield, Boltzmann, deep belief, deconvolutional, generative adversarial, liquid state machine, etc.), and/or other types of machine learning models.

In at least one embodiment, training pipelines 3204 may include AI-assisted annotation, as described in more detail herein with respect to at least FIG. 35B. In at least one embodiment, labeled clinic data 3112 (e.g., traditional annotation) may be generated by any number of techniques. In at least one embodiment, labels or other annotations may be generated within a drawing program (e.g., an annotation program), a computer aided design (CAD) program, a labeling program, another type of program suitable for generating annotations or labels for ground truth, and/or may be hand drawn, in some examples. In at least one embodiment, ground truth data may be synthetically produced (e.g., generated from computer models or renderings), real produced (e.g., designed and produced from real-world data), machine-automated (e.g., using feature analysis and learning to extract features from data and then generate labels), human annotated (e.g., labeler, or annotation expert, defines location of labels), and/or a combination thereof. In at least one embodiment, for each instance of imaging data 3108 (or other data type used by machine learning models), there may be corresponding ground truth data generated by training system 3104. In at least one embodiment, AI-assisted annotation may be performed as part of deployment pipelines 3210; either in addition to, or in lieu of AI-assisted annotation included in training pipelines 3204. In at least one embodiment, system 3200 may include a multi-layer platform that may include a software layer (e.g., software 3118) of diagnostic applications (or other application types) that may perform one or more medical imaging and diagnostic functions. In at least one embodiment, system 3200 may be communicatively coupled to (e.g., via encrypted links) PACS server networks of one or more facilities. In at least one embodiment, system 3200 may be configured to access and referenced data (e.g., DICOM data, RIS data, raw

data, CIS data, REST compliant data, RPC data, raw data, etc.) from PACS servers (e.g., via a DICOM adapter 3202, or another data type adapter such as RIS, CIS, REST compliant, RPC, raw, etc.) to perform operations, such as training machine learning models, deploying machine learning models, image processing, inferencing, and/or other operations.

In at least one embodiment, a software layer may be implemented as a secure, encrypted, and/or authenticated API through which applications or containers may be invoked (e.g., called) from an external environment(s) (e.g., facility 3102). In at least one embodiment, applications may then call or execute one or more services 3120 for performing compute, AI, or visualization tasks associated with respective applications, and software 3118 and/or services 3120 may leverage hardware 3122 to perform processing tasks in an effective and efficient manner.

In at least one embodiment, deployment system 3106 may execute deployment pipelines 3210. In at least one embodiment, deployment pipelines 3210 may include any number of applications that may be sequentially, non-sequentially, or otherwise applied to imaging data (and/or other data types) generated by imaging devices, sequencing devices, genomics devices, etc.—including AI-assisted annotation, as described above. In at least one embodiment, as described herein, a deployment pipeline 3210 for an individual device may be referred to as a virtual instrument for a device (e.g., a virtual ultrasound instrument, a virtual CT scan instrument, a virtual sequencing instrument, etc.). In at least one embodiment, for a single device, there may be more than one deployment pipeline 3210 depending on information desired from data generated by a device. In at least one embodiment, where detections of anomalies are desired from an MRI machine, there may be a first deployment pipeline 3210, and where image enhancement is desired from output of an MRI machine, there may be a second deployment pipeline 3210.

In at least one embodiment, applications available for deployment pipelines 3210 may include any application that may be used for performing processing tasks on imaging data or other data from devices. In at least one embodiment, different applications may be responsible for image enhancement, segmentation, reconstruction, anomaly detection, object detection, feature detection, treatment planning, dosimetry, beam planning (or other radiation treatment procedures), and/or other analysis, image processing, or inferencing tasks. In at least one embodiment, deployment system 3106 may define constructs for each of applications, such that users of deployment system 3106 (e.g., medical facilities, labs, clinics, etc.) may understand constructs and adapt applications for implementation within their respective facility. In at least one embodiment, an application for image reconstruction may be selected for inclusion in deployment pipeline 3210, but data type generated by an imaging device may be different from a data type used within an application. In at least one embodiment, DICOM adapter 3202B (and/or a DICOM reader) or another data type adapter or reader (e.g., RIS, CIS, REST compliant, RPC, raw, etc.) may be used within deployment pipeline 3210 to convert data to a form useable by an application within deployment system 3106. In at least one embodiment, access to DICOM, RIS, CIS, REST compliant, RPC, raw, and/or other data type libraries may be accumulated and pre-processed, including decoding, extracting, and/or performing any convolutions, color corrections, sharpness, gamma, and/or other augmentations to data. In at least one embodiment, DICOM, RIS, CIS, REST compliant, RPC, and/or raw data may be unordered and a pre-pass may be

executed to organize or sort collected data. In at least one embodiment, because various applications may share common image operations, in some embodiments, a data augmentation library (e.g., as one of services **3120**) may be used to accelerate these operations. In at least one embodiment, to avoid bottlenecks of conventional processing approaches that rely on CPU processing, parallel computing platform **3230** may be used for GPU acceleration of these processing tasks.

In at least one embodiment, an image reconstruction application may include a processing task that includes use of a machine learning model. In at least one embodiment, a user may desire to use their own machine learning model, or to select a machine learning model from model registry **3124**. In at least one embodiment, a user may implement their own machine learning model or select a machine learning model for inclusion in an application for performing a processing task. In at least one embodiment, applications may be selectable and customizable, and by defining constructs of applications, deployment and implementation of applications for a particular user are presented as a more seamless user experience. In at least one embodiment, by leveraging other features of system **3200**—such as services **3120** and hardware **3122**—deployment pipelines **3210** may be even more user friendly, provide for easier integration, and produce more accurate, efficient, and timely results.

In at least one embodiment, deployment system **3106** may include a user interface **3214** (e.g., a graphical user interface, a web interface, etc.) that may be used to select applications for inclusion in deployment pipeline(s) **3210**, arrange applications, modify or change applications or parameters or constructs thereof, use and interact with deployment pipeline(s) **3210** during set-up and/or deployment, and/or to otherwise interact with deployment system **3106**. In at least one embodiment, although not illustrated with respect to training system **3104**, user interface **3214** (or a different user interface) may be used for selecting models for use in deployment system **3106**, for selecting models for training, or retraining, in training system **3104**, and/or for otherwise interacting with training system **3104**.

In at least one embodiment, pipeline manager **3212** may be used, in addition to an application orchestration system **3228**, to manage interaction between applications or containers of deployment pipeline(s) **3210** and services **3120** and/or hardware **3122**. In at least one embodiment, pipeline manager **3212** may be configured to facilitate interactions from application to application, from application to service **3120**, and/or from application or service to hardware **3122**. In at least one embodiment, although illustrated as included in software **3118**, this is not intended to be limiting, and in some examples pipeline manager **3212** may be included in services **3120**. In at least one embodiment, application orchestration system **3228** (e.g., Kubernetes, DOCKER, etc.) may include a container orchestration system that may group applications into containers as logical units for coordination, management, scaling, and deployment. In at least one embodiment, by associating applications from deployment pipeline(s) **3210** (e.g., a reconstruction application, a segmentation application, etc.) with individual containers, each application may execute in a self-contained environment (e.g., at a kernel level) to increase speed and efficiency.

In at least one embodiment, each application and/or container (or image thereof) may be individually developed, modified, and deployed (e.g., a first user or developer may develop, modify, and deploy a first application and a second user or developer may develop, modify, and deploy a second application separate from a first user or developer), which

may allow for focus on, and attention to, a task of a single application and/or container(s) without being hindered by tasks of another application(s) or container(s). In at least one embodiment, communication, and cooperation between different containers or applications may be aided by pipeline manager **3212** and application orchestration system **3228**. In at least one embodiment, so long as an expected input and/or output of each container or application is known by a system (e.g., based on constructs of applications or containers), application orchestration system **3228** and/or pipeline manager **3212** may facilitate communication among and between, and sharing of resources among and between, each of applications or containers. In at least one embodiment, because one or more of applications or containers in deployment pipeline(s) **3210** may share same services and resources, application orchestration system **3228** may orchestrate, load balance, and determine sharing of services or resources between and among various applications or containers. In at least one embodiment, a scheduler may be used to track resource requirements of applications or containers, current usage or planned usage of these resources, and resource availability. In at least one embodiment, a scheduler may thus allocate resources to different applications and distribute resources between and among applications in view of requirements and availability of a system. In some examples, a scheduler (and/or other component of application orchestration system **3228**) may determine resource availability and distribution based on constraints imposed on a system (e.g., user constraints), such as quality of service (QoS), urgency of need for data outputs (e.g., to determine whether to execute real-time processing or delayed processing), etc.

In at least one embodiment, services **3120** leveraged by and shared by applications or containers in deployment system **3106** may include compute services **3216**, AI services **3218**, visualization services **3220**, and/or other service types. In at least one embodiment, applications may call (e.g., execute) one or more of services **3120** to perform processing operations for an application. In at least one embodiment, compute services **3216** may be leveraged by applications to perform super-computing or other high-performance computing (HPC) tasks. In at least one embodiment, compute service(s) **3216** may be leveraged to perform parallel processing (e.g., using a parallel computing platform **3230**) for processing data through one or more of applications and/or one or more tasks of a single application, substantially simultaneously. In at least one embodiment, parallel computing platform **3230** (e.g., NVIDIA's CUDA) may enable general purpose computing on GPUs (GPGPU) (e.g., GPUs **3222**). In at least one embodiment, a software layer of parallel computing platform **3230** may provide access to virtual instruction sets and parallel computational elements of GPUs, for execution of compute kernels. In at least one embodiment, parallel computing platform **3230** may include memory and, in some embodiments, a memory may be shared between and among multiple containers, and/or between and among different processing tasks within a single container. In at least one embodiment, inter-process communication (IPC) calls may be generated for multiple containers and/or for multiple processes within a container to use same data from a shared segment of memory of parallel computing platform **3230** (e.g., where multiple different stages of an application or multiple applications are processing same information). In at least one embodiment, rather than making a copy of data and moving data to different locations in memory (e.g., a read/write operation), same data in same location of a memory may be used for any

number of processing tasks (e.g., at a same time, at different times, etc.). In at least one embodiment, as data is used to generate new data as a result of processing, this information of a new location of data may be stored and shared between various applications. In at least one embodiment, location of data and a location of updated or modified data may be part of a definition of how a payload is understood within containers.

In at least one embodiment, AI services **3218** may be leveraged to perform inferencing services for executing machine learning model(s) associated with applications (e.g., tasked with performing one or more processing tasks of an application). In at least one embodiment, AI services **3218** may leverage AI system **3224** to execute machine learning model(s) (e.g., neural networks, such as CNNs) for segmentation, reconstruction, object detection, feature detection, classification, and/or other inferencing tasks. In at least one embodiment, applications of deployment pipeline(s) **3210** may use one or more of output models **3116** from training system **3104** and/or other models of applications to perform inference on imaging data (e.g., DICOM data, RIS data, CIS data, REST compliant data, RPC data, raw data, etc.). In at least one embodiment, two or more examples of inferencing using application orchestration system **3228** (e.g., a scheduler) may be available. In at least one embodiment, a first category may include a high priority/low latency path that may achieve higher service level agreements, such as for performing inference on urgent requests during an emergency, or for a radiologist during diagnosis. In at least one embodiment, a second category may include a standard priority path that may be used for requests that may be non-urgent or where analysis may be performed at a later time. In at least one embodiment, application orchestration system **3228** may distribute resources (e.g., services **3120** and/or hardware **3122**) based on priority paths for different inferencing tasks of AI services **3218**.

In at least one embodiment, shared storage may be mounted to AI services **3218** within system **3200**. In at least one embodiment, shared storage may operate as a cache (or other storage device type) and may be used to process inference requests from applications. In at least one embodiment, when an inference request is submitted, a request may be received by a set of API instances of deployment system **3106**, and one or more instances may be selected (e.g., for best fit, for load balancing, etc.) to process a request. In at least one embodiment, to process a request, a request may be entered into a database, a machine learning model may be located from model registry **3124** if not already in a cache, a validation step may ensure appropriate machine learning model is loaded into a cache (e.g., shared storage), and/or a copy of a model may be saved to a cache. In at least one embodiment, a scheduler (e.g., of pipeline manager **3212**) may be used to launch an application that is referenced in a request if an application is not already running or if there are not enough instances of an application. In at least one embodiment, if an inference server is not already launched to execute a model, an inference server may be launched. Any number of inference servers may be launched per model. In at least one embodiment, in a pull model, in which inference servers are clustered, models may be cached whenever load balancing is advantageous. In at least one embodiment, inference servers may be statically loaded in corresponding, distributed servers.

In at least one embodiment, inferencing may be performed using an inference server that runs in a container. In at least one embodiment, an instance of an inference server may be associated with a model (and optionally a plurality

of versions of a model). In at least one embodiment, if an instance of an inference server does not exist when a request to perform inference on a model is received, a new instance may be loaded. In at least one embodiment, when starting an inference server, a model may be passed to an inference server such that a same container may be used to serve different models so long as inference server is running as a different instance.

In at least one embodiment, during application execution, an inference request for a given application may be received, and a container (e.g., hosting an instance of an inference server) may be loaded (if not already), and a start procedure may be called. In at least one embodiment, pre-processing logic in a container may load, decode, and/or perform any additional pre-processing on incoming data (e.g., using a CPU(s) and/or GPU(s)). In at least one embodiment, once data is prepared for inference, a container may perform inference as necessary on data. In at least one embodiment, this may include a single inference call on one image (e.g., a hand X-ray), or may require inference on hundreds of images (e.g., a chest CT). In at least one embodiment, an application may summarize results before completing, which may include, without limitation, a single confidence score, pixel level-segmentation, voxel-level segmentation, generating a visualization, or generating text to summarize findings. In at least one embodiment, different models or applications may be assigned different priorities. For example, some models may have a real-time (TAT<1 min) priority while others may have lower priority (e.g., TAT<10 min). In at least one embodiment, model execution times may be measured from requesting institution or entity and may include partner network traversal time, as well as execution on an inference service.

In at least one embodiment, transfer of requests between services **3120** and inference applications may be hidden behind a software development kit (SDK), and robust transport may be provide through a queue. In at least one embodiment, a request will be placed in a queue via an API for an individual application/tenant ID combination and an SDK will pull a request from a queue and give a request to an application. In at least one embodiment, a name of a queue may be provided in an environment from where an SDK will pick it up. In at least one embodiment, asynchronous communication through a queue may be useful as it may allow any instance of an application to pick up work as it becomes available. Results may be transferred back through a queue, to ensure no data is lost. In at least one embodiment, queues may also provide an ability to segment work, as highest priority work may go to a queue with most instances of an application connected to it, while lowest priority work may go to a queue with a single instance connected to it that processes tasks in an order received. In at least one embodiment, an application may run on a GPU-accelerated instance generated in cloud **3226**, and an inference service may perform inferencing on a GPU.

In at least one embodiment, visualization services **3220** may be leveraged to generate visualizations for viewing outputs of applications and/or deployment pipeline(s) **3210**. In at least one embodiment, GPUs **3222** may be leveraged by visualization services **3220** to generate visualizations. In at least one embodiment, rendering effects, such as ray-tracing, may be implemented by visualization services **3220** to generate higher quality visualizations. In at least one embodiment, visualizations may include, without limitation, 2D image renderings, 3D volume renderings, 3D volume reconstruction, 2D tomographic slices, virtual reality displays, augmented reality displays, etc. In at least one

embodiment, virtualized environments may be used to generate a virtual interactive display or environment (e.g., a virtual environment) for interaction by users of a system (e.g., doctors, nurses, radiologists, etc.). In at least one embodiment, visualization services **3220** may include an internal visualizer, cinematics, and/or other rendering or image processing capabilities or functionality (e.g., ray tracing, rasterization, internal optics, etc.).

In at least one embodiment, hardware **3122** may include GPUs **3222**, AI system **3224**, cloud **3226**, and/or any other hardware used for executing training system **3104** and/or deployment system **3106**. In at least one embodiment, GPUs **3222** (e.g., NVIDIA's TESLA and/or QUADRO GPUs) may include any number of GPUs that may be used for executing processing tasks of compute services **3216**, AI services **3218**, visualization services **3220**, other services, and/or any of features or functionality of software **3118**. For example, with respect to AI services **3218**, GPUs **3222** may be used to perform pre-processing on imaging data (or other data types used by machine learning models), post-processing on outputs of machine learning models, and/or to perform inferencing (e.g., to execute machine learning models). In at least one embodiment, cloud **3226**, AI system **3224**, and/or other components of system **3200** may use GPUs **3222**. In at least one embodiment, cloud **3226** may include a GPU-optimized platform for deep learning tasks. In at least one embodiment, AI system **3224** may use GPUs, and cloud **3226**—or at least a portion tasked with deep learning or inferencing—may be executed using one or more AI systems **3224**. As such, although hardware **3122** is illustrated as discrete components, this is not intended to be limiting, and any components of hardware **3122** may be combined with, or leveraged by, any other components of hardware **3122**.

In at least one embodiment, AI system **3224** may include a purpose-built computing system (e.g., a super-computer or an HPC) configured for inferencing, deep learning, machine learning, and/or other artificial intelligence tasks. In at least one embodiment, AI system **3224** (e.g., NVIDIA's DGX) may include GPU-optimized software (e.g., a software stack) that may be executed using a plurality of GPUs **3222**, in addition to CPUs, RAM, storage, and/or other components, features, or functionality. In at least one embodiment, one or more AI systems **3224** may be implemented in cloud **3226** (e.g., in a data center) for performing some or all of AI-based processing tasks of system **3200**.

In at least one embodiment, cloud **3226** may include a GPU-accelerated infrastructure (e.g., NVIDIA's NGC) that may provide a GPU-optimized platform for executing processing tasks of system **3200**. In at least one embodiment, cloud **3226** may include an AI system(s) **3224** for performing one or more of AI-based tasks of system **3200** (e.g., as a hardware abstraction and scaling platform). In at least one embodiment, cloud **3226** may integrate with application orchestration system **3228** leveraging multiple GPUs to enable seamless scaling and load balancing between and among applications and services **3120**. In at least one embodiment, cloud **3226** may be tasked with executing at least some of services **3120** of system **3200**, including compute services **3216**, AI services **3218**, and/or visualization services **3220**, as described herein. In at least one embodiment, cloud **3226** may perform small and large batch inference (e.g., executing NVIDIA's TENSOR RT), provide an accelerated parallel computing API and platform **3230** (e.g., NVIDIA's CUDA), execute application orchestration system **3228** (e.g., KUBERNETES), provide a graphics rendering API and platform (e.g., for ray-tracing, 2D graphics,

3D graphics, and/or other rendering techniques to produce higher quality cinematics), and/or may provide other functionality for system **3200**.

In at least one embodiment, in an effort to preserve patient confidentiality (e.g., where patient data or records are to be used off-premises), cloud **3226** may include a registry—such as a deep learning container registry. In at least one embodiment, a registry may store containers for instantiations of applications that may perform pre-processing, post-processing, or other processing tasks on patient data. In at least one embodiment, cloud **3226** may receive data that includes patient data as well as sensor data in containers, perform requested processing for just sensor data in those containers, and then forward a resultant output and/or visualizations to appropriate parties and/or devices (e.g., on-premises medical devices used for visualization or diagnoses), all without having to extract, store, or otherwise access patient data. In at least one embodiment, confidentiality of patient data is preserved in compliance with HIPAA and/or other data regulations.

FIG. **33A** illustrates a data flow diagram for a process **3300** to train, retrain, or update a machine learning model, in accordance with at least one embodiment. In at least one embodiment, process **3300** may be executed using, as a non-limiting example, system **3200** of FIG. **32**. In at least one embodiment, process **3300** may leverage services **3120** and/or hardware **3122** of system **3200**, as described herein. In at least one embodiment, refined models **3312** generated by process **3300** may be executed by deployment system **3106** for one or more containerized applications in deployment pipelines **3210**.

In at least one embodiment, model training **3114** may include retraining or updating an initial model **3304** (e.g., a pre-trained model) using new training data (e.g., new input data, such as customer dataset **3306**, and/or new ground truth data associated with input data). In at least one embodiment, to retrain, or update, initial model **3304**, output or loss layer(s) of initial model **3304** may be reset, or deleted, and/or replaced with an updated or new output or loss layer(s). In at least one embodiment, initial model **3304** may have previously fine-tuned parameters (e.g., weights and/or biases) that remain from prior training, so training or retraining **3114** may not take as long or require as much processing as training a model from scratch. In at least one embodiment, during model training **3114**, by having reset or replaced output or loss layer(s) of initial model **3304**, parameters may be updated and re-tuned for a new data set based on loss calculations associated with accuracy of output or loss layer(s) at generating predictions on new, customer dataset **3306** (e.g., image data **3108** of FIG. **31**).

In at least one embodiment, pre-trained models **3206** may be stored in a data store, or registry (e.g., model registry **3124** of FIG. **31**). In at least one embodiment, pre-trained models **3206** may have been trained, at least in part, at one or more facilities other than a facility executing process **3300**. In at least one embodiment, to protect privacy and rights of patients, subjects, or clients of different facilities, pre-trained models **3206** may have been trained, on-premise, using customer or patient data generated on-premise. In at least one embodiment, pre-trained models **3206** may be trained using cloud **3226** and/or other hardware **3122**, but confidential, privacy protected patient data may not be transferred to, used by, or accessible to any components of cloud **3226** (or other off premise hardware). In at least one embodiment, where a pre-trained model **3206** is trained at using patient data from more than one facility, pre-trained model **3206** may have been individually trained for each

facility prior to being trained on patient or customer data from another facility. In at least one embodiment, such as where a customer or patient data has been released of privacy concerns (e.g., by waiver, for experimental use, etc.), or where a customer or patient data is included in a public data set, a customer or patient data from any number of facilities may be used to train pre-trained model **3206** on-premise and/or off premise, such as in a datacenter or other cloud computing infrastructure.

In at least one embodiment, when selecting applications for use in deployment pipelines **3210**, a user may also select machine learning models to be used for specific applications. In at least one embodiment, a user may not have a model for use, so a user may select a pre-trained model **3206** to use with an application. In at least one embodiment, pre-trained model **3206** may not be optimized for generating accurate results on customer dataset **3306** of a facility of a user (e.g., based on patient diversity, demographics, types of medical imaging devices used, etc.). In at least one embodiment, prior to deploying pre-trained model **3206** into deployment pipeline **3210** for use with an application(s), pre-trained model **3206** may be updated, retrained, and/or fine-tuned for use at a respective facility.

In at least one embodiment, a user may select pre-trained model **3206** that is to be updated, retrained, and/or fine-tuned, and pre-trained model **3206** may be referred to as initial model **3304** for training system **3104** within process **3300**. In at least one embodiment, customer dataset **3306** (e.g., imaging data, genomics data, sequencing data, or other data types generated by devices at a facility) may be used to perform model training **3114** (which may include, without limitation, transfer learning) on initial model **3304** to generate refined model **3312**. In at least one embodiment, ground truth data corresponding to customer dataset **3306** may be generated by training system **3104**. In at least one embodiment, ground truth data may be generated, at least in part, by clinicians, scientists, doctors, practitioners, at a facility (e.g., as labeled clinic data **3112** of FIG. **31**).

In at least one embodiment, AI-assisted annotation **3110** may be used in some examples to generate ground truth data. In at least one embodiment, AI-assisted annotation **3110** (e.g., implemented using an AI-assisted annotation SDK) may leverage machine learning models (e.g., neural networks) to generate suggested or predicted ground truth data for a customer dataset. In at least one embodiment, user **3310** may use annotation tools within a user interface (a graphical user interface (GUI)) on computing device **3308**.

In at least one embodiment, user **3310** may interact with a GUI via computing device **3308** to edit or fine-tune (auto)annotations. In at least one embodiment, a polygon editing feature may be used to move vertices of a polygon to more accurate or fine-tuned locations.

In at least one embodiment, once customer dataset **3306** has associated ground truth data, ground truth data (e.g., from AI-assisted annotation, manual labeling, etc.) may be used by during model training **3114** to generate refined model **3312**. In at least one embodiment, customer dataset **3306** may be applied to initial model **3304** any number of times, and ground truth data may be used to update parameters of initial model **3304** until an acceptable level of accuracy is attained for refined model **3312**. In at least one embodiment, once refined model **3312** is generated, refined model **3312** may be deployed within one or more deployment pipelines **3210** at a facility for performing one or more processing tasks with respect to medical imaging data.

In at least one embodiment, refined model **3312** may be uploaded to pre-trained models **3206** in model registry **3124**

to be selected by another facility. In at least one embodiment, his process may be completed at any number of facilities such that refined model **3312** may be further refined on new datasets any number of times to generate a more universal model.

FIG. **33B** is an example illustration of a client-server architecture **3332** to enhance annotation tools with pre-trained annotation models, in accordance with at least one embodiment. In at least one embodiment, AI-assisted annotation tools **3336** may be instantiated based on a client-server architecture **3332**. In at least one embodiment, annotation tools **3336** in imaging applications may aid radiologists, for example, identify organs and abnormalities. In at least one embodiment, imaging applications may include software tools that help user **3310** to identify, as a non-limiting example, a few extreme points on a particular organ of interest in raw images **3334** (e.g., in a 3D MRI or CT scan) and receive auto-annotated results for all 2D slices of a particular organ. In at least one embodiment, results may be stored in a data store as training data **3338** and used as (for example and without limitation) ground truth data for training. In at least one embodiment, when computing device **3308** sends extreme points for AI-assisted annotation **3110**, a deep learning model, for example, may receive this data as input and return inference results of a segmented organ or abnormality. In at least one embodiment, pre-instantiated annotation tools, such as AI-Assisted Annotation Tool **3336B** in FIG. **33B**, may be enhanced by making API calls (e.g., API Call **3344**) to a server, such as an Annotation Assistant Server **3340** that may include a set of pre-trained models **3342** stored in an annotation model registry, for example. In at least one embodiment, an annotation model registry may store pre-trained models **3342** (e.g., machine learning models, such as deep learning models) that are pre-trained to perform AI-assisted annotation on a particular organ or abnormality. These models may be further updated by using training pipelines **3204**. In at least one embodiment, pre-installed annotation tools may be improved over time as new labeled clinic data **3112** is added.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to determine gaze of a user based on coarse and fine pupil estimates.

Other variations are within spirit of present disclosure. Thus, while disclosed techniques are susceptible to various modifications and alternative constructions, certain illustrated embodiments thereof are shown in drawings and have been described above in detail. It should be understood, however, that there is no intention to limit disclosure to specific form or forms disclosed, but on contrary, intention is to cover all modifications, alternative constructions, and equivalents falling within spirit and scope of disclosure, as defined in appended claims.

Use of terms “a” and “an” and “the” and similar referents in context of describing disclosed embodiments (especially in context of following claims) are to be construed to cover both singular and plural, unless otherwise indicated herein or clearly contradicted by context, and not as a definition of a term. Terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (meaning “including, but not limited to,”) unless otherwise noted. Term “connected,” when unmodified and referring to physical connections, is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand

method of referring individually to each separate value falling within range, unless otherwise indicated herein and each separate value is incorporated into specification as if it were individually recited herein. Use of term “set” (e.g., “a set of items”) or “subset,” unless otherwise noted or contradicted by context, is to be construed as a nonempty collection comprising one or more members. Further, unless otherwise noted or contradicted by context, term “subset” of a corresponding set does not necessarily denote a proper subset of corresponding set, but subset and corresponding set may be equal.

Conjunctive language, such as phrases of form “at least one of A, B, and C,” or “at least one of A, B and C,” unless specifically stated otherwise or otherwise clearly contradicted by context, is otherwise understood with context as used in general to present that an item, term, etc., may be either A or B or C, or any nonempty subset of set of A and B and C. For instance, in illustrative example of a set having three members, conjunctive phrases “at least one of A, B, and C” and “at least one of A, B and C” refer to any of following sets: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, {A, B, C}. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of A, at least one of B, and at least one of C each to be present. In addition, unless otherwise noted or contradicted by context, term “plurality” indicates a state of being plural (e.g., “a plurality of items” indicates multiple items). A plurality is at least two items, but can be more when so indicated either explicitly or by context. Further, unless stated otherwise or otherwise clear from context, phrase “based on” means “based at least in part on” and not “based solely on.”

Operations of processes described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. In at least one embodiment, a process such as those processes described herein (or variations and/or combinations thereof) is performed under control of one or more computer systems configured with executable instructions and is implemented as code (e.g., executable instructions, one or more computer programs or one or more applications) executing collectively on one or more processors, by hardware or combinations thereof. In at least one embodiment, code is stored on a computer-readable storage medium, for example, in form of a computer program comprising a plurality of instructions executable by one or more processors. In at least one embodiment, a computer-readable storage medium is a non-transitory computer-readable storage medium that excludes transitory signals (e.g., a propagating transient electric or electromagnetic transmission) but includes non-transitory data storage circuitry (e.g., buffers, cache, and queues) within transceivers of transitory signals. In at least one embodiment, code (e.g., executable code or source code) is stored on a set of one or more non-transitory computer-readable storage media having stored thereon executable instructions (or other memory to store executable instructions) that, when executed (i.e., as a result of being executed) by one or more processors of a computer system, cause computer system to perform operations described herein. A set of non-transitory computer-readable storage media, in at least one embodiment, comprises multiple non-transitory computer-readable storage media and one or more of individual non-transitory storage media of multiple non-transitory computer-readable storage media lack all of code while multiple non-transitory computer-readable storage media collectively store all of code. In at least one embodiment, executable instructions are executed such that

different instructions are executed by different processors. For example, a non-transitory computer-readable storage medium stores instructions and a main central processing unit (“CPU”) executes some of instructions while a graphics processing unit (“GPU”) executes other instructions. In at least one embodiment, different components of a computer system have separate processors and different processors execute different subsets of instructions.

Accordingly, in at least one embodiment, computer systems are configured to implement one or more services that singly or collectively perform operations of processes described herein and such computer systems are configured with applicable hardware and/or software that enable performance of operations. Further, a computer system that implements at least one embodiment of present disclosure is a single device and, in another embodiment, is a distributed computer system comprising multiple devices that operate differently such that distributed computer system performs operations described herein and such that a single device does not perform all operations.

Use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of disclosure and does not pose a limitation on scope of disclosure unless otherwise claimed. No language in specification should be construed as indicating any non-claimed element as essential to practice of disclosure.

All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

In description and claims, terms “coupled” and “connected,” along with their derivatives, may be used. It should be understood that these terms may be not intended as synonyms for each other. Rather, in particular examples, “connected” or “coupled” may be used to indicate that two or more elements are in direct or indirect physical or electrical contact with each other. “Coupled” may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

Unless specifically stated otherwise, it may be appreciated that throughout specification terms such as “processing,” “computing,” “calculating,” “determining,” or like, refer to action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities within computing system’s registers and/or memories into other data similarly represented as physical quantities within computing system’s memories, registers or other such information storage, transmission or display devices.

In a similar manner, term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory and transform that electronic data into other electronic data that may be stored in registers and/or memory. As non-limiting examples, “processor” may be a CPU or a GPU. A “computing platform” may comprise one or more processors. As used herein, “software” processes may include, for example, software and/or hardware entities that perform work over time, such as tasks, threads, and intelligent agents. Also, each process may refer to multiple processes, for carrying out instructions in sequence or in parallel, continuously or intermittently. Terms “system” and “method” are used herein interchangeably insofar as system may embody one or more methods and methods may be considered a system.

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In present document, references may be made to obtaining, acquiring, receiving, or inputting analog or digital data into a subsystem, computer system, or computer-implemented machine. Obtaining, acquiring, receiving, or inputting analog and digital data can be accomplished in a variety of ways such as by receiving data as a parameter of a function call or a call to an application programming interface. In some implementations, process of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a serial or parallel interface. In another implementation, process of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a computer network from providing entity to acquiring entity. References may also be made to providing, outputting, transmitting, sending, or presenting analog or digital data. In various examples, process of providing, outputting, transmitting, sending, or presenting analog or digital data can be accomplished by transferring data as an input or output parameter of a function call, a parameter of an application programming interface or interprocess communication mechanism.

Although discussion above sets forth example implementations of described techniques, other architectures may be used to implement described functionality, and are intended to be within scope of this disclosure. Furthermore, although specific distributions of responsibilities are defined above for purposes of discussion, various functions and responsibilities might be distributed and divided in different ways, depending on circumstances.

Furthermore, although subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that subject matter claimed in appended claims is not necessarily limited to specific features or acts described. Rather, specific features and acts are disclosed as exemplary forms of implementing the claims.

What is claimed is:

1. One or more processors, comprising:
circuitry to use an amount by which a first image has changed from a second image to select whether to use one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils within the first image, wherein one or more second portions of the one or more neural networks are used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image when the amount is below a threshold amount.
2. The one or more processors of claim 1, wherein the circuitry to use the one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils further comprises passing a full eye image from one or more images of one or more eyes of one or more users through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.
3. The one or more processors of claim 1, wherein the circuitry to use the one or more second portions of one or more neural networks used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image further comprises passing smaller sub-regions of a full eye image through the one or more second portions of one or more neural networks, wherein the full eye image is the prior image that has passed through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

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4. The one or more processors of claim 1, wherein the one or more first portions of one or more neural networks and the one or more second portions of one or more neural networks use a same convolutional neural network (CNN)-based architecture but with different network parameters.

5. The one or more processors of claim 1, wherein the threshold amount comprises an amount of lateral movement of a pupil in an image.

6. The one or more processors of claim 1, wherein the second image includes one or more separate images for both eyes of one of one or more users, and wherein the circuitry is further to concatenate those separate images into a concatenated image with padding between representations of the eyes before providing the concatenated image as input to the one or more first portions of the one or more neural networks.

7. A system comprising:

one or more processors to use an amount by which a first image has changed from a second image to select whether to use one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils within the first image, wherein one or more second portions of the one or more neural networks are used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image when the amount is below a threshold amount.

8. The system of claim 7, wherein the one or more processors to use one or more first portions of the one or more neural networks to generate a coarse estimate of a location of one or more pupils further comprises passing a full eye image from one or more images of one or more eyes of one or more users through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

9. The system of claim 7, wherein the one or more processors to use one or more second portions of the one or more neural networks used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image further comprises passing smaller sub-regions of a full eye image through the one or more second portions of one or more neural networks, wherein the full eye image is the prior image that has passed through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

10. The system of claim 7, wherein the one or more first portions of one or more neural networks and the one or more second portions of one or more neural networks use a same convolutional neural network (CNN)-based architecture but with different network parameters.

11. The system of claim 7, wherein the threshold amount comprises an amount of lateral movement of a pupil in an image.

12. The system of claim 7, wherein the second image includes one or more separate images for both eyes of one of one or more users, and wherein the one or more processors are further to concatenate those separate images into a concatenated image with padding between representations of the eyes before providing the concatenated image as input to the one or more first portions of the one or more neural networks.

13. A method comprising:

using an amount by which a first image has changed from a second image to select whether to use one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more

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pupils within the first image, wherein one or more second portions of the one or more neural networks are used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image when the amount is below a threshold amount.

14. The method of claim 13, wherein using the one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils further comprises passing a full eye image from one or more images of one or more eyes of one or more users through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

15. The method of claim 13, wherein using the one or more second portions of one or more neural networks to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image further comprises passing smaller sub-regions of a full eye image through the one or more second portions of one or more neural networks, wherein the full eye image is the prior image that has passed through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

16. The method of claim 13, wherein the one or more first portions of one or more neural networks and the one or more second portions of one or more neural networks use a same convolutional neural network (CNN)-based architecture but with different network parameters.

17. The method of claim 13, wherein threshold amount comprises an amount of lateral movement of a pupil in an image.

18. The method of claim 13, wherein the second image includes one or more separate images for both eyes of one of one or more users, and further comprising concatenating those separate images into a concatenated image with padding between representations of the eyes before providing the concatenated image as input to the one or more first portions of the one or more neural networks.

19. A non-transitory machine-readable medium having stored thereon a set of instructions, which if performed by one or more processors, cause the one or more processors to at least:

use an amount by which a first image has changed from a second image to select whether to use one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils within the first image, wherein one or more second portions of the one or more neural networks are used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image when the amount is below a threshold amount.

20. The non-transitory machine-readable medium of claim 19, wherein the one or more processors to use one or more first portions of the one or more neural networks to generate a coarse estimate of a location of one or more pupils further comprises passing a full eye image from one or more images of one or more eyes of one or more users through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

21. The non-transitory machine-readable medium of claim 19, wherein the one or more processors to use one or more second portions of the one or more neural networks used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate

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from the second image further comprises passing smaller sub-regions of a full eye image through the one or more second portions of one or more neural networks, wherein the full eye image is the prior image that has passed through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

22. The non-transitory machine-readable medium of claim 19, wherein the one or more first portions of one or more neural networks and the one or more second portions of one or more neural networks use a same convolutional neural network (CNN)-based architecture but with different network parameters.

23. The non-transitory machine-readable medium of claim 19, wherein the

threshold amount comprises an amount of lateral movement of a pupil in an image.

24. The non-transitory machine-readable medium of claim 20, wherein the second image includes one or more separate images for both eyes of one of one or more users, and wherein the one or more processors are further to concatenate those separate images into a concatenated image with padding between representations of the eyes before providing the concatenated image as input to the one or more first portions of the one or more neural networks.

25. A gaze estimation system, comprising:

one or more processors to use an amount by which a first image has changed from a second image to select whether to use one or more first portions of one or more neural networks to generate a coarse estimate of a location of one or more pupils within the first image, wherein one or more second portions of the one or more neural networks are used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image when the amount is below a threshold amount.

26. The gaze estimation system of claim 25, wherein the one or more processors to use one or more first portions of the one or more neural networks to generate a coarse estimate of a location of one or more pupils further comprises passing a full eye image from one or more images of one or more eyes of one or more users through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

27. The gaze estimation system of claim 25, wherein the one or more processors to use one or more second portions of the one or more neural networks used to generate a fine estimate of the location of the one or more pupils within the first image using a prior estimate from the second image further comprises passing smaller sub-regions of a full eye image through the one or more second portions of one or more neural networks, wherein the full eye image is the prior image that has passed through the one or more first portions of one or more neural networks to generate smaller sub-regions of the full eye image.

28. The gaze estimation system of claim 25, wherein the one or more first portions of one or more neural networks and the one or more second portions of one or more neural networks use a same convolutional neural network (CNN)-based architecture but with different network parameters.

29. The gaze estimation system of claim 25, wherein the threshold amount comprises an amount of lateral movement of a pupil in an image.

30. The gaze estimation system of claim 25, wherein the second image includes one or more separate images for both eyes of one of one or more users, and wherein the one or more processors are further to concatenate those separate images into a concatenated image with padding between

representations of the eyes before providing the concatenated image as input to the one or more first portions of the one or more neural networks.

31. The one or more processors of claim 1, wherein the circuitry to select whether to use the one or more first 5 portions of the one or more neural networks, is further to use only the one or more second portions of the one or more neural networks based on the threshold amount, wherein the threshold amount comprises the amount by which the pupil location changes from the first image to the second image. 10

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